Another Brick in the Wall?
The Educational Effects of Repurposed Mafia Properties

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Abstract

Italy's anti-Mafia legislation allows confiscated Mafia properties to be converted into educational, cultural, and welfare facilities where local NGOs offer various social activities specifically targeting the youth and other vulnerable groups. This study provides the first causal evidence on how exposure to these repurposed spaces affects students' dropout rates by changing their attitudes toward educational and criminal pathways. Using school-level geo-referenced data from 2015 to 2022 and exploiting the staggered timing of property repurposing, I investigate changes in local dropout rates and students' aspirations. Results find a significant reduction in dropout rates of approximately 29% for students living near repurposed properties. Students are 24% less likely to expect to find occupation through Mafia connections, and 34% more likely to value education investment. Students also report greater confidence in State capacity, with reduced distrust in local police and politicians. Sentiment analysis of students' definitions of the Mafia corroborates these results, with the strongest effects among students from lower socioeconomic backgrounds. The effects are not explained by gentrification, additional educational support, or civic engagement levels.

Keywords: Mafia; property; Italy; education; NGOs; perception; State.

JEL Codes: R23, H72, I25, K42.

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1 Introduction

The detrimental effects of organised crime on education are well-documented, particularly in areas controlled by Mafia-type groups where these effects are highly localised. Juveniles can be directly recruited by Mafia affiliates, but they can also be affected by criminal social norms and behavioural codes (van Dijk, Kleemans and Eichelsheim, 2019; Kleemans and de Poot, 2008). In Italy, Mafia groups have been shown to systematically distort youth's aspirations about investing in education versus pursuing criminal alternatives (Acemoglu, De Feo and De Luca, 2020; Coniglio, Celi and Scagliusi, 2010; Caglayan, Flamini and Jahanshahi, 2017), prompting policymakers to develop targeted interventions for Mafia-affected communities. Among these interventions, the most significant is the Mafia Confiscated and Reallocated Assets Reuse Policy (CRR) implemented nationwide since 1996. Under this policy, the confiscated Mafia properties, former symbols of territorial control and criminal activity, are converted into educational, cultural and welfare facilities, where local NGOs offer various social activities specifically targeting young people and other vulnerable groups. These repurposed spaces aim to transform former Mafia strongholds into community assets that foster legitimate social bonds and provide alternative role models (Falcone, Giannone and Iandolo, 2016). The question remains, however, whether this symbolic transformation translates into measurable social and economic benefits for the youth. Existing case studies document positive effects on communities' social participation (Nazzaro, 2021; Martone, 2020), while establishing that Mafia social bonds may prove resilient to these policy efforts. I address this empirical gap by documenting the evolution of repurposing practices and their subsequent impacts on educational outcomes.

In this paper, I construct a novel dataset and leverage the quasi-experimental variation in the timing and use of repurposed Mafia properties to provide the first empirical evidence on their impact on local education. Moreover, I investigate the mechanisms driving these effects by examining several potential channels. I focus on the ten main urban centres in historically Mafia-ridden regions, which concentrate the majority of confiscated Mafia properties[1]. Young people's educational outcomes tend to be especially low in these regions, with dropout rates reaching concerning levels: 14.6% in Campania and Apulia, and 18.8% in Sicily against a national average of 12.7% (Save The Children, 2022). Local NGOs and municipalities manage these repurposed properties to operate programs focused on education, culture, skills training, and social services, with 45% of them explicitly targeting the youth. In this setting I exploit the staggered repurpose of Mafia properties, conditioning on neighbourhood and municipality-level attributes.

I construct a novel database combining information scraped from the website *Confiscatibene*, internal surveys from the anti-Mafia Association *Libera*, municipal records, and administrative data on the precise locations of confiscated properties. I digitised Antimafia Investigative Directory

[1]The cities I investigate include Naples, Salerno, Bari, Taranto, Foggia, Reggio di Calabria, Palermo, Messina, Catania, and Siracusa. Focusing on the major Mafia-ridden urban centres allows me to leverage the necessary variation in Mafia real estate reallocation and repurposing practices.

maps measuring the baseline Mafia presence, and I draw on Ministry of Education records covering high school enrolment, graduation rates, and student ages. I further collect NGO location data from the Ministry of Labour, survey data about the students' perception of the Mafia, municipal expenditure and transparency records, and the 2011 census data to account for baseline socio-economic characteristics.

To estimate the effect of this policy on educational outcomes, I employ a DiD design that compares educational outcomes in areas where Mafia properties have been repurposed to those located in unaffected areas over the same period. My main outcome variable tracks dropout rates by following the same student cohorts across high school grades. This allows me to focus on students older than 16, which corresponds to the end of compulsory schooling age in Italy. This exercise allows me to measure students' dropout decisions when continued schooling becomes voluntary. Given the absence of official high school districts in Italy, I construct school catchment areas based on the distance between high schools and census blocks[2], defining them as treated if they contain at least one repurposed Mafia property and as control otherwise. In this setting, I estimate the average treatment effect on the treated (ATT) with traditional TWFE models as well as modern DiD estimators following Sun and Abraham (2021).

The effective repurpose of former Mafia properties significantly reduces dropout rates by 1.6 to 1.7 percentage points, approximately 29% relative to the mean, among students who have reached the age when formal education is no longer compulsory. The effect becomes significant from the second year after the first repurpose and persists for up to six years. It is consistent with previous research on the effect of the Mafia on education: Caglayan, Flamini and Jahanshahi (2021) find that Mafia presence reduces graduation rates by 25% in Northern Italy. Additionally, I test the intensive margin of the treatment. The effect is concentrated in catchment areas where more than one property has been repurposed, where dropout falls by 2.2 to 2.5 percentage points (36% to 40% relative to the mean), while it is small and insignificant where only one property has been repurposed. Using the cumulative number of repurposed properties, weighted by the catchment area's student-age population, as a continuous treatment in the de Chaisemartin and D'Haultfœuille (2024) estimator, I find that each additional weighted repurposed property reduces dropout by 0.59 percentage points (10% relative to the mean), with an effect that builds up over time. The consistency across binary and continuous treatment specifications reinforces confidence in the validity of the main results.

Moreover, the heterogeneity analysis reveals several important patterns. First, technical high schools emerge as the primary beneficiaries of the repurposing interventions, exhibiting sustained reductions in dropout rates of two to three percentage points that become statistically significant from the third year after the first repurpose. This contrasts sharply with academic and vocational high schools, which display small and statistically insignificant effects. This heterogeneous pattern aligns with the distinct institutional contexts and student populations served by each educational track.

[2]Census blocks are defined by the Italian Institute of Statistics and typically contain on average 200-400 inhabitants.

Unlike vocational students, who can enter the labour market with a three-year professional certificate, technical high school students need the full five-year diploma to meaningfully improve their employment prospects, exposing them to a longer period of risk during which dropout becomes increasingly likely. Second, the estimated effects are entirely driven by the repurpose of *operational Mafia properties*, namely villas and apartments that symbolised the local control of the Mafia family, while properties previously used for commercial purposes show no significant effect. Third, the effect is concentrated in properties hosting cultural and welfare activities, while properties offering education and employment services show smaller and insignificant reductions. This is consistent with the interpretation that what drives the reduction is less the specific service offered than the signal the reclaimed space itself sends. Together, these heterogeneous effects point to a common mechanism: signalling the retreat of criminal influence and the return of legitimate institutions is particularly effective in supporting students who face the greatest structural barriers to completing their education.

My empirical analysis suggests that the key mechanism is that the repurposing of formerly Mafia-owned properties provides a signal that the Mafia is weaker, redirecting youth preferences from illegal to legal pathways for career development. Survey data on student perceptions reveal that after their area receives treatment, students are more likely to regard asking the Mafia or a politician as a very useless way to find a job, and more likely to regard education as a very useful one. The timing of the effect indicates that students revise their views about the usefulness of the Mafia around the second and third years after treatment, on the same horizon at which dropout rates decline. This suggests students may be adjusting their perception of the returns to criminality in their local area[3]. I also test whether students perceive a higher cost of joining the Mafia through a renewed perception of State capacity. Students are less likely to agree that the Mafia is strong because it infiltrates the State, and less likely to name the police and local politicians among the institutions they trust least. The dynamic estimates for these outcomes are imprecise, but suggest that they emerge in the first years after repurposing and then fade, consistent with repurposed properties providing a short-term local signal of State capacity.

I test the robustness of these results against several alternative explanations. First, I address social desirability bias, since the surveys are administered by teachers in school. Sentiment analysis of students' written definitions of the Mafia shows an increase in negative terms, in both quantity and intensity. These effects are strongest among students producing the least linguistically complex definitions, a proxy for lower socioeconomic background and the group least equipped to reproduce

[3]I cannot observe the realised returns to education or to criminal activity, which would require following these cohorts into the labour market. What I observe instead is what students believe about their prospects at the point they decide whether to stay in school: students' beliefs about which pathways lead to employment. Students who previously expected to secure work through Mafia or political connections come to expect that education and formal application processes will do so instead. Since the dropout margin depends on perceived rather than realised returns, a shift in these beliefs is the proximate mechanism through which any change in returns would operate, and the coincident timing of the belief shift and the dropout response is consistent with this interpretation.

a socially approved framing, consistent with the interpretation that repurposed properties alter the trade-off facing youth who would otherwise encounter greater barriers to legitimate career advancement. These findings complement anecdotal evidence gathered through interviews on the effects of repurposing activities on young people (Nazzaro, 2021; Falcone et al., 2016). I also test whether students' answers simply reflect a change in the views of their households, and find null effects. Second, I test whether the effect operates through incapacitation. Confiscation of Mafia properties follows the arrest of the controlling family, removing them from the local area. If students were simply facing reduced recruitment opportunities, the response should appear at the earlier stages of the policy rather than at repurposing. Confiscation takes place on average more than a decade before repurposing, well outside the estimation window, and I find a null effect of reallocation on dropout: the decline appears only once the property gets repurposed, which is difficult to reconcile with incapacitation but consistent with the visible repurpose of these properties. This is also consistent with the fact that a share of confiscated assets are left vacant for years before any reassignment, and many are never reassigned at all (Falcone et al., 2016); seizure alone therefore leaves little visible trace in the neighbourhood, whereas repurposing sends a distinct signal. Finally, I test whether the results reflect a change in the perceived cost of education. I do this in two steps. Given that the effect is not driven by properties specifically offering after-school and homework-support activities, I first test whether additional educational or social programmes at the municipality level absorb part of the effect and whether the effect is visible on students' graduation grades, finding null results. Second, I find that the treatment effect does not vary with neighbourhood rental prices, making it unlikely to be driven by concurrent gentrification in specific areas.

Estimating the effects of repurposing former Mafia properties poses non-trivial challenges. The primary endogeneity concern is that repurposed properties are not randomly assigned across schools and neighbourhoods: their allocation may systematically reflect pre-existing differences in educational outcomes or local institutional capacity. Second, the multi-stage nature of the policy, spanning confiscation, reallocation to local authorities, and eventual repurposing, raises the question of whether the estimated effects capture repurposing itself or earlier stages of the process. A third concern is that the effect may reflect the provision of a local amenity rather than anything specific to the property's history, in which case any facility run by an NGO would produce the same result. To address these concerns, I conduct a battery of robustness and placebo exercises. I first show that there is no evidence of differential pre-trends between treated and control schools, ruling out mean reversion as an alternative explanation. I then show that the effects are not driven by reallocation prior to repurposing. Treated areas host more NGOs and have higher grade retention at baseline; the results are robust to reweighting control schools by inverse probability weights on these characteristics, and to controlling for NGO presence in the previous year and for trends in the municipal characteristics that predict repurposing. NGO presence itself shows no association with dropout, which suggests that the effect is specific to activities hosted in former Mafia properties rather than

to civil society presence in general.
Finally, I address selection and spillovers. The results are robust to restricting the sample to treated and not-yet-treated schools, to capacity-weighted catchment area definitions, and to controlling for spillovers between neighbouring catchments. I find no evidence of anticipation in the year before repurposing, and no effect of repurposing on student migration and grade retention.

To the best of my knowledge, this is the first paper providing causal estimates on the effects of the repurpose of Mafia properties on local education and social outcomes. Prior research established that the confiscation of Mafia properties increases the prices of commercial real estate located nearby as well as firms' performance, turnover and local market competition (Calamunci, Ferrante and Scebba, 2022; Calamunci and Drago, 2020; Ferrante, Fontana and Reito, 2021a; Operti, 2018). Moreover, the reallocation of such properties increases both market and electoral competition (Ferrante et al., 2021a; Ferrante, Reito, Spagano and Torrisi, 2021b). Boeri, Di Cataldo and Pietrostefani (2023) examines neighbourhood-level effects of Mafia properties' confiscation and reallocation on housing prices. Their findings indicate that while confiscations lead to a decline in nearby housing prices, reallocation drives their increase. I take a step forward from these works in two key directions. First, I leverage the staggered effect of the reuse of Mafia properties, demonstrating how this process translates policy into action by providing social activities, unlike reallocation, which remains a purely bureaucratic step. Second, I explore the impact of these activities on educational outcomes and the students' social outcomes, an area that has been entirely unexplored until now. For this purpose, the creation of a novel database of the reuse practices of Mafia properties represents an essential contribution. Furthermore, this paper builds on the insights of Nazzaro (2021) and Martone (2020), who offer qualitative case studies on the social effects of reusing Mafia properties in Apulia and Campania, respectively. These studies focus on outcomes related to education and the labour market, enabling me to identify credible mechanisms behind the effects I estimate.

Second, I contribute to the extensive literature investigating the relationship between neighbourhood-specific factors and human capital accumulation. Previous studies from Ludwig, Duncan, Gennetian, Katz, Kessler, Kling and Sanbonmatsu (2013), Chetty and Hendren (2018), and Bergman, Chetty, DeLuca, Hendren, Katz and Palmer (2024) show that community environment profoundly shapes children's educational trajectories and life outcomes. These studies have been typically focused on poverty, segregation, or physical neighbourhood quality. My contribution shows that public policies reshaping the institutional and social environment can make education a more viable pathway for the local youth. This extends our understanding of neighbourhood effects to contexts where criminal governance is a key social barrier to human capital investment.

Last, this study provides new evidence on how educational and social activities shape juveniles' perceptions of future opportunities, ultimately influencing their choices and life trajectories. Prior work in experimental economics shows that localised, tailor-made programs can play a transformative role not only in improving educational outcomes but also in reducing crime (Heller, Shah, Guryan, Ludwig, Mullainathan and Pollack, 2017; García, Heckman and Ziff, 2019), as well as

in strengthening students' interpersonal trust and self-esteem (Kenney and Godson, 2002). Other studies highlight how targeted policies can counteract parental incentives to involve children in illegal markets, thereby reducing pathways into criminality. For instance, Conditional Cash Transfer (CCT) programs increase the opportunity cost of a criminal career and enhance the relative benefits of pursuing education (Sviatschi, 2022a,b). This study advances the literature by focusing on Mafia-ridden areas as a novel setting, showing how educational and social programs can alter juveniles' beliefs about the trade-offs between illicit involvement and educational opportunities.

This paper offers policy implications for young people in Mafia-affected communities. First, the results show that repurposing operates through information rather than through service provision. The effects are not driven by properties offering after-school or homework support, nor absorbed by other educational programmes at municipality level. What changes is what students believe about local order in their area: reliance on Mafia and political connections falls, while education is valued more highly. In contexts where the Mafia holds long-standing social influence, visible evidence that legitimate pathways exist may do work that additional educational support alone does not. Second, the timing of these effects carries a distinct implication. The signal of State capacity appears short-lived, emerging in the first years of treatment and fading thereafter, whereas the shift in beliefs about the Mafia's usefulness emerges around the second and third years after repurposing and coincides with the decline in dropout. The behavioural response therefore depends on the sustained visible repurpose of the property rather than on the announcement of its repurpose. This suggests that one-off interventions, or properties reassigned and subsequently left inactive, are unlikely to produce lasting effects. Third, the paper underscores that the earlier stages of the policy are insufficient on their own. I find no effect on dropout at the reallocation stage; the decline appears only once the property is repurposed. Given that a substantial share of confiscated assets remain vacant or are never reassigned, the binding constraint is not seizure but the capacity and funding to bring these properties into active community use. Greater support for the local organisations undertaking this work would contribute directly to educational outcomes. Finally, the heterogeneity results indicate that the policy is most effective where it is most needed: among students in technical schools, who face a longer period at risk of dropout. The sentiment analysis shows the same pattern, with the strongest effects among students with proxied lower socioeconomic backgrounds. This is consistent with the interpretation that repurposed properties alter the trade-off facing youth who would otherwise encounter the greatest barriers to legitimate advancement, and suggests the policy might be operating as an equalising force.

The remainder of the paper is structured as follows: Section 2 illustrates the relevant institutional background, while Section 3 frames the conceptual framework that informs the key mechanisms. Section 4 describes the data, and the empirical strategy is developed in Section 5. Section 6 presents the effects on dropout, and Section 7 the effects on students' beliefs. Section 8 discusses alternative explanations, and Section 9 the robustness checks. Finally, Section 10 summarises the

conclusions and the policy implications of the main findings.

2 Institutional background

Mafia groups have been substantially affecting Italy's economy and social development for more than a century. As criminal syndicates, Mafia groups mainly reinforce their power through the accumulation of both economic resources and social consensus (Sciarrone, 1998). Until the 1980s, the economic foundations of Mafia groups were largely overlooked.[4] This changed in 1982, when the politician Pio La Torre and the Ministry of the Interior Virginio Rognoni drafted a new Bill targeting the Mafia's economic assets. In 1976, Pio La Torre described the purpose of their effort, stating that

> The path of simple repression - which strikes at the outgrowth, but does not change the economic, social and political humus in which the Mafia has its roots - did not and could not lead to definitive results.

Following the murder of Pio La Torre by the Sicilian Mafia in April 1982, his proposed bill was finally approved in September of the same year, becoming a milestone in Italy's fight against the Mafia. The law introduced two key innovations: first, it established the crime of Mafia association under the article 416-bis of the Italian Criminal Code, making Mafia groups directly liable for specific criminal activities[5]; second, since Mafia families' economic assets serve to reinforce their criminal sovereignty (Operti, 2018; Mosca, 2017), the La Torre-Rognoni law strategically enforced the confiscation of these symbols of power. By targeting criminal assets, this Bill was intended to send the message that the State is stronger than the Mafia and can hinder its economic resources.

Yet, the Rognoni-La Torre law did not provide any regulations for the management of the confiscated Mafia assets, especially real estate and land properties (Menditto, 2013; Nazzaro, 2021). In 1995, after an intense period of Mafia killings, several CSOs joined a new Italian network to counter the Mafia: *Libera. Associazioni, nomi e numeri contro le mafie. Libera* proposed the new Bill n. 109 of 1996 to allow the confiscated Mafia properties to be repurposed for urban and social recovery, specifically targeting the most Mafia-infiltrated areas of the country (Nazzaro, 2021; Falcone et al., 2016). This proposal represented a significant shift from repressive anti-Mafia policies towards policies aimed at compensating local

[4] Some preventive measures against people connected to Mafia groups have been previously introduced in 1965 through the Bill 575/1965: *Provisions against mafia-type criminal organisations, including foreign ones*. These measures only targeted criminals without addressing the management of their economic assets.

[5] The Rognoni-La Torre Bill (n. 646 in 1982) stated that *"Anyone who is part of a Mafia-type association consisting of three or more persons shall be punished by imprisonment from three to six years. Those who promote, direct or organise the association are punished with imprisonment from four to nine years."*

communities by returning confiscated assets to them and offering visible examples of legality. This way *Libera* put into practice the beliefs of the Sicilian judge Paolo Borsellino, who was murdered by the Sicilian Mafia in 1992:

> The fight against the Mafia must first and foremost be a cultural movement that accustoms everyone to smell the beauty of the fresh scent of freedom, which is opposed to the stench of moral compromise, indifference, contiguity and therefore complicity.

Following the formulation of the Bill n. 109 of 1996, a standardised procedure for repurposing confiscated Mafia properties was established under the CRR policy. As shown in Figure 1, such policy consists of three main phases. It is useful to distinguish them, as this paper treats them as conceptually and empirically distinct. At the time of *Confiscation*, the State takes ownership of properties belonging to Mafia members upon conviction. This legal act makes the State their sole owner but often leaves the properties physically abandoned and unused. During *Reallocation*, the confiscated property is administratively transferred from the National Agency for Confiscated Assets (ANBSC)[6] to the local authority in whose territory it is located. This is a bureaucratic step towards the property's reuse but, on its own, produces no visible change on the ground. At the *repurposing* stage, the properties are finally put back into use. Some properties are converted into community spaces hosting social, cultural, and educational activities, typically managed by local associations and NGOs. Others are used by public institutions for administrative or institutional purposes. While the confiscation of properties might signal what the State has seized from the Mafia, it is only at the repurposing stage that the property become a visible, functioning part of local life rather than an empty mark of the Mafia's defeat. This paper isolates the effect of the repurposing of these properties, the stage at which prior work leaves off.

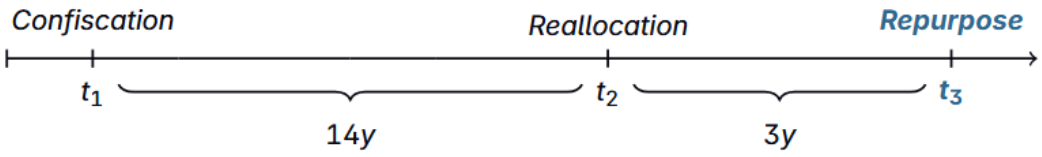


Figure 1 The CRR policy timeline.

[6]The National Agency for the Administration and Destination of Properties Seized and Confiscated from Organised Crime (ANBSC) is an executive body which determines how to repurpose confiscated properties, with decisions aimed at compensating local communities. Being a direct government agency, its accounts are subject to oversight by the Court of Auditors(ANBSC, 2019).

Figure 2 shows the number of confiscated, reallocated, and repurposed properties in the historically Mafia-ridden municipalities populating my sample. Confiscations and reallocations begin rising from the 1990s, while repurposing practices, first established in the late 1990s, increase dramatically in the late 2010s. On average, it takes 14 years for confiscated Mafia properties to be effectively reallocated, and a further 3 years for them to be repurposed once reallocated.

This paper focuses on the final stage of the Mafia Confiscated and Reallocated Assets Repurpose (CRRA) Policy. While existing work shows that the reallocation of Mafia properties affects economic outcomes such as housing prices and market concentration (Boeri et al., 2023; Ferrante et al., 2021b), a main gap persists in the study of the socio-economic effects of the repurposing of these properties (Ferrante et al., 2021a). The confiscation administrative transfer from the ANBSC to local authorities may signal the removal of Mafia control, but it does not by itself constitute a tangible renewal of the local social fabric (Falcone et al., 2016); that renewal occurs only at the repurposing stage. The staggered timing between reallocation and repurposing allows me to disentangle the two effects and assess whether the effects I measure are driven by the confiscation and reallocation of properties or by their active repurposing. I exploit this timing directly in Section 6, where I show that reallocation alone does not affect school-level outcomes while the repurposing does.

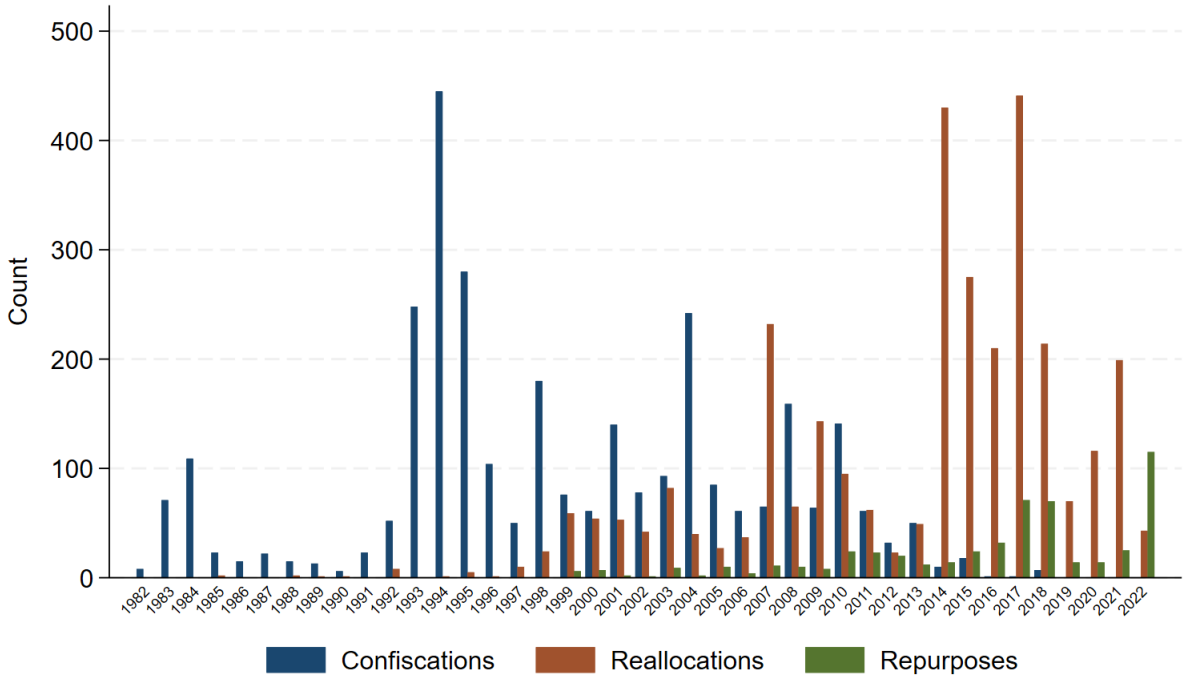


Figure 2 Number of confiscations, reallocations and repurposes from 1982 to 2022.

Having established that the repurposing of properties is the stage at which the intervention becomes tangible, I follow by describing what the repurposing involves and why it is likely to affect young people in particular.

Once repurposed for social use, the properties previously owned by the Mafia families transform into public spaces and community centres managed by local NGOs. These spaces serve as hubs for social activities, youth programs, and community development initiatives, involving direct citizen participation by promoting trust, social cohesion, and sustainable local development (Nazzaro, 2021; Falcone et al., 2016). Residents of all ages participate in recreational and educational activities aimed at reclaiming neighbourhood spaces, rebuilding social bonds, and fostering anti-Mafia resilience. The properties hold particular educational significance as centres that represent safe spaces for young people during after-school hours and weekends, with participants often sharing their positive experiences in schools and throughout the wider community (Nazzaro, 2021). The transformative power of the repurpose practices is perhaps most strikingly demonstrated by the behaviour of Mafia affiliates, as they frequently attempt to occupy or vandalise their former properties, revealing their deep concerns about the potential for these spaces to start serving as community assets (Italian Ministry of Justice, 2018).

To measure how students value the criminal path, I use the share of students in the Survey of the Perception of the Mafia who rate asking the Mafia as a very useless way to find a job in their town, following the extreme-category coding of Villa (2025). Because it is a share of respondents, the measure captures how students assess the Mafia channel rather than survey participation. Panel C shows that this share rose from X% in 2011 to Y% in 2022, as repurposing expanded across the sample. These stylised facts motivate the subsequent empirical analysis.

3 Conceptual framework

In this section, I develop a conceptual framework consistent with my empirical analysis and the anecdotical evidence. I first draw on the opportunity cost framework from the economics of crime literature to explain how Mafia presence affects juveniles' educational incentives, particularly in historically Mafia-ridden neighbourhoods. I then review the existing evidence on how Mafia groups leverage their properties to sustain juvenile recruitment and establish alternative authority structures. Finally, I introduce the potential channels through which the repurpose of Mafia properties can affect juveniles trajectories. Rather than presenting formal model predictions, this section aims to motivate the economic mechanisms underlying

the estimated treatment effects.

Building on the previous literature about the trade-off between crime and education (Lochner, 2004, 2011; Lochner and Moretti, 2004; Machin and Meghir, 2004), which extends the work from Becker (1968), I discuss how students weigh their propensity to invest in education against joining Mafia networks. At critical educational junctures, students weigh the legitimate pathway of human capital accumulation against the alternative offered by the Mafia. The costs and returns of each depend on the local opportunity structure and on beliefs about the relative strength of the Mafia and the State. I argue that the repurpose confiscated Mafia properties shifts these beliefs locally.

On the one hand, juveniles weigh the costs and benefits of joining the Mafia. The costs relate to the probability of being arrested and the related punishment, while returns not only relate to monetary gains, but also to non-monetary benefits such as local power, reputation, and social status, which are essential for the Mafia to maintain territorial and social control (Gambetta, 1993; Sciarrone, 1998). Indeed, young people growing up in Mafia-dominated areas often see the organisation as a fast and viable route to success, with powerful bosses serving as role models (Caglayan et al., 2017). This perception is reinforced by family and kinship ties that bind households to criminal groups. Because Mafia organisations are deeply embedded in local social networks (Sciarrone, 1998), juveniles raised in families or communities with existing criminal connections come to see affiliation not as an unlikely choice but as an ordinary, normalised life path (Sviatschi, 2022a).

On the other hand, juveniles weigh the returns to education against its costs. The benefits of schooling accrue largely in the future, through access to higher legitimate wages. By raising expected earnings, human capital also raises the opportunity cost of unskilled, street-level crime(Lochner, 2004): the more a young person can earn legally, the more they give up by turning to crime. The costs, by contrast, are immediate: staying in school means delaying paid work. The balance between these future benefits and present costs depends critically on the local labour market. Where legitimate jobs are scarce, the expected returns to education shrink, and so does the opportunity cost of joining the Mafia. Juveniles with the weakest labour market prospects therefore have the least to lose from affiliation. Consistent with this, falls in wages at the bottom of the distribution have been shown to directly increase crime (Machin and Meghir, 2004). For young people growing up in poverty, the income and status promised by the Mafia become one of the few visible routes to social mobility, as vividly depicted in several towns around Naples (Balestrini, 2004). In such contexts, Mafia affiliation can seem a quicker and more reliable route to career advancement than education in the short term (Catozzella, 2011).

Through what mechanism does the repurpose of such properties weakens the social influence of Mafia networks? The effect is likely to operate through several channels, which I test and discuss in greater detail in Section 7. First, anti-Mafia policies can alter the cost-benefit calculation of joining the Mafia by updating juveniles beliefs.Repurposing directly targets one of the key functions these assets serve within Mafia organisations: that of *positional goods*, namely visible symbols of territorial control and legitimacy (Hirsch, 1976; Baldascino and Mosca, 2012). As tangible displays of power and wealth, these properties help Mafia families project authority and sustain their status as role models within local communities (Mosca, 2017). When the State confiscates them and hands them over to the community, it publicly reverses this symbolism, signalling both the vulnerability of the Mafia and the capacity of the State. This can lower the perceived returns to criminality by undermining the Mafia's image as a legitimate and aspirational force.

Second, juveniles may update their beliefs about the State's capacity to confront the Mafia. A confiscated property visibly and durably repurposed for public use is a credible signal that the State can confiscate criminal assets and restore them for collective use, a more demanding demonstration of enforcement than the arrest of Mafia groups alone (Chiodo, 2021; Mosca, 2017; Heller et al., 2017). As trust in institutional capacity rises, juveniles might perceive a higher expected cost of Mafia affiliation, facing what appears to be a stronger and more present State.

Last, the repurpose of properties may reshape the local opportunity structure. First, confiscations accompany the arrest of Mafia families, reducing both the organisation's capacity to recruit juveniles locally and the criminal earnings it can offer, and thereby lowering the material returns to affiliation. Second, properties hosting education-oriented activities can provide direct support that keeps juveniles in school through critical educational junctures, lowering the effective cost of staying enrolled (Heller, 2014; García et al., 2019). Third, properties repurposed for cultural and social use can serve as additional local amenities that engage young people and signal visible urban regeneration (Datcher, 1982; Ludwig et al., 2013). Together, these channels may lower the returns to joining the Mafia and reduce the cost of investing in education.

Each channel has distinct empirical implications, which I exploit in Section 7 to distinguish between them. If the effect operated by lowering the material returns to affiliation through reduced recruitment capacity, it should emerge at confiscation, when the family is removed, rather than at the time of repurposing. If it operated by lowering the cost of remaining in education through additional support provided, it should be concentrated in properties delivering education-oriented activities and should be visible in academic attainment as well as in enrolment. If it operated through generic amenity provision or neighbourhood up-

grading, it should not depend on the property's criminal origin and should vary with local property values. The belief-based channel, by contrast, acts on the perceived rather than the realised terms of the trade-off. It predicts effects concentrated in properties that most visibly symbolised Mafia authority, emerging only once repurposing is complete and the property is returned to community use. It also implies that these effects should be accompanied by shifts in how juveniles view their own prospects and the relative standing of the Mafia and the State. While I cannot observe whether repurposing changes realised returns in the local labour market, I can observe what students currently think and expect about their future. This is precisely the margin on which the belief-based channel operates.

4 Data

Schools data. I use school-level data covering the upper secondary schools in the main Italian urban areas with a significant historical presence of the Mafia. School-level data were provided by the Italian Ministry of Education, allowing me to geocode each school's location to cover the period from 2015 to 2022. To provide clarity to the analysis, I first omit schools that changed address across the years, unless they were relocated within the same specific census block[7]. Second, to reduce measurement error, I double-check that schools that merged at some point in the panel were located at the same address. Additionally, the data present information about the number of students completing each academic year in each grade, the graduating grades at the end of the educational cycle, as well as their age and gender. Additionally, the data provide information about the schools' educational tracks, distinguishing among *Licei* or Academic High Schools, which focus on academic subjects and mainly prepare students to enrol in tertiary education, *Istituti Tecnici* or Technical High Schools, which provide a more practical and technical education, and *Istituti Professionali* or Vocational High Schools, which provide vocational training. From each track, we know students can further specialise in different subjects; the specialisations offered to date count to 14 and are summarised in Table 1, as outlined by the Italian Ministry of Education[8]. The universe of all existing high school tracks is described in more detail in Figure A.1. I exclude private high schools from the sample, as they cater to an endogenous segment of the population in Italy and account for only 5% of total student enrollment (**?**)istat$_i$*struzione*$_2$021.

[7] As explained later in this Section, since the treatment is defined by aggregating census blocks into schools' catchment areas, any change of the school's location occurring within the same census block is insignificantly affecting my identification strategy.

[8] The organisation of Italian high schools can be found online on the website of the of Education (2024): Organisation of upper secondary education in Italy.

Table 1 Types of high school tracks in the Italian educational system.

HS Tracks	Description	Specialisations	Sample %
Academic	Provide students with the skills and knowledge for higher education in classical studies, scientific studies, linguistic studies, humanities, music, and fine arts.	6	0.411
Technical	Provide practical and technical skills related to key sectors such as administration, marketing, and industrial development.	2	0.371
Vocational	Provide vocational skills for industrial, commercial, and social development, as well as agricultural, maritime and hospitality skills.	6	0.101

Italian upper-secondary education follows a five-year cycle, from grade 9 to grade 13. Students typically enter grade 9 at age 14 or 15 and graduate at age 18 or 19, depending on their birth month. The academic year begins in September, with students born in the same calendar year enrolling together, while grade completion is measured in May. Consequently, students born between January and May are 15 upon completing grade 9 and 19 at graduation, whereas those born between June and December are 14 and 18, respectively [9].

Compulsory education lasts 10 years, from ages 6 to 16, creating a critical threshold for school attendance. [10]. However, once students turn 16, they gain legal freedom to discontinue their studies. This legal framework leads to specific dropout patterns, which typically happen in the transition between the third and the fourth year of the upper secondary cycle, namely from grade 11 to grade 12. Indeed, as shown in Figure 3, after completing grade 11, all students are at least 16 years old. In 2015, 3.3% of students dropped out before 16 and 8.2% after; in 2022, the figures were 2% and 5.41% (ISTAT, 2023). I therefore focus on dropout after grade 11, when staying in school becomes a choice weighed against immediate alternatives (Angrist and Krueger, 1991) and local context matters most (Oreopoulos, 2006). This is when the repurpose of former Mafia properties is most likely to shape students' attitudes, and focusing on it separates that effect from the constraints of compulsory schooling.

[9]Each academic year starts in September and students born in the same calendar year get enrolled regardless their month of birth. The number of students who completed each academic year is measured in May.

[10]In Italy, Law 296/2006 extended compulsory education from ages 6–15 to 6–16 and raised the minimum working age accordingly; therefore, leaving school before 16 is both legally prohibited and actively monitored, making it very unlikely for students to not attend school before turning 16 without facing institutional scrutiny within an entire academic year.

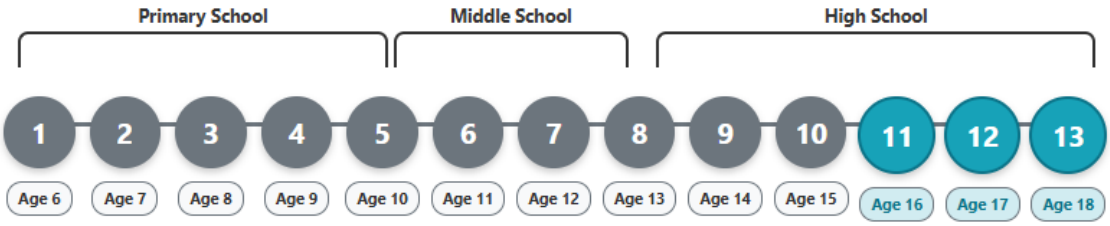


Figure 3 The timeline of compulsory schooling in the Italian educational system. This Figure represents a scenario where students do not start school later and do not repeat any year.

The main outcome variable is the cumulative dropout rate among students aged 16 and older, computed by following the same cohorts from grade 11 to grade 13:

$$D_{g11-13_{ct}} = \frac{C_{g11-13_{ct-1}} - G_{g13_{ct}} - C_{g12-13_{ct}}}{C_{g11-13_{ct-1}}} \tag{1}$$

where C_{g11-13} is the number of students enrolled in grades 11 to 13 in catchment area c in year $t-1$, G_{g13} is the number of students who complete grade 13 and graduate in year t, and C_{g12-13} is the number of students from the same cohorts still enrolled in grades 12 and 13 in year t. Because it follows the same cohorts and treats graduation separately, the measure reflects genuine dropout rather than demographic differences between cohorts or successful completion of the five-year programme. Section 5 discusses remaining measurement concerns, including exam-related attrition, dropout before graduation, and school transfers.

Mafia properties data. I built a new database covering the universe of repurposed Mafia properties, combining data scraped from several institutional sources with records from the Italian antimafia association *Libera*. These sources were previously fragmented and had never been systematically compiled or used for empirical analysis.

I start from the list of reallocated Mafia properties published by ANBSC (2023), which reports for each property a unique identifier, the type of real estate, its use before confiscation, the recipient and purpose of the reallocation, and the dates of confiscation and reallocation. I geocode 3,119 of the 3,667 properties.[11] Of these, 42% were reallocated for social and 41% for institutional purposes. Crucially, 66% were assigned to the nearest municipality, as only these can be fully transformed into community resources. Most are housing such as villas and

[11]Most of the properties I could not geocode, owing to incomplete information, are rural land, which is less likely to affect dropout in the urban areas I study.

flats (83%), followed by commercial and industrial estate (10%) and land (12%). Appendix B reports further attributes of the repurposed properties.

Next, I gathered information on how each property has been repurposed. I scraped *Confiscatibene*, a platform run by *Libera* with the ANBSC, where *Libera* and other users report whether each property has been repurposed for social activities, describe these activities, name the managing NGO and, in a few cases, give the year and month the repurposing began. As the website only covers the period up to 2018, I complement it with *Libera*'s yearly survey of the NGOs managing repurposed properties, to which its national managers gave me access. Finally, to minimise missing information, I cross-reference these data with the lists that municipalities publish on the current status of the properties under their jurisdiction.[12]

Between 2015 and 2022, 346 of the 3,119 properties were reused for social purposes by municipalities and local NGOs, mostly in the largest municipalities such as Palermo and Naples. Figure B.1 maps Mafia real estate across the ten urban areas I study, and Table B.2 describes the local context where properties were repurposed, including population, educational attainment, labour market conditions, real estate conditions and the presence of NGOs.

To classify the social activities at each site, I manually categorise them using the labels from previous *Libera* reports (Falcone et al., 2016) into five types: educational, welfare, cultural, employment-related and other. Table 2 the definition and representative examples of each, and Figure 4 the frequency of each category. Figure B.2 shows the 100 most frequent words in the activity descriptions.

[12] Article 48 of the Antimafia Code (2011) requires municipalities to publish a monthly updated list of these properties and their current use. Some repurposed properties may still be missing from both municipal archives and *Libera*'s website, so my measure of reuse is likely an undercount; any resulting attenuation bias implies that my estimates are a lower bound of the true effect.

Table 2 Types of reuse activities described.

Activity	Description	Reused Mafia properties %
Education	After-school activities and schooling support, socio-educational activities for at-risk juveniles	9.5%
Culture	Cultural activities to raise awareness about organised crime, the rule of law and self-responsibility	29.8%
Welfare	Places for housing emergency, activities to assist the homeless and people in need, social activities targeting vulnerable groups	41.1%
Employment	Support and training activities for unemployed people or fragile groups	9.2%
Other	Activities in the agricultural industries aiming to sell organic products and other	8.9%

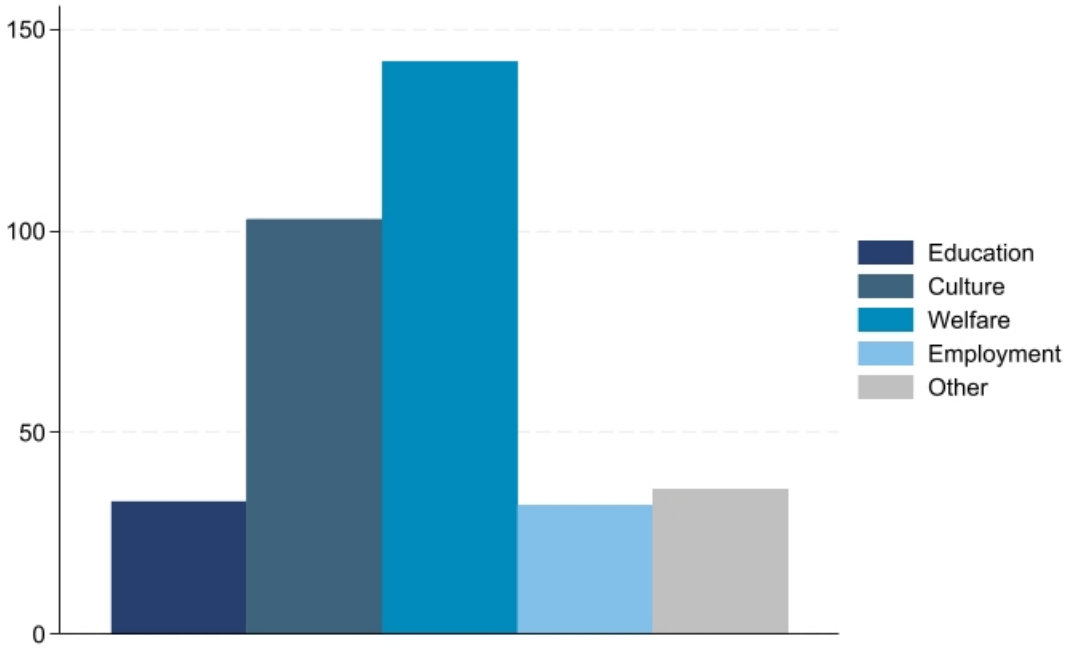


Figure 4 Repurpose categories of activities.

Survey on youth's perception of the Mafia. I use survey data from the *Survey of the Perception of the Mafia* to examine changes in perceptions of the Mafia and the State,

and their role in shaping youth attitudes. The survey data are created and provided by the Sicilian NGO Centro Studi Pio La Torre (2025), targeting students enrolled from grade 11 to grade 13, roughly aged between 16 and 19. Some responses are aggregated at the school level, which I can identify with a unique code, while others provide a unique identifier per respondent. The survey, which spans from 2009 to 2024, is administered to students by the teachers thanks to a collaboration between the schools and the NGO. While this creates a selected sample, it allows me to directly test whether changes in the perception of the Mafia serve as a mediating channel for the treatment effect[13]. The students' questionnaire includes a battery of questions focusing on several key dimensions of the perception of the Mafia and the State: the students' beliefs about whether the Mafia could be useful or not for career advancement, their perception of Mafia' strength and State's strength, their trust in several public institutional figures, student-level definitions of the Mafia, and questions about how their family perceives the Mafia. Table A.1 offers a complete list of the answers I employ. For multiple-choice questions, I construct a dummy variable for each response option, such that the dummies sum to the total number of responses. For agreement scales, I use the top and bottom of the distribution to capture strong agreement and strong disagreement with each statement. For the student-level answers, I perform a simple text analysis and calculate sentiment scores for the words used by the students to describe the Mafia. I use these indicators to assess whether the repurposing of Mafia properties leads students to change their perception and sentiments towards the Mafia and educational investments.

Other data. I collected information about the socio-economic conditions of the local community around the schools; understanding these contextual factors is essential, as local socio-economic conditions can affect educational outcomes and criminal behaviour through multiple pathways. I first collected information about rental prices from 2016 to 2022 from Immobiliare.it as a proxy for deprivation at the neighbourhood level; second, from the National Registry of the Third Sector[14], I scraped information about local NGOs, such as their legal address, the time the NGOs were created, the type of activity they pursue and the number of people which engage in their activities. As with the Mafia properties, I geocode the NGOs' addresses to map them at the street level. Moreover, I digitise the historical distribution of Mafia families at the neighbourhood level published by the Italian Antimafia Directorate DIA (2015a,b).

[13]Results reported in Section 7 also account for the baseline measure of students' socio-economic background to address any related concern.

[14]The registry is offered online by the Italian Ministry of Labour and Social Policies (MLPS)

4.1 Treatment allocation

Given the localised nature of the analysis, students must be linked to the schools they attend via their residential addresses and school locations. This would require student-level data, which are rarely publicly released. In the absence of formally defined school districts, a common challenge related to the use of educational data, I build school catchment areas using an adapted distance-based location-allocation method, following established practice in spatial economics for defining catchment areas when administrative boundaries are unavailable (Pearce, 2000; Singleton, Longley, Allen and O'Brien, 2011). The two main approaches are weighted-Voronoi polygons, which treat the problem geometrically, and location-allocation, which frames it in terms of student demand (based on residential distribution) and school supply. Pearce (2000) shows that the former is computationally simpler, but the latter better predicts real-world catchment dimensions.

I apply an adapted location-allocation method in three steps. First, I compute the Euclidean distance between each school and each residential census block, using block geometries from the 2011 Italian census (ISTAT, 2011), and select the closest schools for each block, since distance is one of the main driver of enrollment preferences (Mandic, Sandretto, García Bengoechea, Hopkins, Moore, Rodda and Wilson, 2017). Second, because families' and students' choices also depend on the track offered and preferred, I differentiate catchment areas by specialisation, creating them for the 14 sub-tracks that map onto the three main tracks in Table 1. Matching each census block to its nearest school within each sub-track determines catchment boundaries without arbitrary distance thresholds, leaves no block unassigned, and reflects that students' choices are defined by program availability.

I work with 10,556 census blocks, with an average area of 0.05 km^2 and an average population of 200, of whom 11.64% are aged 14-19. I compute 334 school catchment areas, each aggregating on average 580 census blocks[15] and around 10,000 individuals, 570 of them aged 14-19. I also construct a second set of catchment areas that accounts for each school's baseline enrollment, capturing that allocation depends on school capacity as well as proximity. Appendix D details this procedure, and Section 6 reports results under this alternative construction.

To identify treated catchment areas, I use the precise address of each repurposed Mafia property and compute two treatment measures. For the extensive margin, I create a dummy equals 1 when at least one property has been repurposed within the catchment area in a given year; areas treated before the start of the panel are coded as always treated. For the intensive margin, I count the properties reused in each catchment area each year, giving a continuous

[15]Because catchment areas are calculated for each school sub-track, census blocks sum only within each sub-track, not across the entire dataset.

measure that varies over space and time. To better control for demographic confounders, I also compute area-specific weights based on the local student population. As a robustness check, I weight each catchment by the average distance of repurposed properties from its population-weighted centroid[16]. These weighted measures capture both the localised nature of property reuse and the expected decay of its effect with distance from treated sites (Boeri et al., 2023; Damm and Dustmann, 2014).

Figure 5 gives a stylised example of the construction. Each school (filled coloured circle) is assigned the census blocks (grey dots, block centroids) that are closest to it by Euclidean distance within its sub-track; grey lines link each block to its assigned school. The coloured regions are the resulting catchment areas of School A (blue), B (green) and C (orange), and purple diamonds mark Mafia properties repurposed in a given year. School A is a *control*, while Schools B and C are *treated* on the extensive margin but differ in intensity: two properties are reused in School C's catchment area and one in School B's.

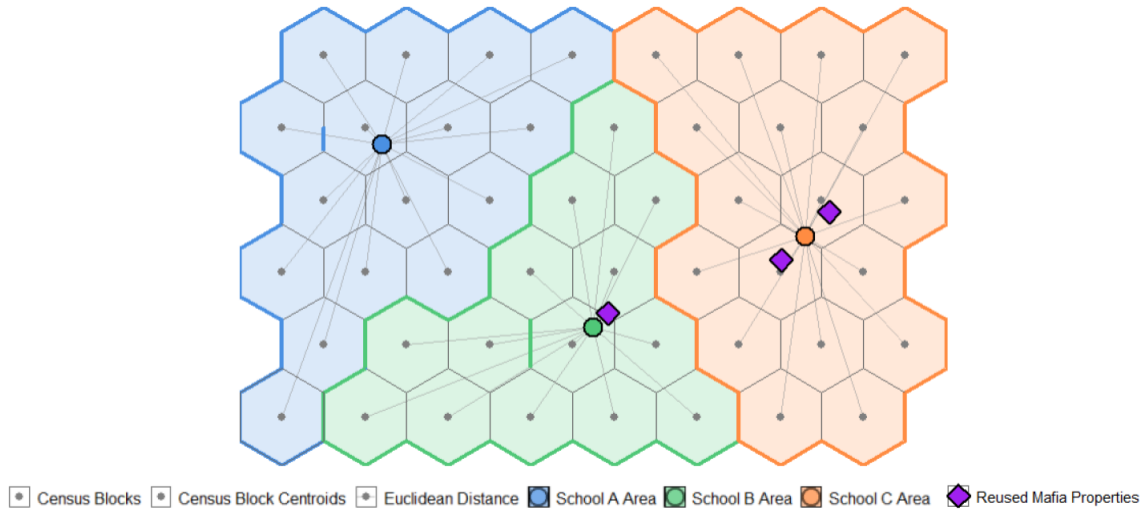


Figure 5 Ad-hoc example of schools' catchment areas construction.

Summary Statistics. Table 3 compares school-years with and without a repurposed Mafia property in the school catchment area, for the main covariates of the analysis. Treated and control areas are similar in Mafia presence, rental prices, and student migration, which is virtually identical across the two groups. They differ, however, in two dimensions. First, areas with repurposed properties host substantially more NGOs, about 129 against 59 on average, consistent with the role of civil society organisations in the reuse process, since

[16]Population-weighted centroids are the weighted average of the coordinates of all census block centroids in each catchment, weighted by each block's high school-age (14–19) population, yielding one centroid per catchment that reflects the spatial distribution of resident students.

repurposed properties are assigned to local associations. Second, grade retention in grades 11 and 12 is higher in treated areas, by 1.5 and 1.8 percentage points respectively, suggesting that these schools serve students with weaker academic trajectories. Both differences remain statistically significant when standard errors are clustered at the school level.

These are differences in levels, which the school fixed effects absorb; identification relies instead on the assumption of parallel trends, which I assess with the event study in Section 6. I nonetheless address these differences directly in three ways. First, I re-estimate the main results by inverse probability weighting, reweighting control schools by the inverse of their propensity score of being treated, estimated on baseline NGO presence and grade retention (Hirano, Imbens and Ridder, 2003). The reweighted estimates are virtually identical to the main result (Table C.1). Because few control areas have NGO presence comparable to the most NGO-dense treated areas, the weights concentrate on a small number of control schools. The direction of the effect is therefore robust to reweighting, while its precision depends on the comparison with the few control areas most similar to the treated ones. Second, I show that repurposing does not affect grade retention or student migration (Table A.3), and I therefore control for both in the main specifications. Third, I estimate a linear probability model of the determinants of repurposing (Table C.2). Repurposing is more likely to happen in areas with more NGOs in the previous year and in municipalities with higher expenditure on youth policies and urban planning, and less likely in municipalities with higher police expenditure and in those that did not publish the list of confiscated properties. I then re-estimate the main specification controlling for NGO presence in the previous year and for linear time trends interacted with each of these baseline municipal predictors (Table C.3). The estimates are robust to this test, and NGO presence shows no independent association with dropout. Appendix C reports these analyses in detail.

Table 3 Descriptive statistics.

	Repurposed Mafia property in the school's area							
	No			Yes			Δ	
Covariates	N	mean	sd	N	mean	sd	Robust SE	Clustered SE
Mafia presence	427	0.632	0.483	993	0.648	0.478	-0.015	-0.015
							(0.028)	(0.063)
NGOs presence	427	59.30	60.14	993	128.95	120.66	-69.648***	-69.648***
							(4.809)	(10.898)
Rental prices	427	7.693	2.567	993	7.680	2.019	0.014	0.014
							(0.140)	(0.317)
Students' migration	427	0.077	0.115	993	0.080	0.128	-0.003	-0.003
							(0.007)	(0.011)
Grade retention 3rd year	427	0.050	0.067	993	0.065	0.079	-0.015***	-0.015**
							(0.004)	(0.008)
Grade retention 4th year	427	0.050	0.068	993	0.068	0.081	-0.018***	-0.018**
							(0.004)	(0.008)

Notes: School-years in the estimation sample. Δ is the mean of the school-years with no repurposed property minus the mean of those with at least one. The last column presents standard errors clustered at the school level.

5 Empirical strategy

Estimating the effect of reusing confiscated Mafia properties on dropout is complicated by the many institutional actors involved in the reuse process. As shown in Table 3, areas with repurposed properties have more NGOs and higher grade retention at baseline, while Mafia presence, rental prices and student migration are similar. Since unobserved municipal characteristics may also shape reuse decisions, I identify the determinants of reuse, control for them, and show that, conditional on them, the timing of reuse is plausibly unrelated to other determinants of dropout.[17]

When a Mafia property is repurposed for social scopes, such as a community centre, an after-school facility or a social services centre, it turns from a symbol of Mafia power into a symbol of community resistance (Mosca, 2017). Reuse may therefore reduce dropout through two channels: by improving local educational and social services, which raises the returns to education, and by making criminal careers less attractive as symbols of Mafia power become symbols of State authority.

The empirical strategy proceeds in three steps. First, I estimate a difference-in-differences

[17]Reused properties may still differ from unused ones in unobserved ways, such as acts of resistance or vandalism, or local political dynamics. Greater unobserved criminal resistance or social tension around reuse would bias the estimates towards zero; stronger unobserved community mobilisation or anti-Mafia political commitment would bias them away from zero.

model comparing school catchment areas exposed and not exposed to reuse. Second, I address the baseline differences between treated and control areas and the determinants of reuse (Appendix C). Third, I estimate the intensive margin of the treatment.

5.1 DiD model

I estimate a TWFE model in a difference-in-differences design with staggered adoption. *Treated* catchment areas have at least one reused Mafia property within their boundaries; *control* areas have none.[18] The sample comprises 1,393 school-year observations from 251 catchment areas with non-missing dropout, of which 54 are never treated and 197 are treated at some point.[19] The model is:

$$Dropout_{G11\text{-}13,cnmt} = \beta_1 Repurpose_{ct} + \beta_2 X'_{ct} + \delta_c + \eta_t + \epsilon_{cnmt} \tag{2}$$

where $Dropout_{G11\text{-}13}$ is the share of students in grades 11 to 13 who leave school, in catchment area c, neighbourhood n, municipality m and year t; $Repurpose_{ct}$ equals 1 from the first year in which at least one Mafia property in the catchment area is repurposed. β_1 captures the average treatment effect on the treated, comparing treated areas with never- and not-yet-treated ones. X'_{ct} includes student migration and grade retention rates in grades 11 and 12, which capture educational disruptions that the dropout measure could otherwise pick up.[20] δ_c and η_t are catchment-area and year fixed effects. Standard errors are clustered at the catchment-area level, to account for serial correlation within schools.

The baseline differences in NGO presence and grade retention are differences in levels, which the catchment-area fixed effects absorb. Identification relies on parallel trends: absent the treatment, dropout in treated and control areas would have evolved similarly. I test this assumption with an event study. Since TWFE event studies can be biased under staggered adoption and heterogeneous effects (Goodman-Bacon, 2021; Sun and Abraham, 2021; Callaway and Sant'Anna, 2021), my preferred estimator is the interaction-weighted estimator of Sun and Abraham (2021), which interacts each treatment cohort with its time since treatment.[21]

[18] In an additional model I restrict the sample to schools whose catchment area already hosts at least one reallocated property, so that areas are comparable in the policy process. Results are reported in Figure A.3.

[19] Dropout cannot be computed for 2015 and for some 2016 cohorts, because it follows the same cohorts from grade 11 to graduation and requires the previous year's enrolment. The staggered implementation of the policy still provides sufficient variation for identification.

[20] The treatment does not affect student migration or grade retention, so these controls capture pre-existing school characteristics rather than outcomes of the treatment. Results are reported in Table A.3.

[21] The Sun and Abraham (2021) estimator is also well suited to the large number of time trends and fixed effects included in the exercises in Appendix B.

I address the baseline differences and the determinants of reuse directly in three ways, reported in Appendix C. First, I re-estimate the main results by inverse probability weighting, reweighting control schools by the inverse of their propensity score of being treated, estimated on baseline NGO presence and grade retention (Hirano et al., 2003); the estimate is almost unchanged (Table C.1).[22] Second, I control for grade retention after showing that repurposing does not affect it (Table A.3). Third, I estimate a linear probability model of the determinants of reuse, which shows that reuse is more likely in areas with more NGOs in the previous year and depends on the municipality's transparency and expenditure on youth policies, urban planning and municipal police (Table C.2). I then re-estimate the main specification controlling for NGO presence in the previous year and for linear time trends interacted with these baseline municipal predictors; the estimates are stable (Table C.3).

Student migration is the share of students in grades 9 and 10 at $t-1$ who are not in grades 10 and 11 at t. These transitions occur at ages 14 to 16, during compulsory schooling, when students who leave are more likely to transfer to another school than to drop out. Grade retention is the share of students who are at least two years older than the standard age for their grade: aged 18 or older in grade 11 and 19 or older in grade 12. Both can affect measured dropout and may change within catchment areas over time. Appendix D describes their construction.

Two further concerns remain. First, dropout in grade 13 is rare, but the measure may pick up students who fail or skip the final exam; Figure A.2 re-estimates the main specification using only the transitions from grade 11 to 12 and from 12 to 13. Second, students born between January and May may turn 16 and leave during grade 10, escaping the measure; Section 6 shows that the results are unchanged when dropout is measured from grade 10 to 13. Some exits may also be transfers rather than dropout, but transfers and other motivated exits account for only 6.4% of departures from upper secondary school, against 89.1% for unmotivated abandonment (Italian Ministry of Education, 2017), and the results are robust to controlling for student migration and grade retention.

5.2 DiD dynamic model

The effect of reuse likely depends on its intensity. To capture the intensive margin, I use the cumulative number of repurposed properties in each catchment area as a continuous treatment, with the dynamic difference-in-differences estimator of de Chaisemartin and

[22] Because few control areas have NGO presence comparable to the most NGO-dense treated areas, the weights concentrate on a small number of control schools. Trimming observations with propensity scores outside $[0.1, 0.9]$ (Crump, Hotz, Imbens and Mitnik, 2009) reduces the estimate to -1.0 percentage points, which is no longer statistically significant.

D'Haultfœuille (2020). The estimator compares changes in outcomes across schools exposed to different doses of treatment, allows for heterogeneous and dynamic effects, and does not impose a linear dose-response relationship, unlike TWFE. As robustness checks, I weight the treatment intensity in two ways: by the catchment area's population aged 15 to 19, relative to the average catchment area, since the effect depends both on how many properties are reused and on how many students they can reach; and by the inverse of the distance between the properties and the population-weighted centroid of the catchment area, since closer properties are more accessible to students.

6 Effects on the Dropout Rate

After the first Mafia property is repurposed within the school catchment area, dropout rates for students in grades 11 to 13 fall. The estimated ATTs are presented in Table A.2; all specifications include school and time fixed effects, and standard errors are clustered at the school level. Column (1) shows that having at least one repurposed Mafia property within the catchment area reduces the dropout rate by 1.7 percentage points, a 29% reduction relative to the mean dropout rate. Columns (2), (3), and (4) show that the estimate is stable, between 1.6 and 1.7 percentage points, when controlling for student migration and grade retention.

Figure 6 shows the event study for my preferred specification, Column (4), estimated with the interaction-weighted estimator of Sun and Abraham (2021). There is no evidence of pre-trends: I cannot reject that the pre-treatment coefficients are jointly zero (p-value 0.80). The effects are predominantly long-term. Dropout rates become significantly lower from the second year after the first repurpose, and the reduction persists, and slightly grows, up to six years after the start of the first activities. This gradual pattern is consistent with the timing of the treatment variable: the indicator switches to 1 when local NGOs and municipalities sign the agreement to reuse the property, which precedes the time needed to develop and deliver activities for the local community. It is therefore reasonable to expect the full impact to appear only once such activities are established. This is in line with Nazzaro (2021), who documents the substantial obstacles that NGO projects in confiscated properties face before becoming fully operational, and with the view that cultural change driven by anti-Mafia policies builds up gradually through sustained community engagement (Nazzaro, 2021).

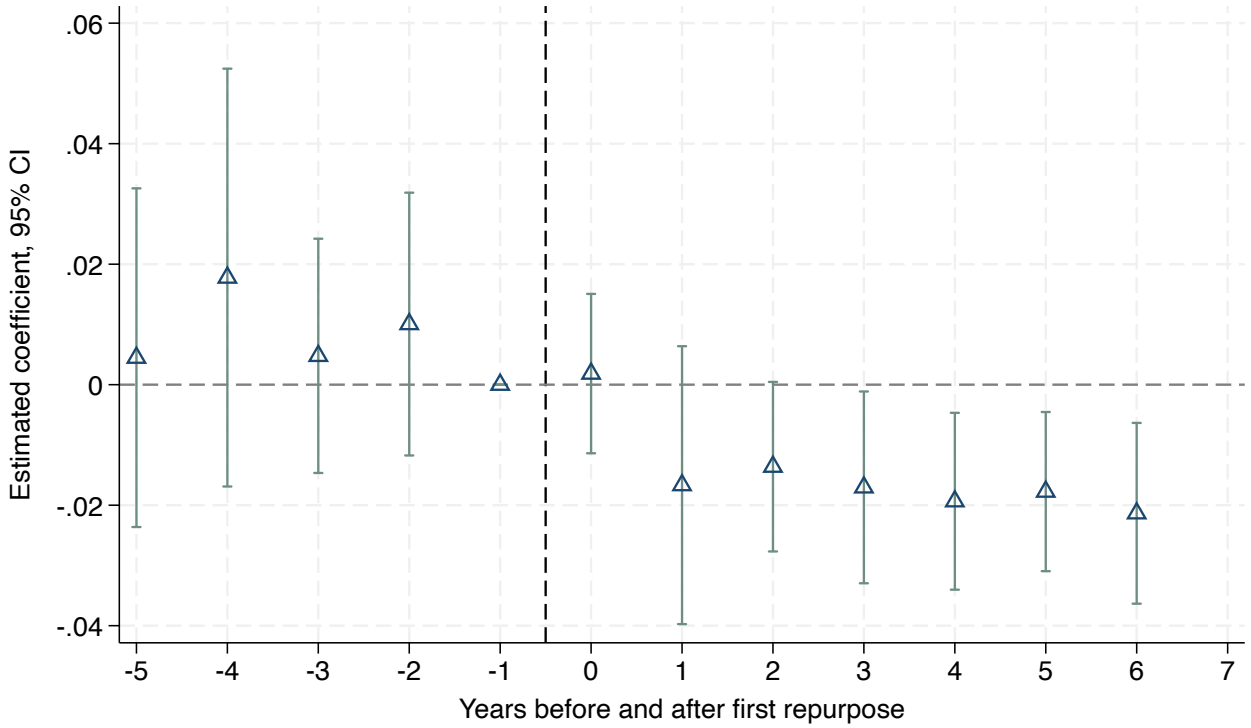
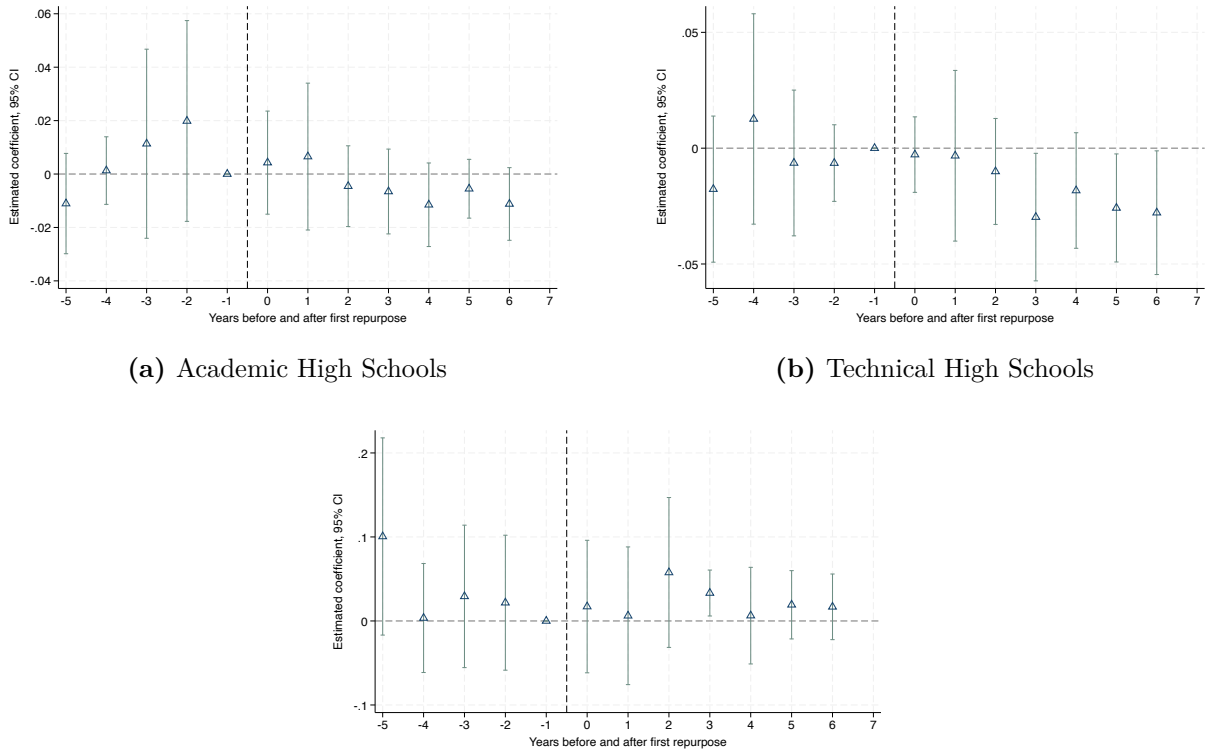


Figure 6 Estimated effects of the repurpose of Mafia properties on dropout rate before and after the first repurpose starts.

I also explore heterogeneity by school track. As explained in Section 4, Italian upper secondary schools are divided into 14 sub-tracks, which can be grouped into academic, technical and vocational schools. I estimate a separate event study for each track; in none of them can I reject that the pre-treatment coefficients are jointly zero.

The results reveal substantial heterogeneity across tracks. Academic high schools (Figure 7a) show small and statistically insignificant coefficients throughout the observation window. Vocational schools (Figure 7c) show no reduction in dropout either, consistent with their institutional structure, whereby students can obtain a professional qualification after the third year rather than completing a five-year diploma. Technical high schools (Figure 7b), by contrast, show a pronounced and sustained response, with reductions in dropout of two to three percentage points that become statistically significant from the third year after the first repurpose. This differential response likely reflects the different student populations and institutional contexts of each track. Technical schools occupy an intermediate position between academic preparation and vocational training, and technical schools have experienced sharper declines in tertiary transition rates than other school types (Contini and Salza, 2020). Unlike vocational students, who can enter the labour market with a three-year professional

certificate, technical students need the full five-year diploma to meaningfully improve their employment prospects, making early dropout especially costly for this group. The improved incentives generated by repurposing may therefore be especially valuable for students in technical programmes. Table A.4 shows, however, that repurposing also affects enrollment: the share of first-year students declines in vocational and, to a lesser extent, technical schools, while it slightly increases in academic schools. Since dropout is measured within schools and cohorts, changes in enrollment do not mechanically affect the dropout rate of students already enrolled, but they may change the composition of later cohorts. The effect on technical schools should therefore be interpreted with this caveat in mind.



(a) Academic High Schools

(b) Technical High Schools

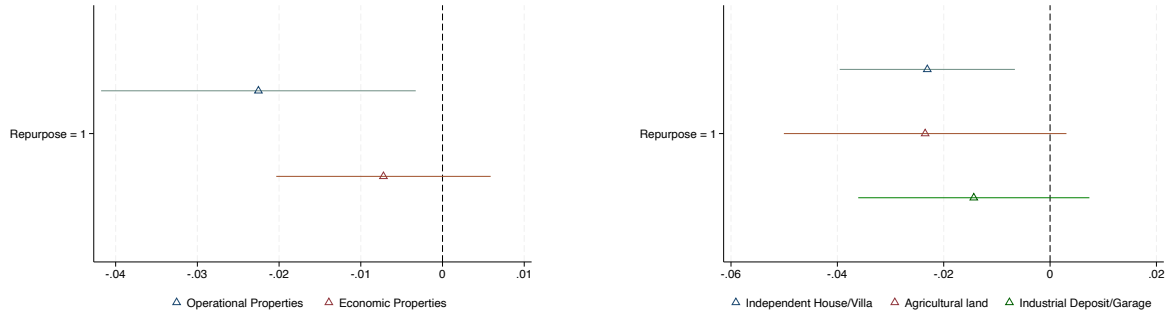
(c) Vocational High Schools

Figure 7 Estimated effects of the repurpose of Mafia properties on dropout rates before and after the first repurpose starts by school track.

Beyond school characteristics, the treatment effect may also vary with the characteristics of the repurposed Mafia properties.

I start with the basic attributes of the repurposed properties. Following Operti (2018), I define as *operational properties* the residential real estate that belonged to Mafia families and signalled their local sovereignty, and as *economic properties* those used for commercial purposes. Figure 8 shows that the effect is driven by catchment areas with at least one

repurposed operational property, while areas treated only by economic properties show no statistically significant effect.



(a) Property use before confiscation

(b) Property type

Figure 8 Estimated effects of the repurpose of Mafia properties on dropout rates before and after the first reuse starts by properties' features.

Second, I investigate whether the effect depends on the type of activity hosted in the repurposed property. If the intervention worked by adding educational and training resources, properties dedicated to education and employment should generate larger effects than those used for cultural or welfare purposes. Since a catchment area may contain properties offering several types of activity, I estimate a separate specification for each category. In each regression, I compare treated areas where at least one property offers a given type of activity with never- and not-yet-treated areas, controlling for treated areas whose properties offer no activity of that type. Appendix D describes the specification and Figure 9 reports the coefficients.[23] The effect is concentrated in properties hosting cultural and welfare activities. Areas with a property hosting cultural activities show a reduction of 1.4 percentage points (25% relative to the mean), with a confidence interval lying entirely below zero, and areas with welfare activities a reduction of 1.6 percentage points (28%), significant at the 10% level. Properties offering education or employment services, by contrast, are associated with smaller and statistically insignificant reductions, of 1.1 and 0.8 percentage points respectively. This pattern is difficult to reconcile with a mechanism operating through the provision of educational resources: had the effect been driven by educational inputs, properties delivering education and employment services should have produced the largest effects. Instead, the reduction is concentrated where repurposed properties host cultural and community activities, pointing to a channel that operates through the visible reclaiming of the property and its role in the community, rather than through the services delivered within it.

[23] Regression results are reported in Table A.5.

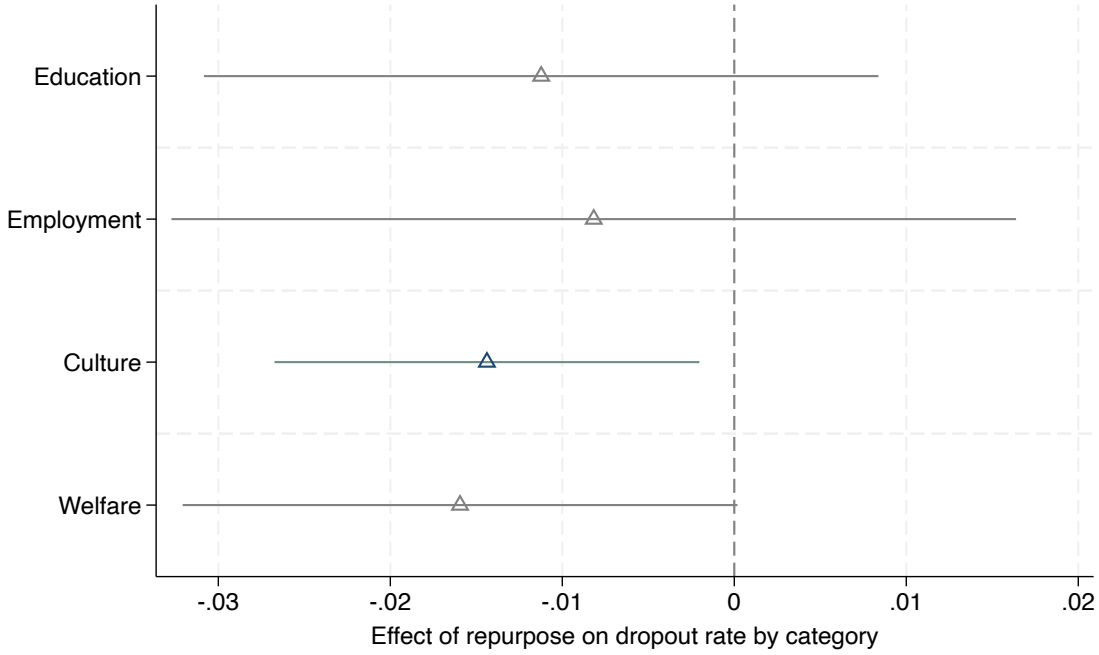


Figure 9 Estimated effects of the repurpose of Mafia properties on dropout rates before and after the first reuse starts by type of activity in the repurposed property.

Last, the treatment effect may vary with the number of properties repurposed within a catchment area. I therefore examine whether the effect is driven by a single repurpose event or intensifies with multiple ones. To investigate this, I split the sample into schools whose catchment area experienced the repurpose of more than one property in the same year, and those that did not. Table 4 reports the results. Columns (1) and (2) show that schools in the first group exhibit small and statistically insignificant effects. By contrast, schools exposed to multiple repurposed properties at once, in Columns (3) and (4), display larger and precisely estimated reductions of 2.5 and 2.2 percentage points, corresponding to declines of 40% and 36% relative to the mean. This pattern suggests that the effect intensifies with the scale of repurposing within the catchment area.

Table 4 Impact of repurposing Mafia real estate on the share of students dropping out from grades 11 to 13 by intensity of exposure.

	(1)	(2)	(3)	(4)
Dep: Dropout rates G11-G13	**Properties repurposed = 1**		**Properties repurposed > 1**	
Repurpose = 1	-0.0089	-0.0086	-0.0245***	-0.0221***
	(0.0117)	(0.0119)	(0.0090)	(0.0070)
clustered SE	yes	yes	yes	yes
school FE	yes	yes	yes	yes
time FE	yes	yes	yes	yes
migration	no	yes	no	yes
retention	no	yes	no	yes
Observations	685	685	708	708
Number of schools	124	124	127	127
Mean dep. var.	0.0541	0.0541	0.0617	0.0617

Notes: TWFE model. The treatment variable is equal to 1 whenever there is at least one repurposed Mafia property within the school catchment area. Columns (1) and (2) include the schools whose catchment area has one repurposed property, Columns (3) and (4) those with more than one. Columns (1) and (3) do not include controls, while Columns (2) and (4) control for students' migration and grade retention rates in grades 11 and 12. Standard errors, clustered at the school level.

I then directly estimate the dynamic evolution of the treatment effect with respect to the intensity of repurposing. I use the estimator of de Chaisemartin and D'Haultfœuille (2024), with a continuous treatment equal to the cumulative number of repurposed properties in the catchment area, weighted by the catchment area's population aged 15 to 19 relative to the average catchment area. While the count alone captures the intensity of repurposing, it does not account for the size of the population that could potentially benefit from it; weighting by the student-age population gives more weight to properties repurposed where more students live. This allows me to follow the effect of an additional weighted repurposed property over time. Figure 10 shows no differential pre-trends: I cannot reject that the placebo coefficients are jointly zero (p-value 0.79). The effect grows over time: it is small in the first year, about 3 percentage points in the second and third years, and reaches 8 and 13 percentage points in the fourth and fifth years, when it becomes statistically significant. The average total effect per additional weighted repurposed property is a reduction of 0.59 percentage points (10% relative to the mean), significant at the 5% level. This increasing pattern suggests that the returns to repurposing build up over time, as programmes in repurposed properties become established.

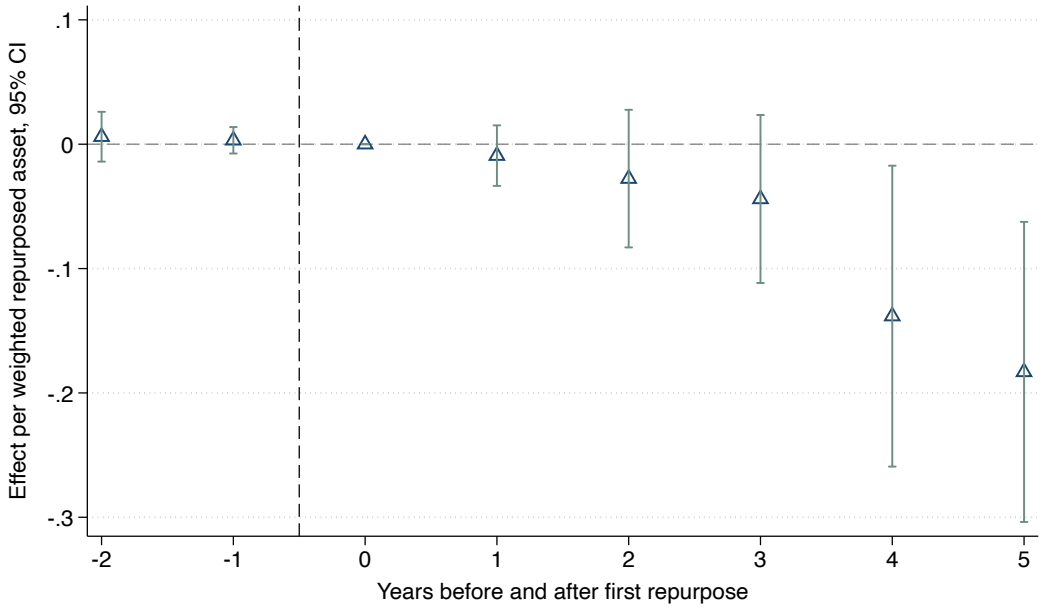


Figure 10 Event study using the cumulative count of repurposed Mafia properties weighted by the population aged 15 to 19 of the school catchment area, relative to the average catchment area. Effects of an additional weighted repurposed property on dropout rates by years since the first repurposing, estimated with de Chaisemartin and D'Haultfœuille (2024); placebo estimates before the first repurposing. 95% confidence intervals.

Finally, I weight the cumulative number of repurposed properties by the inverse of their average distance from the population-weighted centroid of the catchment area, so that properties closer to where students live receive more weight. Figure A.6 in Appendix A shows effects of the same sign and a similar increasing pattern, which are, however, not statistically significant. Taken together, the results are consistent with the effect increasing with exposure to repurposed properties, although the evidence on the role of their spatial distribution is less precise.

7 Effects on Students' Beliefs

Repurposing Mafia properties for social use decreases the levels of dropout rate for schools located nearby. These effects are particularly pronounced for technical high schools and lower-performing schools. The main decrease is driven by the repurpose of properties that were previously used as luxurious Villas and apartments formerly owned by Mafia families, while prior to repurpose we do not find any significant effect of confiscations and realloca-

tions on educational outcomes. Notably, the reduction is concentrated in properties hosting cultural activities, rather than those providing direct educational or vocational resources, pointing to a community-level rather than a purely educational-input channel. Moreover, a dose-response pattern emerges, with each additional repurposed property amplifying the reduction in dropouts. In this Section I investigate the mechanisms behind the estimated effects presenting both empirical and anecdotical evidence; the results suggest that the decrease in dropout is driven by a change in students' beliefs about the importance of meritocratic channels for job-seeking, as well as the reasons why people should turn to the Mafia. Additionally, I find that students believe the Mafia is a weaker force, while they perceive an increase in their own empowerment as civil society and the State's capacity to face the Mafia. Last, I don't observe significant changes in the local opportunity structure which might drive the effects, such as Mafia recruitment levels, additional educational support delivered in repurposed properties, and a concurrent ongoing gentrification process.

7.1 Students' Aspirations

The central aim of repurposing former Mafia-owned properties is to change the local opportunity structure local residents face, both considering what they experience and what they perceive of their surrounding reality. Acknowledging that arresting Mafia groups is only the first enforcement step of the policy, the State seeks to send a strong message by building capacity in these infiltrated areas, replacing criminal governance with visible institutions that serve the community. According to this, there are two main mechanisms that could be driving the effect: a change in the way juveniles perceive their future options, and a change in what their options are. On the one hand, it could be the case that after seeing a physical change and after getting aware about the socio-economic effect of the Mafia, juveniles change their perspective about Mafia's reputation and social status and they might prefer to follow meritocratic channels for job-seeking and career advancement. This would reduce the perceived benefits of joining the Mafia, while push juveniles to substitute for alternative job-seeking channel. On the other hand, the results could be driven by an increase in the educational support provided to juveniles on these properties, such as after-school programmes and educational mentoring activities which account for around the 10% of the existing activities carried out. This way, once facilitated in their educational journey, juveniles will face a lower cost of staying in education.

Recent conversations with *Libera*'s repurpose team managers and social workers from the NGOs working in the repurposed properties, together with the interviews conducted by

Nazzaro (2021) and Falcone et al. (2016), lend support to the idea that these interventions reshape local attitudes toward the Mafia. NGOs' workers describe these places as gathering centres for the local community, attracting people of all ages to take part and put efforts into transforming the previously infiltrated area; renovating social bonds and reclaiming the local area through the participatory redevelopment of these properties seems to address a local change of perspective.
Interviews conducted by Nazzaro (2021) highlight that

> Directly involving citizens in various activities for the redevelopment of former Mafia properties is important in the perception of that space as a newly acquired common space that had previously been taken away for criminal purposes. - Interview from the NGO *AGESCI*, Apulia, Italy

Moreover, especially considering the juveniles' experiences, Nazzaro (2021) reports that

> "On properties and lands confiscated from the Mafia [...] we find experiences featuring many young people, who were the first to decide to participate, aware of the importance of what was at stake." - *Carlo Borgomei, President of Fondazione Con il Sud*

On the same topic, Falcone et al. (2016) reports that

> This policy has served to develop NGOs gathering juveniles who work in the sector of sustainable agriculture and develop new markets that did not exist until the day before, and has introduced and consolidated in our society a strong and open message: it is indeed possible to build a different future, where many and many different people in terms of culture, profession and age participate to make it happen. - *Lucio Cavazzoni, president of Alce Nero,* interview from Libera

To provide more nuanced empirical evidence, I examine how these interventions alter students' perceptions about educational and criminal pathways to career development in their communities. I employ TWFE models using data from the survey *The Perception of the Mafia*, which was collected from a subsample of schools. The survey covers 54 schools with an average of 110 students per school. Out of the total sample, 24 schools are treated while 30 serve as controls. I first exploit the answers given to the question: *What of this options you think will be more useful for you to find a job in your town?*. Students were asked to rate several job-seeking channels on a scale from 1 - as the most useful - to 7 as the most useless. The options include several meritocratic channels, such as embark in an educational training, refer to a job centre, take a competitive examination for civil service, and several informal channels such as asking the Mafia, a politician, family, or a friend. If the repurpose of Mafia properties reduces the attractiveness of criminal pathways and the expected returns of joining the Mafia, students in treated areas should place greater value on other employment chan-

nels, whether formal or informal, depending on whether they substitute toward legal or illegal alternatives. Figure 11 presents the ATT estimated under a DiD design that includes school fixed effects and time fixed effects[24]; I also weight each school by the number of students replying to the survey to account for varying school sizes in the survey sample. The outcome variables represent the percentage of students for each school who rated each pathway as a *very useless* or *very useful* option, allowing me to identify whether repurposed Mafia properties systematically alter the distribution of student perceptions within schools. Following previous literature (Villa, 2025), I focus on the extreme categories of student responses to better capture meaningful shifts in perceptions[25] Finally, standard errors are clustered at the school level across all specifications.

The results reveal compelling evidence of altered student perceptions consistent with a weakened criminal influence and a strengthened belief in legitimate pathways. The left panel (a) examines the effect of reusing activities on the probability of rating each pathway as *very useful*. Most notably, students in areas with repurposed Mafia properties are significantly more likely to view education training as very useful, with a substantial 34% increase relative to the baseline, indicating that the intervention meaningfully enhances students' perception of education as a pathway to employment success. A similar pattern emerges for how students perceive civil-service examinations: students in treated areas are more likely to rate this channel as very useful, a reduction of roughly 84% relative to the baseline mean of 0.163, though the effect is only marginally significant (t = 1.65). The right panel (b) examines the effect of reusing activities on the probability of rating each pathway as *very useless*. Most notably, students in areas with repurposed Mafia properties are 24% more likely to rate the option to ask for help from the Mafia as very useless, while they are 26% more likely to rate the option to ask a politician for help as very useless. Overall the results suggest that students come to view Mafia connections as a weaker and less useful route to employment, the expected benefit of that pathway falls, reducing the incentive to leave school early to pursue it. The observed reduction in dropout rates is therefore consistent with a mechanism that works chiefly by eroding the attractiveness of the criminal channel, with the strengthened perception of education as a legitimate alternative reinforcing, rather than driving, the effect.

[24]Regressions table results are reported in Table A.6

[25]I measure the most useful choices as the percentage of students that rate each option with 1 or 2, and the most useless choices as the percentage of students that rate each option with 6 ot 7.

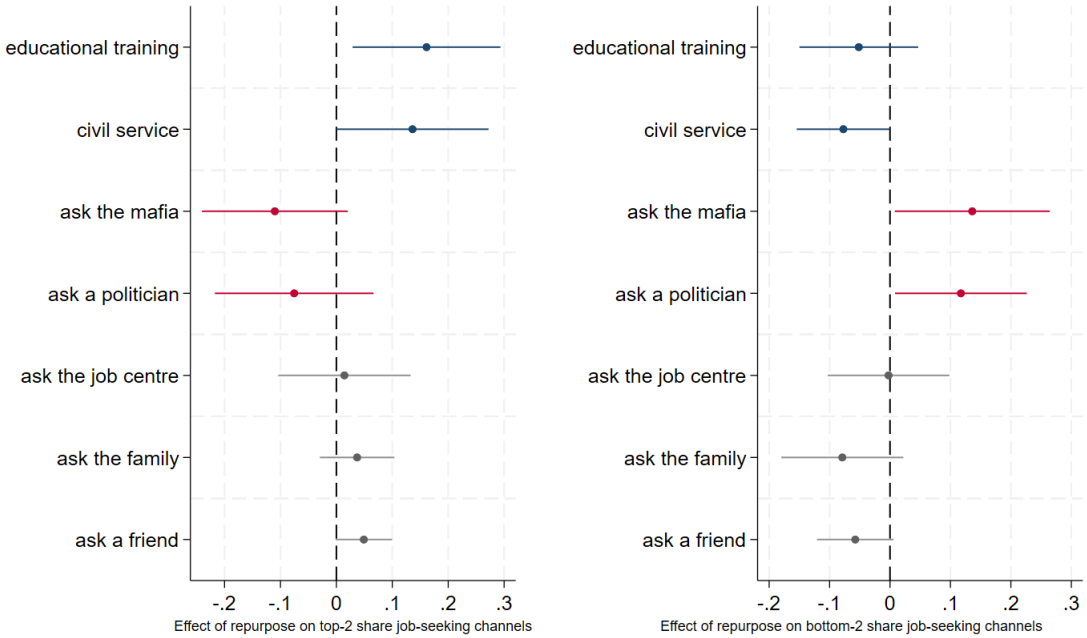


Figure 11 Estimated effects of repurposing practices on students' survey answers about career development. The left panel (a) reports the effect of reuse on the share of students ranking each channel among the two *most* important routes to career success; the right panel (b) reports the effect on the share ranking each channel among the two *least* important. Coefficients are from separate regressions with school and year fixed effects, weighted by the number of students and with standard errors clustered at the school level; results are significant at the 90% confidence level.

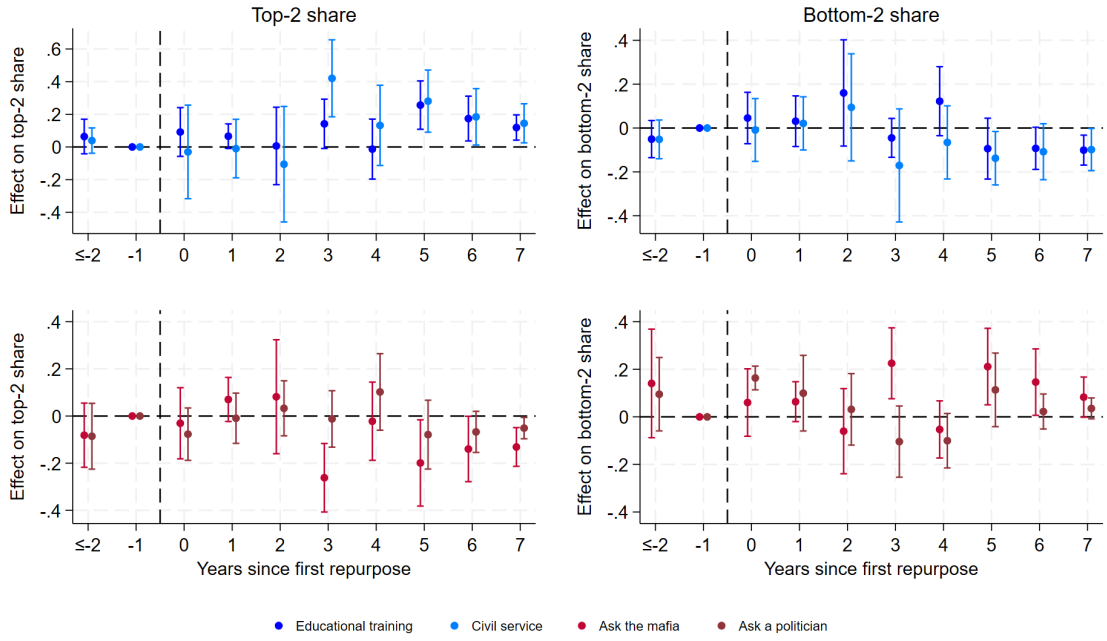


Figure 12 Event studies of repurposing practices on students' survey answers about career development. The left panels report the effect of repurpose on the share of students ranking each channel among the two *most* important routes to career success; the right panels report the effect on the share ranking each channel among the two *least* important. Coefficients are from separate regressions with school and year fixed effects, weighted by the number of students and with standard errors clustered at the school level; results are significant at the 90% confidence level.

The event study estimates in Figure 12 speak directly to the timing of the mechanism. Across the four channels, the pre-treatment coefficients are flat and statistically indistinguishable from zero[26], supporting the parallel-trends assumption underlying the design. The post-treatment movements are concentrated in the two channels most central to the mechanism and emerge on a common horizon: the share of students ranking the Mafia among the most useful employment routes falls, and its share among the least useful rises, while the standing of the civil-service examination among the most useful routes. Both effects materialise around the third year fo treatment, moving over the same period. The remaining channels move less precisely, consistent with the intervention acting chiefly on the perceived returns to the criminal pathway and its most direct meritocratic substitutes.

This timing is informative for the concern that a change in perceptions should either precede or coincide with a behavioural change. The belief shift materialises only from the second-to-third year after repurpose, and it does so on the same horizon at which the reduction in dropout becomes statistically significant in Figure 6. The co-timing of the two

[26]Due to a small sample, I am unable to retrieve more information about pre-trends.

responses is what the proposed mechanism predicts. Repurposing does not change the local opportunity structure overnight: the property must be assigned, an NGO must take it over, and its activities must become an established and visible presence in the community before residents' assessment of the relative attractiveness of criminal and legal pathways begins to shift. The delay of roughly three years is therefore not evidence against the mechanism but a signature of it, as a gradual erosion of the perceived returns to the Mafia channel, reflecting in students' stated beliefs and in their enrolment decisions at the same time.

7.2 Trust in State Capacity

The evidence so far suggests that repurposing properties reduces the level of dropout by shifting how students evaluate criminal versus legal pathways to career advancement. It is possible that the interventions raise the expected returns from education by strengthening students' belief that education leads to legitimate employment, and it could lower the expected return to a criminal career, as students come to see Mafia connections as a weaker and less rewarding route to advancement. However, it is also possible that the repurpose of Mafia properties raise the perceived cost of joining the Mafia as the visible reclaiming of a property signals a more present and capable State and thus a higher perceived probability of enforcement. These interpretations are not mutually exclusive, but distinguishing between them clarifies what the policy actually changes.

Two further survey questions let me examine whether repurposing shifts students' perceptions of the State's capacity. The first asks the students' level of agreement with a series of statements about the relative strength of the State and the Mafia. The second asks the students to rate the most and least trusted institutional actors among several choices, allowing me to measure whether the intervention raises confidence in the formal institutions that embody State capacity. Figure 13 shows that students' are less likely to agree with the fact that the Mafia is stronger because it infiltrates the State, and that the State does too little to fight the Mafia, respectively by 4.3% and 24% relative to the mean. I further test the dynamics effects of repurpose on these two perception measures. The estimates are imprecise and most individual coefficients cannot be distinguished from zero. The point estimates are nonetheless suggestive: for both measures coefficients turn negative in the first years after repurpose before returning toward zero at longer horizons, consistent with a shift toward greater confidence in State capacity. While we interpret these results cautiously given the absence of estimable pre-treatment coefficients, the pattern is consistent with the static estimates. Notably, the shift in perceived State capacity appears earlier than the other mechanism responses: it emerges in the first years after repurpose, whereas the change in students'

preference of employment channels and the reduction in dropout become pronounced only from around the third year. This ordering is consistent with the proposed mechanism, in which the visible reclaiming of the property first alters students' perception of the State's presence and capacity, and the shift in students' beliefs about legal and illegal pathways follows as that perception takes hold. I therefore read these dynamics as corroborating the estimates from Figure 13, and as suggestive that the State-capacity channel might operate upstream of the other responses.

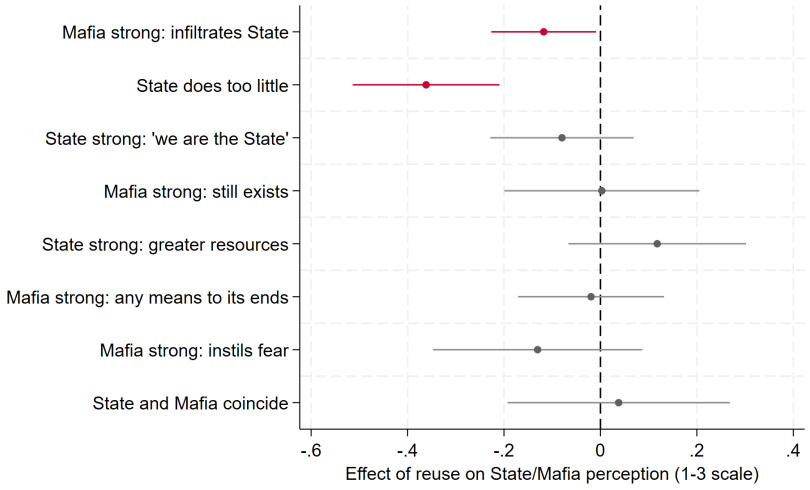


Figure 13 Estimated effects of repurposing practices on students' agreement with statements about the strength of the Mafia and the State. Each point reports the effect of repurpose on the share of students rank of statements, estimated from separate regressions that include school and year fixed effects, and are weighted by the number of students. Standard errors are clustered at the school level. Results are significant at the 90% confidence level.

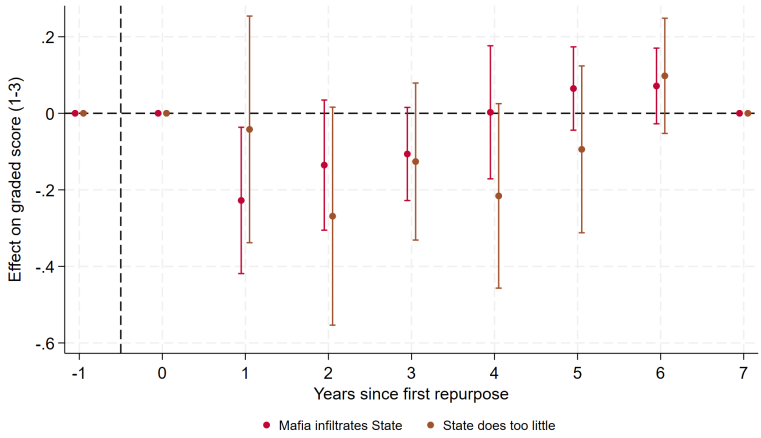


Figure 14 Event studies of repurposing practices on students' agreement with statements about the strength of the Mafia and the State. Each point reports the effect of repurpose on the share of students rank of statements, estimated from separate regressions that include school and year fixed effects, and are weighted by the number of students. Standard errors are clustered at the school level. Results are significant at the 90% confidence level.

If repurposing strengthens perceived State capacity and raises the perceived cost of joining the Mafia, we should also observe changes in how much students trust the institutional bodies that represent the State. Figure 15 reports the effect of reuse on the share of students naming each institution as their most and least trusted. The effects are concentrated on the least-trusted margin and among the institutions most directly associated with the State's local enforcement and governance. Students in treated areas are significantly less likely to name the police and local politicians among the institutions they trust least — by 30% and 42% relative to the mean, respectively. As the most-trusted margin shows no comparable shift, the intervention appears to work by eroding active distrust of core State institutions rather than by generating enthusiasm for them: a more modest but arguably more plausible change, since reclaiming a single property is more likely to soften scepticism about the State's presence than to produce positive endorsement of it.

This reading fits the state-capacity channel. Police and local politicians are precisely the actors whose credibility is undermined where the Mafia governs in the State's place; a fall in distrust for police and local politicians is consistent with students perceiving the State as a more present and capable authority following the intervention. Figure 16 investigates the dynamic effects of the treatment.Despite the small sample size, the point estimates line up in an intuitive way: the share of students naming these institutions as least trusted falls first, in the first year after repurpose, while any rise in the share naming them most trusted appears only later, around the third year. This ordering is consistent with the interpretation above for which distrust of core State institutions softens soon after the property is reclaimed, and

more positive endorsement, where it emerges at all, follows only once the State's presence is established.

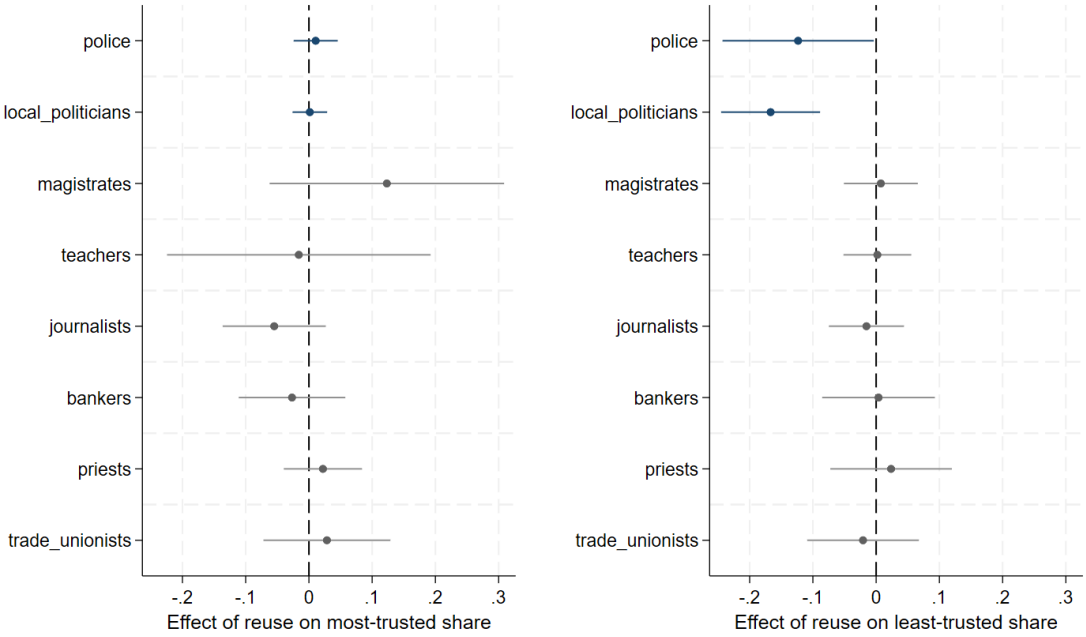


Figure 15 Estimated effects of repurposing practices on students' level of trust in several institutional bodies. Each point reports the effect of repurpose on the share of students level of trust, estimated from separate regressions that include school and year fixed effects, and are weighted by the number of students. Standard errors are clustered at the school level. Results are significant at the 90% confidence level.

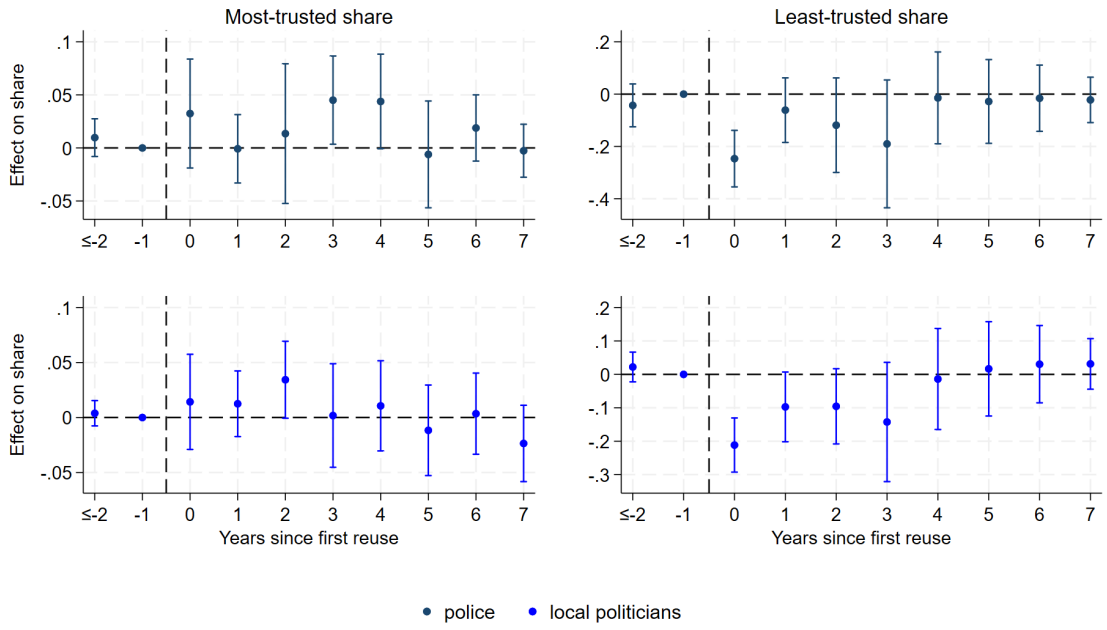


Figure 16 Event studies of repurposing practices on students' level of trust in several institutional bodies. Each point reports the effect of repurpose on the share of students level of trust, estimated from separate regressions that include school and year fixed effects, and are weighted by the number of students. Standard errors are clustered at the school level. Results are significant at the 90% confidence level.

8 Alternative Explanations

The results so far are consistent with a belief-based channel and the repurposed properties signaling greater State's capacity. However, alternative mechanisms could produce the same pattern. I discuss and rule out here four alternative explanations. The first is that the dropout reduction reflects a lower cost of education, through the provision of additional educational activities at the repurposed sites. The second is that repurposing projects are part of a concurrent process of gentrification might be removing neighbourhood barriers to schooling, such as distance and transport costs. The remaining two concern the belief mechanism: even if we assume that the effect operates through students' perceptions, such a shift could reflect social-desirability pressure rather than a genuine change in perceived returns, or it could occur at the household level rather than at the student level.

Provision of educational support. It is possible that the estimated reduction in the dropout rate reflects a decrease in the costs related to education. If repurposed properties offer specific educational activities such as after-school clubs and educational mentoring training, the decrease we observe in dropout rate might be the direct consequence of higher investments

in keeping the youth in school in treated areas. Similar effects could be observed if repurposing projects are implemented as a part of broader educational or socially-focused programmes funded at the municipality or neighbourhood level. Figure 9 already suggest this might not be the case. The activity decomposition shows that the effect is driven by properties hosting cultural activities, not those offering education or employment services. Here I take a further step and test whether repurposing projects affects students' academic performance, as measured by graduating grades. If the intervention operates by lowering the cost of schooling through the provision of additional educational resources, we could also expect an improvement in graduating grades alongside the reduction in dropout.[27]

[27]The prediction is not unambiguous. If the intervention retains students who would otherwise have dropped out, and these tend to be lower-performing, their inclusion in the pool of graduates could push average grades downward.

Table 5 Alternative explanations: the effect of repurposing properties on graduating grades.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dep. Variable	**Grades from 60-69**				**Grades from 70-79**			
Repurpose = 1	-0.0062	-0.0064	-0.0052	-0.0055	0.0148	0.0147	0.0146	0.0144
	(0.0208)	(0.0210)	(0.0208)	(0.0209)	(0.0131)	(0.0131)	(0.0130)	(0.0130)
Observations	1,371	1,371	1,371	1,371	1,371	1,371	1,371	1,371
R-squared (within)	0.381	0.383	0.383	0.384	0.154	0.154	0.156	0.156
Number of schools	251	251	251	251	251	251	251	251
clustered SE	yes	yes	yes	yes	yes	yes	yes	yes
school FE	yes	yes	yes	yes	yes	yes	yes	yes
time FE	yes	yes	yes	yes	yes	yes	yes	yes
migration	no	yes	no	yes	no	yes	no	yes
retention	no	no	yes	yes	no	no	yes	yes
Mean dep. var.	0.2987	0.2987	0.2987	0.2987	0.2541	0.2541	0.2541	0.2541
Dep. Variable	**Grades from 80-89**				**Grades from 90-100**			
Repurpose = 1	0.0021	0.0022	0.0013	0.0015	-0.0066	-0.0065	-0.0072	-0.0071
	(0.0128)	(0.0128)	(0.0128)	(0.0128)	(0.0122)	(0.0123)	(0.0121)	(0.0121)
Observations	1,371	1,371	1,371	1,371	1,371	1,371	1,371	1,371
R-squared (within)	0.087	0.089	0.090	0.092	0.312	0.314	0.314	0.316
Number of schools	251	251	251	251	251	251	251	251
clustered SE	yes	yes	yes	yes	yes	yes	yes	yes
school FE	yes	yes	yes	yes	yes	yes	yes	yes
time FE	yes	yes	yes	yes	yes	yes	yes	yes
migration	no	yes	no	yes	no	yes	no	yes
retention	no	no	yes	yes	no	no	yes	yes
Mean dep. var.	0.1941	0.1941	0.1941	0.1941	0.1277	0.1277	0.1277	0.1277

Notes: TWFE model. The treatment variable is equal to 1 whenever there is at least one repurposed Mafia property within the school catchment area. The dependent variable is the share of graduates in each grade bracket. Columns (1) and (5) include school and time fixed effects; Columns (2) and (6) add students' migration, Columns (3) and (7) grade retention rates in grades 11 and 12, and Columns (4) and (8) both. All columns are estimated on the same sample. Standard errors, clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 5 reports the effect of repurposing on the distribution of graduating grades, estimated separately for four grade brackets using the same specification as for dropout. The coefficient on repurposing is statistically insignificant in every bracket and across all specifications, indicating no detectable effect of the intervention on academic attainment. The pattern of signs is nonetheless informative for the composition concern raised above. Were

the reduction in dropout retaining weaker students who would otherwise have left, their inclusion among graduates should raise the share in the lowest grade band. Instead, the point estimate for the lowest band (60–69) is, if anything, negative, while those for the higher bands are marginally positive. There is thus no evidence that the dropout effect operates by adding low-performing students to the graduate pool, nor that it raises attainment among those it retains. Altogether, the absence of any grade effect is inconsistent with an educational-inputs channel, which would predict measurable gains in performance, and consistent with a mechanism that operates on the decision to remain in school rather than on the students' performance itself.

Another related possibility is that the effect reflects not the activities on the repurposed properties themselves, but concurrent public investment in education and social inclusion in the same areas, which could reduce dropout through the same cost-of-schooling channel. I test this directly by controlling for other educational and social programmes implemented at the municipal level.

I collect data on the number of educational and social-cohesion projects implemented in each municipality and year from the Italian Department for Cohesion Policy [28]. The data cover projects targeting a range of developmental goals for local administrations between 2007 and 2022; I focus on education and social-inclusion projects, measured as the share of completed and ongoing projects relative to the total implemented in each municipality per year. Adding these as controls tests whether the estimated effect of repurposing reflects coincident policy activity rather than the intervention itself. Table 6 reports the results. The coefficient of the effect of repurposing properties is virtually unchanged across all specifications, ranging from 1.9 to 2.1 percentage points and remaining significant throughout, while neither educational nor social-inclusion projects show any detectable association with dropout rates (coefficients close to zero with large standard errors). The stability of the treatment coefficient indicates that the dropout reduction is specific to the repurposing intervention rather than a reflection of broader municipal development activity. These results suggest that repurposing effect might not be operating through the same channel as general educational provision. This is consistent with the activity decomposition, where the effect was driven by cultural rather than educational programmes.

[28] The data have been collected from opencoesione.gov.it

Table 6 Alternative explanations: controlling for concurrent municipal programmes.

	(1)	(2)	(3)	(4)
	Dropout rate G11-G13			
Repurpose = 1	-0.0170**	-0.0168**	-0.0169**	-0.0165**
	(0.0079)	(0.0075)	(0.0079)	(0.0075)
Educational programmes	-0.0036	-0.0036	-0.0037	-0.0036
	(0.0089)	(0.0090)	(0.0090)	(0.0091)
Social cohesion programmes	0.0036	0.0016	0.0034	0.0012
	(0.0106)	(0.0106)	(0.0106)	(0.0104)
Clustered SE	yes	yes	yes	yes
School FE	yes	yes	yes	yes
Time FE	yes	yes	yes	yes
Migration controls	no	yes	no	yes
Retention controls	no	no	yes	yes
Observations	1,335	1,335	1,335	1,335
Number of schools	251	251	251	251
Mean dep. var.	0.0588	0.0588	0.0588	0.0588

Notes: TWFE model. The treatment variable is equal to 1 whenever there is at least one repurposed Mafia property within the school catchment area. Educational and social cohesion programmes are the shares of ongoing education and social-inclusion projects in the municipality and year (OpenCoesione). Column (1) does not include controls, Column (2) controls for students' migration, Column (3) for grade retention rates in grades 11 and 12, and Column (4) for both. All columns are estimated on the same sample. Standard errors, clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Mafia presence and recruitment. The repurpose of Mafia properties represents the final stage of the CRR policy, which requires first confiscating and then reallocating confiscated assets under the jurisdiction of local authorities. Boeri et al. (2023) demonstrate that the confiscation of nearby Mafia properties depresses housing prices by signalling the local presence of the Mafia, while reallocation sends the opposite signal, restoring property values. A potential concern is therefore that my treatment variable captures a recovery process following multiple confiscations that negatively affected neighbourhoods, reduced local economic activity, and contributed to higher dropout rates.

A related concern is that the repurpose treatment merely captures dropout reductions driven by the reallocation stage rather than the repurposing itself. I argue this is implausible for two reasons. First, reallocation is a purely administrative process whereby confiscated properties are formally transferred to local authorities. While this may signal a municipality's intention to repurpose former Mafia assets, it generates no tangible changes at the community level and no social activity that could plausibly alter youth behaviour or educational

decisions. Second, Table 7 provides direct empirical evidence: Columns (3) and (4) show that reallocation alone has no independent effect on dropout rates, with coefficients close to zero and consistently insignificant across specifications. Columns (5) and (6) show that the reuse coefficient remains stable in magnitude and statistically significant at the 5% level even after controlling for reallocations. Taken together, this evidence supports the interpretation that meaningful educational impacts arise when properties are actively repurposed for community use, rather than simply transferred to public ownership. The event study reported in Figure **??** corroborates these findings, showing no differential pre-trends and estimates closely in line with the main specification.

Table 7 Impact of repurposing Mafia real estate on the share of students dropping out from grades 11 to 13, including the previous policy stage.

	(1)	(2)	(3)	(4)	(5)	(6)
	Dropout rate G11-G13					
Repurpose = 1	-0.0168**	-0.0159*			-0.0165**	-0.0156*
	(0.0083)	(0.0081)			(0.0083)	(0.0081)
Reallocation = 1			0.0066	0.0057	0.0059	0.0050
			(0.0069)	(0.0068)	(0.0069)	(0.0068)
clustered SE	yes	yes	yes	yes	yes	yes
school FE	yes	yes	yes	yes	yes	yes
time FE	yes	yes	yes	yes	yes	yes
migration	no	yes	no	yes	no	yes
retention	no	yes	no	yes	no	yes
Observations	1,393	1,393	1,393	1,393	1,393	1,393
Number of schools	251	251	251	251	251	251
Mean dep. var.	0.0580	0.0580	0.0580	0.0580	0.0580	0.0580

Notes: TWFE model. Repurpose = 1 whenever there is at least one repurposed Mafia property within the school catchment area; Reallocation = 1 from the year in which at least one confiscated property in the catchment area is reallocated to the local authority. All columns are estimated on the same sample. Standard errors, clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Gentrification and neighbourhood effects. A second explanation is that repurposing former Mafia properties is part of a broader gentrification process occurring in treated areas. Regenerating a neighbourhood could lower local barriers to schooling, and the estimated effect would then reflect this upgrading rather than the Mafia-related intervention itself.
I test this using a triple difference-in-differences design that interacts the treatment with local property values, proxied by average rental prices per square metre at the neighbourhood level. If the effect operates through neighbourhood upgrading, it should be stronger where

property values are rising. The specification is:

$$\begin{aligned} DropoutG11\text{-}G13_{cnmt} = \beta_1 Repurpose_{ct} \times Prices_{nt} \\ + \beta_2 Repurpose_c + \beta_3 Prices_n \\ + \beta_4 X'_{ct} + \delta_c + \eta_t + \epsilon_{cnmt} \end{aligned} \tag{3}$$

where $DropoutG11 - G13_{cnmt}$ is the main outcome as specified before, and $Reuse$ is the main treatment dummy equal to 1 whenever there is at least one reused property in the school catchment area c in time t, and 0 otherwise. $Prices$ measures the average rental prices in neighbourhood n at time t, while β_1 is the coefficient of interest which captures how the impact of repurposed Mafia properties on dropout rates varies with local property price dynamics. $X'_{c,t}$ is a vector of time-varying community-level variables controlling for students' migration and grade retention dynamics. δ_c is school fixed effects and η_t is time fixed effects. If the main effect operates through the upgrading and gentrification of the neighbourhood, I would expect the effect of treatment to be stronger in areas experiencing increases in rental prices. Table 8 shows the results; the coefficient of the interaction term is very close to zero and not statistically significant, making the effect of rental price changes on dropout rates virtually identical in areas with and without repurposed properties. Moreover, the results are robust to the inclusion of migration and grade retention controls. These results support the argument that specific activities and programs do not reduce dropout rates through a general gentrification effect at the neighbourhood level.

Table 8 Impact of repurposing Mafia real estate on dropout rates for G11-G13 interacted with rental prices.

	(1)	(2)	(3)	(4)
	Dropout rate G11-G13			
Repurpose = 1	-0.0108	-0.0135	-0.0111	-0.0139
	(0.0341)	(0.0345)	(0.0337)	(0.0339)
Rental Prices	0.0036	0.0028	0.0039	0.0032
	(0.0044)	(0.0046)	(0.0045)	(0.0047)
Repurpose = 1 × Rental Prices	-0.0007	-0.0004	-0.0006	-0.0002
	(0.0033)	(0.0033)	(0.0033)	(0.0033)
clustered SE	yes	yes	yes	yes
school FE	yes	yes	yes	yes
time FE	yes	yes	yes	yes
migration	no	yes	no	yes
retention	no	no	yes	yes
Observations	1,393	1,393	1,393	1,393
Number of schools	251	251	251	251
Mean dep. var.	0.0580	0.0580	0.0580	0.0580

Notes: TWFE model. The treatment variable is equal to 1 whenever there is at least one repurposed Mafia property within the school catchment area. Rental prices are average rents per square metre in the neighbourhood. Column (1) does not include controls, Column (2) controls for students' migration, Column (3) for grade retention rates in grades 11 and 12, and Column (4) for both. All columns are estimated on the same sample. Standard errors, clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

9 Robustness Checks

The estimated effects are robust to a series of specification checks and alternative treatment assumptions. This section proceeds as follows. First, I test for additional confounding factors that may represent sources of omitted variable bias. Second, I investigate whether pre-treatment reallocations or anticipation effects could be confounding the estimated results. Third, I assess whether the treatment allocation is subject to spatial selection bias. Fourth, I test for spatial spillover effects onto neighbouring schools.

9.1 Selection Bias

Treated versus not-yet-treated schools. A primary concern is that never-treated schools may be fundamentally different from eventually-treated schools. Although Table 3 shows broadly comparable pre-treatment characteristics across groups, the never-treated areas may differ along unobserved dimensions, such as the degree of institutional engagement with confiscated assets and the level of local support for antimafia policies, making them an inappropriate control group. To address this, I only exploit the timing of the treatment by restricting the comparison to treated and not-yet-treated schools, excluding never-treated units entirely. This ensures that the control group consists only of schools that eventually receive treatment, and therefore plausibly belong to the same population of Mafia-affected areas undergoing institutional repurpose processes.

Table 9 reports the results from this restricted sample. The estimates are remarkably stable and, if anything, slightly larger in magnitude than the main specification, while remain statistically significant at the 5 percent level throughout. This stability is reassuring on two fronts. First, it suggests that the main results are not inflated by the inclusion of never-treated schools as being structurally different from treated areas. Second, the fact that the effect is preserved and marginally strengthened when identification comes solely from variation in treatment timing among eventually-treated schools lends further credibility to the causal interpretation. The reduction in the number of schools from 232 to approximately 182 does not meaningfully affect precision, indicating that the never-treated units are not driving the main statistical power of the baseline estimates.

Table 9 Impact of repurposing Mafia real estate on dropout rates for G11-G13: treated and not-yet-treated schools only.

	(1)	(2)	(3)	(4)
	Dropout rate G11-G13			
Repurpose = 1	-0.0188**	-0.0187**	-0.0188**	-0.0185**
	(0.0084)	(0.0082)	(0.0086)	(0.0083)
clustered SE	yes	yes	yes	yes
school FE	yes	yes	yes	yes
time FE	yes	yes	yes	yes
migration	no	yes	no	yes
retention	no	no	yes	yes
Observations	1,099	1,099	1,099	1,099
R-squared (within)	0.185	0.240	0.186	0.241
Number of schools	197	197	197	197
Mean dep. var.	0.0617	0.0617	0.0617	0.0617

Notes: TWFE model on the schools whose catchment area has at least one repurposed Mafia property during the sample period (never-treated schools excluded). The treatment variable is equal to 1 from the first year with at least one repurposed property within the school catchment area. Column (1) includes school and time fixed effects; Columns (2) and (3) control for students' migration and grade retention rates in grades 11 and 12, respectively; Column (4) reports the complete specification. All columns are estimated on the same sample. Standard errors, clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

The presence of amenities. In the main specification, I control for unit-specific and time-specific characteristics, as well as the time-varying rates of retention and migration among students. By controlling for these factors, I find no pre-trends in any of the estimated event study. Nevertheless, previous trends in the presence of NGOs and civic engagement might play a substantial role in how and when the policy is implemented locally. A natural question arises when interpreting the effects of the policy on dropout rate: is there something uniquely valuable about conducting activities in former Mafia properties, or would any NGO operating in the area produce similar reductions in dropout rates? To disentangle the specific contribution of repurposed properties from the broader effect of civil society presence, I test whether the main estimates are sensitive to controlling for the number of NGOs active within

each school's catchment area. If the observed effects were simply driven by the availability of social services nearby, regardless of whether they operate from repurposed Mafia properties, the main coefficient should diminish substantially once I account for the local presence of NGOs. Conversely, if the main coefficient remains stable while the proxy for the presence of NGOs shows little independent effect, this would suggest that the symbolic and contextual significance of operating within former Mafia properties generates distinctive value that conventional NGO activities might not be able to replicate. Table 10 supports this last interpretation. Columns (2) and (5) show that the presence of other NGOs in school catchment areas has negligible and statistically insignificant effects on dropout rates, indicating that general NGO activity alone does not meaningfully reduce educational dropout. Moreover, the repurposing effect remains remarkably stable when controlling for NGO presence. Comparing the baseline specifications in columns (1) and (4) with the full specifications in columns (3) and (6), the reuse coefficients are virtually unchanged in magnitude and remain statistically significant at the 5% level throughout. These results provide compelling evidence that the observed effects stem not from general social service provision in the area, but from the particular significance of conducting activities within former Mafia properties. I further confirm the absence of differential pre-trends while controlling for the presence of NGOs, with the corresponding event study presented in Figure A.5.

Table 10 Impact of repurposing Mafia real estate and NGO presence on dropout rates for G11-G13.

	(1)	(2)	(3)	(4)	(5)	(6)
			Dropout rate G11-G13			
Repurpose = 1	-0.0168**		-0.0161*	-0.0159*		-0.0150*
	(0.0083)		(0.0084)	(0.0081)		(0.0082)
NGOs presence		0.00012	0.00010		0.00014	0.00013
		(0.00009)	(0.00009)		(0.00009)	(0.00009)
clustered SE	yes	yes	yes	yes	yes	yes
school FE	yes	yes	yes	yes	yes	yes
time FE	yes	yes	yes	yes	yes	yes
migration	no	no	no	yes	yes	yes
retention	no	no	no	yes	yes	yes
Observations	1,393	1,393	1,393	1,393	1,393	1,393
Number of schools	251	251	251	251	251	251
Mean dep. var.	0.0580	0.0580	0.0580	0.0580	0.0580	0.0580

Notes: TWFE model. The treatment variable is equal to 1 whenever there is at least one repurposed Mafia property within the school catchment area. NGOs presence is the number of NGOs registered in the census sections of the catchment area. Columns (1)–(3) include school and time fixed effects only; Columns (4)–(6) also control for students' migration and grade retention rates in grades 11 and 12. All columns are estimated on the same sample. Standard errors, clustered at the school level.

Capacity-weighted catchment areas. A further concern relates to spatial selection bias arising from the method used to construct school catchment areas. Treatment allocation may be sensitive to the baseline assumption that students attend their geographically closest school regardless of its size or capacity. Since catchment areas are constructed by assigning each census block to the nearest school, the boundaries between adjacent catchment areas are determined purely by proximity. This rule may misclassify treatment status for schools near those boundaries: if larger schools in practice serve students living closer to a neighbouring smaller school, then a reused property near such a boundary could be assigned to the wrong catchment area, generating a measurement error in treatment that is correlated with school characteristics. I re-estimate the main results under alternative assumptions about the construction of schools' catchment areas. If the effect were spuriously driven by the specific rule used to assign treatment status, changing that rule should produce meaningfully different treatment allocations and consequently different estimates. Specifically, I weight the Euclidean distance between each census block and the closest school by the school capacity, namely the enrollment in the first year for each school measured at the baseline. This

capacity-weighted approach allows larger schools to draw students from greater distances and have more extensive catchment areas; this provides a more realistic representation of how students are distributed across urban areas (Pearce, 2000).[29] Under the capacity-weighted construction, only 18 schools switch treatment status relative to the baseline assignment. The small number of switchers is itself reassuring, suggesting that the two approaches yield largely consistent treatment allocations. To further ensure that the main results are not driven by schools with inherently ambiguous treatment assignment, I re-estimate the main specification excluding these 18 schools. Table 11 shows that the coefficients of interest are virtually unchanged across all specifications, whether estimated on the full sample or on the restricted sample. The stability of the estimates across both the alternative weighting scheme and the restricted sample jointly suggests that the treatment effect is not an artifact of how catchment areas are constructed. The somewhat wider confidence intervals in the restricted sample are expected given the reduction in sample size, and if anything underscore the importance of retaining the full sample for statistical precision without compromising identification.

Table 11 Impact of reusing Mafia real estate on the share of students dropping out at the age of 16 - capacity-weighted catchment areas

	(1)	(2)	(3)	(4)
Dep Var: Dropout rate G11-13	**All sample**	**Non-switchers only**	**All sample**	**Non-switchers only**
Reuse = 1	-0.0192**	-0.0206*	-0.0196**	-0.0207*
	(0.00934)	(0.0122)	(0.00943)	(0.0123)
clustered SE	yes	yes	yes	yes
school FE	yes	yes	yes	yes
time FE	yes	yes	yes	yes
migration	no	no	yes	yes
retention	no	no	yes	yes
Observations	1,296	1,192	1,272	1,174
R-squared	0.093	0.117	0.098	0.123
Number of schools	235	217	234	216
Mean dep. var.	0.0537	0.0558	0.0531	0.0552

Notes: TWFE model. The treatment variable is equal to 1 whenever there is at least one reused Mafia property within the school catchment area. Columns (1) and (2) report the full-sample results, while Columns (3) and (4)report the results for the sub sample of schools that have a consistent catchment area across the two methods. Standard errors are clustered at the school level

[29] Appendix D explains in detail how the areas have been computed.

9.2 Treatment Timing

Confounding from earlier policy stages. A distinct concern arises from the staged nature of the CRR policy. Boeri et al. (2023) show that confiscation itself alters local outcomes, depressing housing prices by signalling Mafia presence. If confiscation and repurposing occur close together in time, the pre-treatment period in my event study would be contaminated by an earlier shock, and the post-repurposing decline in dropout could reflect recovery from that shock rather than the intervention itself.

Two features of the setting limit this concern. First, the interval between confiscation and repurposing is substantial: the average gap in my sample is 14 years, so the confiscation shock is well outside the estimation window around repurposing. Second, and more directly, the null effect of the reallocation effects reported in Section 8 means there is no earlier movement in dropout rates from which a recovery could be measured. The flat pre-treatment coefficients in Figure 6 are consistent with this: had confiscation raised dropout in treated catchment areas, we would observe an upward drift in the years preceding the repurpose of confiscated properties.

Anticipation effects. A potential threat to the identification strategy is that families and students may anticipate the repurpose of confiscated Mafia properties before it formally takes place. If information about upcoming activities spreads through informal networks or institutional channels, households could adjust their educational decisions in advance, and the estimated treatment effect would partly capture these anticipatory responses rather than the direct impact of the intervention. To test for this, I examine whether the reuse of confiscated properties in period $t+1$ affects dropout rates measured in period t. A significant coefficient on the lead of treatment would indicate that behavioural changes precede the intervention, undermining the causal interpretation of the main results. Table 12 shows that the estimated coefficients are consistently small and statistically insignificant across all specifications, providing no evidence of anticipation effects. This reinforces the credibility of the main identification strategy: the reduction in dropout rates occurs only after properties are effectively repurposed and local NGOs begin offering social activities in the treated area.

Table 12 Anticipation effects: dropout in the year before the first repurposing.

	(1)	(2)	(3)	(4)
	Dropout rate G11-G13			
Year before repurpose $(t-1)$	-0.0054	-0.0088	-0.0045	-0.0079
	(0.0084)	(0.0087)	(0.0082)	(0.0085)
Repurpose = 1	-0.0261**	-0.0269**	-0.0253*	-0.0256*
	(0.0131)	(0.0133)	(0.0136)	(0.0138)
clustered SE	yes	yes	yes	yes
school FE	yes	yes	yes	yes
time FE	yes	yes	yes	yes
migration	no	yes	no	yes
retention	no	no	yes	yes
Observations	1,143	1,143	1,143	1,143
Number of schools	250	250	250	250
Mean dep. var.	0.0551	0.0551	0.0551	0.0551

Notes: TWFE model. "Year before repurpose" equals 1 in the year immediately preceding the first repurposing of a Mafia property in the school catchment area; Repurpose = 1 from the first year with at least one repurposed property. The sample excludes 2022, for which the following year is not observed. Column (1) includes school and time fixed effects; Columns (2) and (3) control for students' migration and grade retention rates in grades 11 and 12, respectively; Column (4) reports the complete specification. All columns are estimated on the same sample. Standard errors, clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

9.3 Spillover Effects

In the main results, I compare treated and control schools without accounting for potential spillover effects. However, it is plausible that the treatment also reduces dropout rates in nearby schools that are not formally treated. In such cases, the estimates would violate the Stable Unit Treatment Value Assumption (SUTVA), which requires that each unit's potential outcomes are independent of other units' treatment status (Cunningham, 2021). To address this concern, I follow a standard approach in the difference-in-differences literature: either excluding potentially affected controls or introducing an indicator for spillover exposure to separate the direct and spillover components of the treatment effect (Kline and Moretti, 2014; Butts, 2023). Because my sample includes only a limited number of control schools located far from treated areas, I opt for the second option and construct a time-varying spillover dummy. This variable equals 1 if a school's catchment area shares at least one boundary

with a treated area after treatment occurs, and zero otherwise.

Table 13 reports the results. Column (1) is the baseline, which ignores spillover effects. Column (2) controls for spillovers, estimating a direct effect which is slightly smaller, around 1.6 percentage points; the spillover effect appears to be insignificant, but with a substantial reduction of the dropout rate of nearby untreated areas of 1.6 percentage points. In Columns (3) and (4), I re-estimate the same specifications but including controls, showing similar patterns. This test indicates that ignoring spillovers leads to a modest overestimation of the direct treatment effect, but the overall results remains robust.

Table 13 Impact of repurposing Mafia real estate on dropout rates for G11-G13, controlling for spillover effects.

	(1)	(2)	(3)	(4)
	Dropout rate G11-G13			
Repurpose = 1	-0.0168**	-0.0156*	-0.0159*	-0.0148*
	(0.0083)	(0.0080)	(0.0081)	(0.0078)
Spillover = 1		-0.0114		-0.0105
		(0.0117)		(0.0117)
clustered SE	yes	yes	yes	yes
school FE	yes	yes	yes	yes
time FE	yes	yes	yes	yes
migration	no	no	yes	yes
retention	no	no	yes	yes
Observations	1,393	1,393	1,393	1,393
Number of schools	251	251	251	251
Mean dep. var.	0.0580	0.0580	0.0580	0.0580

Notes: TWFE model. Repurpose = 1 whenever there is at least one repurposed Mafia property within the school catchment area. Spillover = 1 if the school catchment area shares at least one boundary with a treated catchment area, after treatment occurs. Columns (1) and (2) include school and time fixed effects; Columns (3) and (4) also control for students' migration and grade retention rates in grades 11 and 12. All columns are estimated on the same sample. Standard errors, clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

9.4 Validity of Survey Results

Social desirability bias. Having ruled out the educational-inputs and gentrification channels, the reduction in dropout is most consistent with the belief-based mechanism documented in Section 7.1. Yet a main concern remains about social desirability bias. Students in treated schools are frequently surveyed while at school, an environment that makes the socially approved response salient. In this context, a student can easily register pro-State and anti-Mafia answers at little cost, whether or not it reflects a genuine change in personal beliefs. This would bias the structured results in precisely the direction I document, as school and year fixed effects cannot account for this. To relax this concern, I analyse open-ended survey responses where students were asked to give their own definition of the Mafia. This approach provides deeper insights into students' emotional and cognitive associations with the Mafia, moving beyond structured survey questions to capture more nuanced attitudinal changes. The survey collected written definitions of the Mafia from students across 46 schools, yielding 3,242 observations across 11 years. I employ natural language processing (NLP) techniques to perform sentiment analysis of students' definitions. I show the results employing two established sentiment analysis algorithms, namely Syuzhet and AFINN, and I extract sentiment scores that capture the overall emotional valence of each response. While Syuzhet scores are well-suited for analysing the narrative structure of students' written definitions, the AFINN lexicon provides fine-grained intensity measures with distinct methodological advantages. Syuzhet assigns sentiment values between -1 and +1 with 16 gradients, and it has been developed specifically for narrative text analysis (Silge and Robinson, 2017). In contrast, AFINN scores words on a scale from -5 to +5; originally developed for Twitter sentiment analysis, it enables stronger differentiation between extreme positive and negative expressions (Isasi, 2021; Kim, 2022). The aggregated sentiment score for each Mafia definition is calculated by summing the individual word scores, with AFINN's broader range providing greater sensitivity to intensity variations in student responses. Following the same approach used for analysing the survey answers, I employ a TWFE estimator to investigate how the implementation of reuse activities affects the sentiments extracted from Mafia definitions. If the reuse of Mafia properties alters students' perceptions of the Mafia, we would expect to observe systematic differences in how students in treated versus control areas conceptualise the Mafia. Specifically, successful interventions might lead to more negative sentiment scores. Table 14 reports the results where all standard errors are clustered at the school level and both school and time FEs are included. Students seem to have more negative sentiments towards the Mafia after the implementation of reuse activities in their area. Column (1) reveals a statistically significant decrease in the syuzhet score of 37%

relative to the mean, representing a meaningful shift towards more negative narrative construction about the Mafia. The AFINN algorithm, which provides more granular intensity measures, shows an even stronger effect, suggesting students in areas with repurposed Mafia properties express more intense negative emotions when describing the Mafia. The event studies reported in Figures A.7 and A.8 confirm these results are driven by the treatment itself rather than pre-existing trends: coefficients in the pre-treatment period are statistically indistinguishable from zero and show no systematic pattern, while post-treatment effects emerge sharply following repurposing and persist over time. This dynamic pattern reinforces that the shift in narrative sentiment reflects a causal response to property repurposing rather than differential pre-trends between treated and control areas.

Table 14 Impact of reusing Mafia properties on students' sentiment scores

	Sentiment analysis outcomes	
	(1)	(2)
	Syuzhet Score	AFINN Score
Reuse = 1	-0.266**	-0.417**
	(0.112)	(0.195)
Clustered SE	yes	yes
Time FE	yes	yes
Schools FE	yes	yes
Observations	3,242	3,242
Number of schools	46	46
Mean dep. var.	-0.710	-2.349

Notes: TWFE models. Standard errors are clustered at the school level.

A further piece of evidence points the same way. If the sentiment shift reflected students producing the answers they believed were expected, it should be concentrated among the more linguistically able students, who are best placed to recognise and reproduce the socially approved framing. In Figure 17 I show that the opposite holds: the shift is strongest among students with the least linguistically complex responses, the group that we expect as least equipped to manage their answers strategically. This is difficult to reconcile with social desirability and more consistent with a genuine change in perception. Since further information

on household income or educational level are not accessible at the student level, I construct a proxy of socio-economic background based on the level of each student's linguistic complexity. I build on rich evidence from developmental linguistics showing that children from lower socio-economic backgrounds tend to produce shorter and less lexically diverse sentences, reflecting differences in the quantity and quality of language input received during early development (Huttenlocher, Waterfall, Vasilyeva, Vevea and Hedges, 2010; Spencer, Clegg and Stackhouse, 2012). For simplicity, I use the word count of each student's open-ended definition of the Mafia as a measure of syntactic complexity. Specifically, I compute the number of words in each response, standardise the measure into z-scores using the sample mean and standard deviation, and assign each student to a quartile rank from 0 (bottom 25%) to 3 (top 25%). Students with empty responses are assigned to the lowest rank. I then split the sample at the median, classifying students in ranks 0–1 as proxied lower socio-economic status and those in ranks 2–3 as proxied higher socio-economic status. I report some examples of what type of definitions fall in which rank in Table A.11. Figure 17 presents the treatment effects on sentiment scores separately for the two groups, estimated with school and time fixed effects. A clear pattern emerges: students with lower syntactic complexity express significantly more negative sentiment toward the Mafia following the repurpose intervention, with both the Syuzhet and AFINN scores shifting downward.[30] By contrast, students with higher syntactic complexity show no significant change in either sentiment measure. Since the specification includes school fixed effects, these results capture within-school variation, meaning that the differential response is not driven by differences across schools but rather by heterogeneity among students attending the same institutions. Overall, the results show that the intervention is most effective in reshaping perceptions among students from more disadvantaged backgrounds, precisely the group most exposed to Mafia influence and for whom the provision of alternative role models and community engagement is likely to carry the greatest marginal value.

[30] Although the Syuzhet coefficient does not reach conventional significance levels (β = -0.21 and p-value = 0.150), it displays a more precise estimate than the corresponding coefficient for higher socio-economic background students (β = -0.19 and p-value = 0.497).

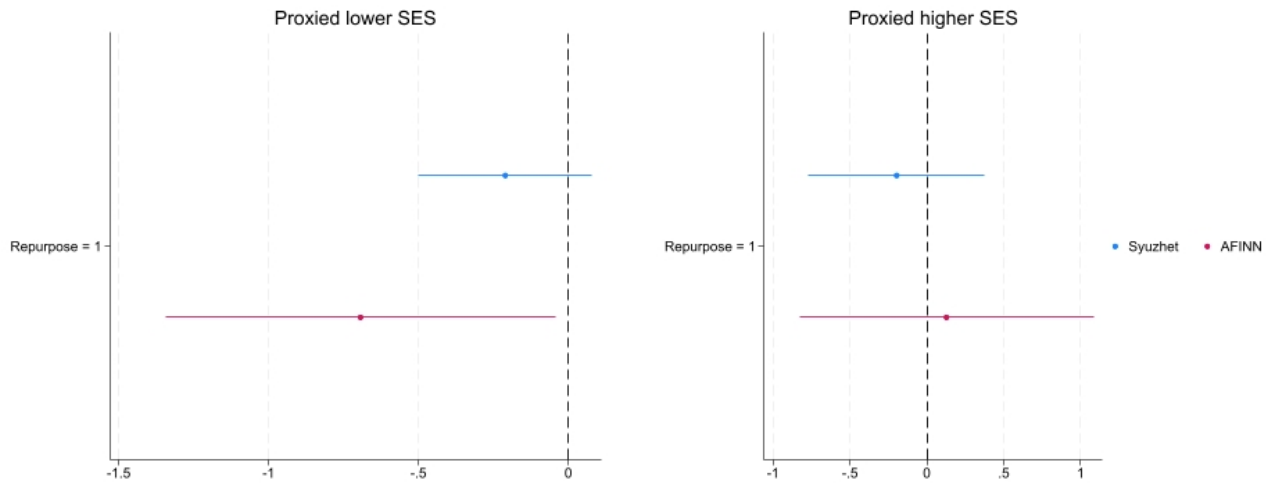


Figure 17 Estimated effects of reusing practices on students' survey emotions in defining the mafia by proxied SES.

Household-level effect. A final concern is that the cultural activities on the repurposed properties reshape the attitudes of the wider community rather than of students directly, and that the shift I document among students is a mediated reflection of their families' changing views. If so, the mechanism would operate at the household level, and attributing it to a change in students' own perceptions would be misleading.

I test this using a survey question asking students to report their family's opinion about the Mafia. If repurposing shifts attitudes at the household level, treated areas should show a change in the distribution of reported family opinions. Figure 18 reports the effect of reuse on the share of students attributing each stance to their family. The estimates are close to zero and statistically insignificant across every category: there is no detectable shift in how students describe their families' views of the Mafia following the intervention. The absence of any effect on reported family opinions indicates that the attitudinal change operates at the level of the students themselves rather than being transmitted through the household, consistent with a mechanism in which the repurposed property acts directly on juveniles through the alternative role models and community engagement it provides.

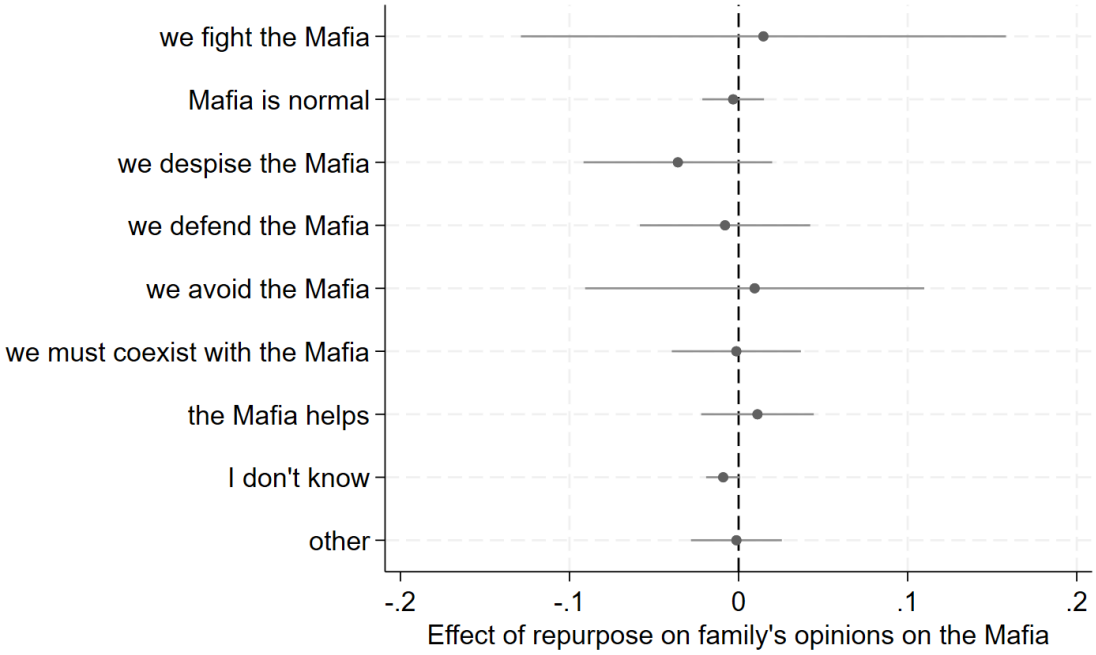


Figure 18 Estimated effects of repurposing practices on students' families' perception of the Mafia reported by students. Each point reports the effect of repurpose on the share of students agreeing with specific statements. Standard errors are clustered at the school level. Results are significant at the 90% confidence level.

10 Conclusions

11 Conclusions

This study provides the first causal evidence that converting confiscated Mafia properties into community spaces yields tangible educational benefits for youth in historically Mafia-affected areas. I construct a comprehensive dataset tracking the implementation of the CRR policy across major urban areas, combining scraped records on confiscated assets, internal surveys from *Libera*, municipal records, digitised Antimafia Investigative Directorate maps of baseline Mafia presence, Ministry of Education administrative records, and survey data on students' perceptions of the Mafia. I follow student cohorts from grade 11 to graduation and build school catchment areas by matching census blocks to their closest high school for each schooling track. Catchment areas are defined as treated whenever at least one Mafia property has been repurposed within their boundaries, and I compare treated areas to those unaffected or not yet affected, before and after the first repurposing event.

Using a difference-in-differences design, I find that transforming former criminal strongholds

into hubs for social activities reduces high school dropout rates by 1.6 to 1.7 percentage points, approximately 29% relative to the mean, among students beyond the age of compulsory schooling. The effect becomes significant from the second year after the first repurposing and persists for up to six years. Its magnitude scales with intensity: the effect is concentrated in catchment areas where more than one property has been repurposed, and each additional repurposed property, weighted by the catchment area's student-age population, reduces dropout by a further 0.59 percentage points, or 10% relative to the mean, with an effect that builds up over time. Weighting properties by their proximity to where students live yields effects of the same sign and a similar pattern, although less precisely estimated.

The heterogeneity analysis reveals a consistent pattern. The effects are concentrated among students in technical high schools, who need the full five-year diploma to improve their employment prospects and therefore face a longer period at risk of dropping out, and are entirely driven by the repurposing of *operational properties*, the villas and apartments that symbolised the local control of Mafia families, rather than by commercial or industrial assets. Across activity types, the effect is concentrated in properties hosting cultural and welfare activities, while properties offering education and employment services show smaller and insignificant reductions. That the effect does not increase with the educational services a property offers, but does vary sharply with what the property once represented, points toward a channel operating through the visible reclaiming of the space rather than through the services delivered within it.

Survey evidence on students' perceptions supports this interpretation. Following repurposing, students are more likely to regard asking the Mafia or a politician as a very useless way to find a job, and more likely to regard education as a very useful one. This shift emerges around the second and third years after treatment, on the same horizon at which dropout declines. I also find evidence of a change in perceived State capacity: students become less likely to agree that the Mafia is strong because it infiltrates the State, and less likely to name the police and local politicians among the institutions they trust least. The dynamic estimates for these outcomes are imprecise, but suggest that they emerge in the first years after repurposing and fade thereafter, consistent with an initial signal of State presence operating upstream of the slower revision of beliefs about the returns to each pathway.

I rule out several alternative channels. Reallocation, the administrative stage that precedes repurposing, has no effect on dropout, and confiscation, when the controlling family is arrested and removed, takes place on average more than a decade before repurposing, well outside the estimation window. The decline in dropout therefore appears only once the property is repurposed, which is difficult to reconcile with an incapacitation mechanism. The effect is not accompanied by any change in graduating grades, nor absorbed by

concurrent municipal educational and social-inclusion programmes, as an educational-inputs channel would predict. Triple-difference estimates interacting the treatment with neighbourhood rental prices are close to zero, ruling out concurrent gentrification. Controlling for the presence of other NGOs in the catchment area leaves the estimate unchanged while the NGO variable itself is insignificant, indicating that the activities hosted in repurposed Mafia properties are not readily substitutable by conventional civil-society provision. The results are also robust to reweighting control schools on their baseline NGO presence and grade retention, and to controlling for trends in the municipal characteristics that predict repurposing. Sentiment analysis of students' open-ended definitions of the Mafia shows a shift toward more negative language, in both frequency and intensity, and this shift is strongest among students producing the least linguistically complex responses, the group least equipped to reproduce a socially approved framing, which is difficult to reconcile with social desirability. Reported family attitudes show no change, locating the response with students themselves rather than with their households.

These findings carry implications for anti-Mafia policy design, in Italy and in comparable settings internationally. Most fundamentally, they show that effective prevention requires moving beyond asset confiscation toward the active construction of legitimate alternatives within affected communities. The binding constraint is not seizure but reuse: a substantial share of confiscated assets remain vacant or are never reassigned, and the evidence here indicates that the earlier stages of the policy leave little visible trace in the neighbourhood. The timing of the effects reinforces this. Because the behavioural response depends on sustained and visible use rather than on the announcement of reassignment, properties transferred and subsequently left inactive are unlikely to generate lasting benefits. Accelerating the transition from reallocation to active reuse, and providing sustained financial support to the local organisations that manage these spaces, would amplify the policy's reach. That the effects are largest among students in technical schools and, in the sentiment analysis, among students with proxied lower socioeconomic backgrounds suggests the policy may operate as an equalising force, directing its benefits toward those with the fewest alternatives.

A Additional Figures and Tables

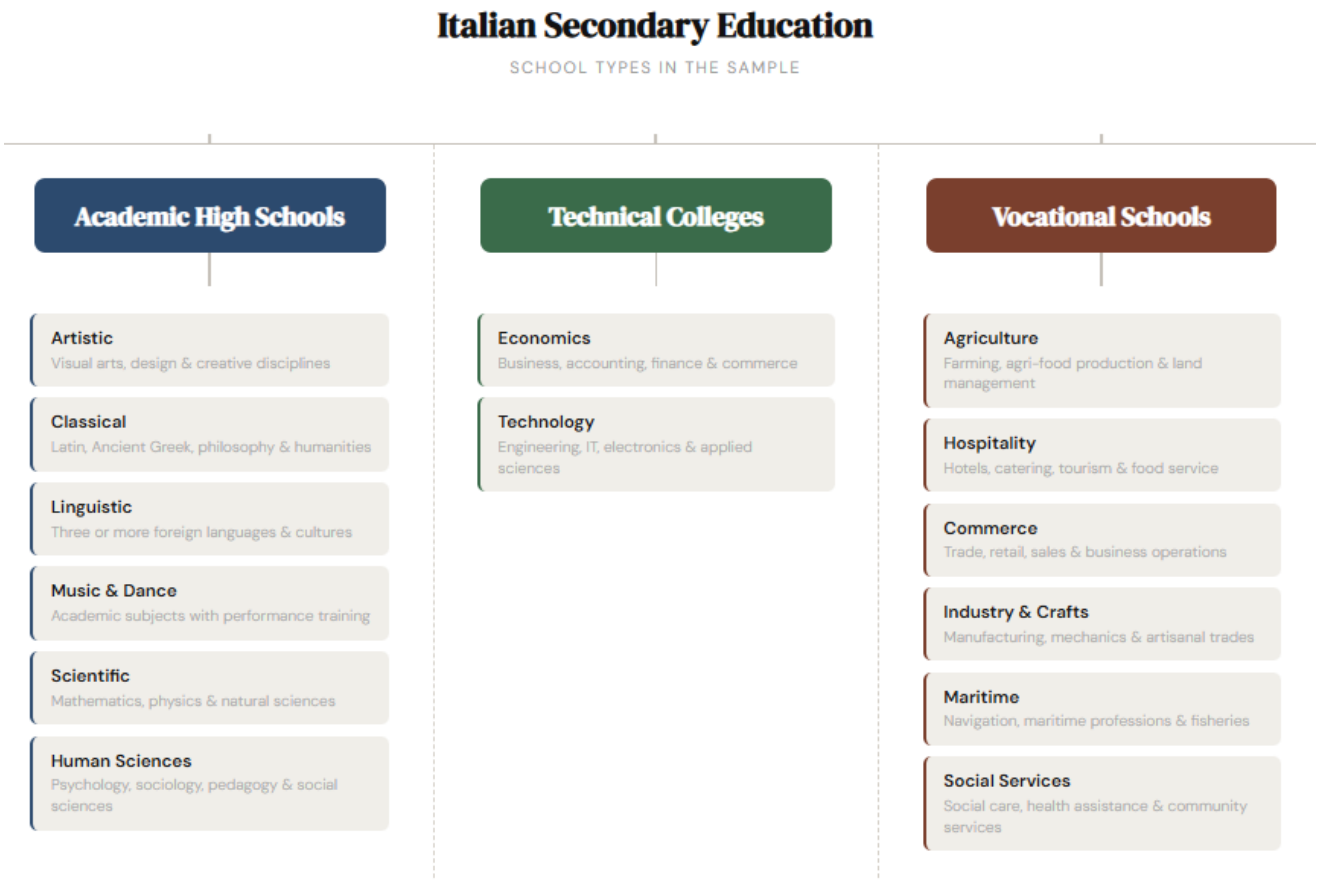


Figure A.1 The universe of possible high school sub-tracks in Italy.

Table A.1 Survey questions from the Survey on the Perception of the Mafia

ID	Topic	Question	Response	Unit
V12	Definition	How would you define the Mafia?	Open-ended	Student
V21	Family view	How is the Mafia regarded within your family?	9 options	School
V28	Job-seeking	In your town, what is most useful to do when looking for work?	7 items, ranked 1–7	School
V33	State vs. Mafia	Do you agree with each of the following statements?	9 items, yes/no/don't know	School
V45	Trust	How much trust do you place in each of the following?	10 items, scale 1–4	School

Notes: Survey responses provided by Centro Studi Pio La Torre (2025). Full response items are listed in Appendix D. Outcomes are school-level shares of students selecting each category; for V28 I use the top two (1–2) and bottom two (6–7) ranks. V12 is analysed at student level.

Table A.2 Impact of repurposing Mafia real estate on the share of students dropping out from grades 11 to 13.

	Dropout rates G11-G13			
	(1)	(2)	(3)	(4)
Repurpose = 1	-0.0168**	-0.0165**	-0.0163*	-0.0159*
	(0.0083)	(0.0081)	(0.0083)	(0.0081)
clustered SE	yes	yes	yes	yes
school FE	yes	yes	yes	yes
time FE	yes	yes	yes	yes
migration	no	yes	no	yes
retention	no	no	yes	yes
Observations	1,393	1,393	1,393	1,393
Number of schools	251	251	251	251
Mean dep. var.	0.058	0.058	0.058	0.058

Notes: TWFE model. The treatment variable is equal to 1 whenever there is at least one reused Mafia property within the school catchment area. Column (1) represents the baseline accounting for school and time fixed effects, while in Columns (2) and (3) I control for students' migration and grade retention rates in grades 11 and 12, respectively. Column (4) reports the complete specification.

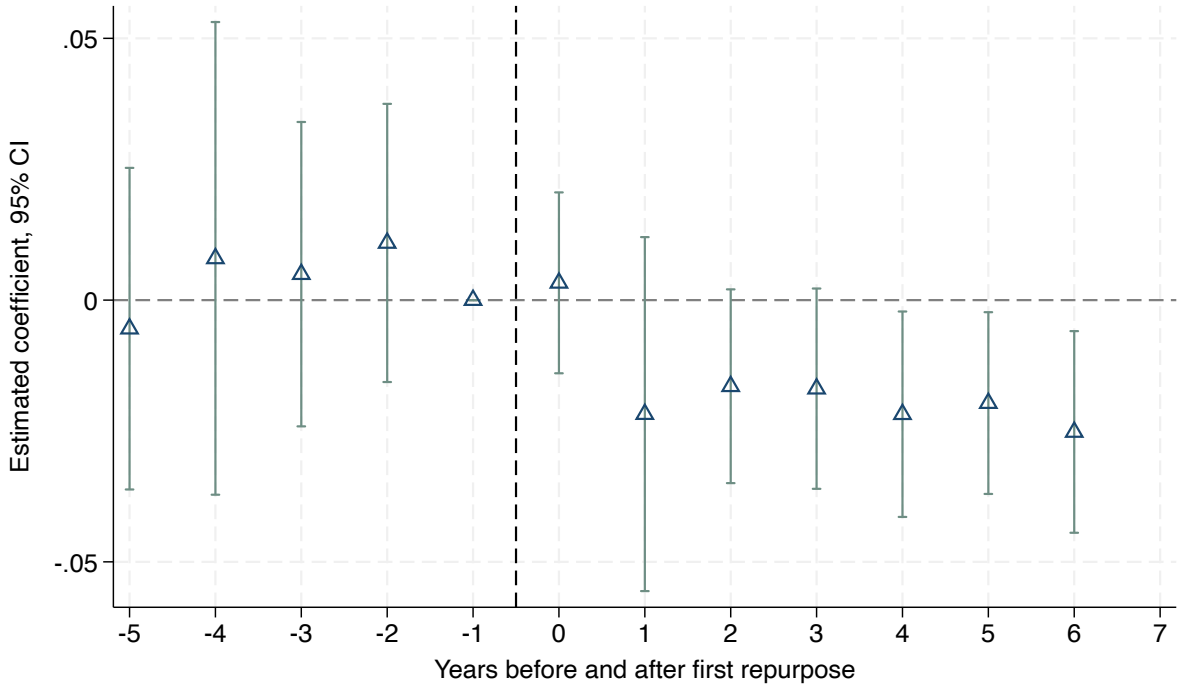


Figure A.2 Estimated effects of reusing practices on dropout rate before and after the first reuse starts - excluding dropout from grade 13

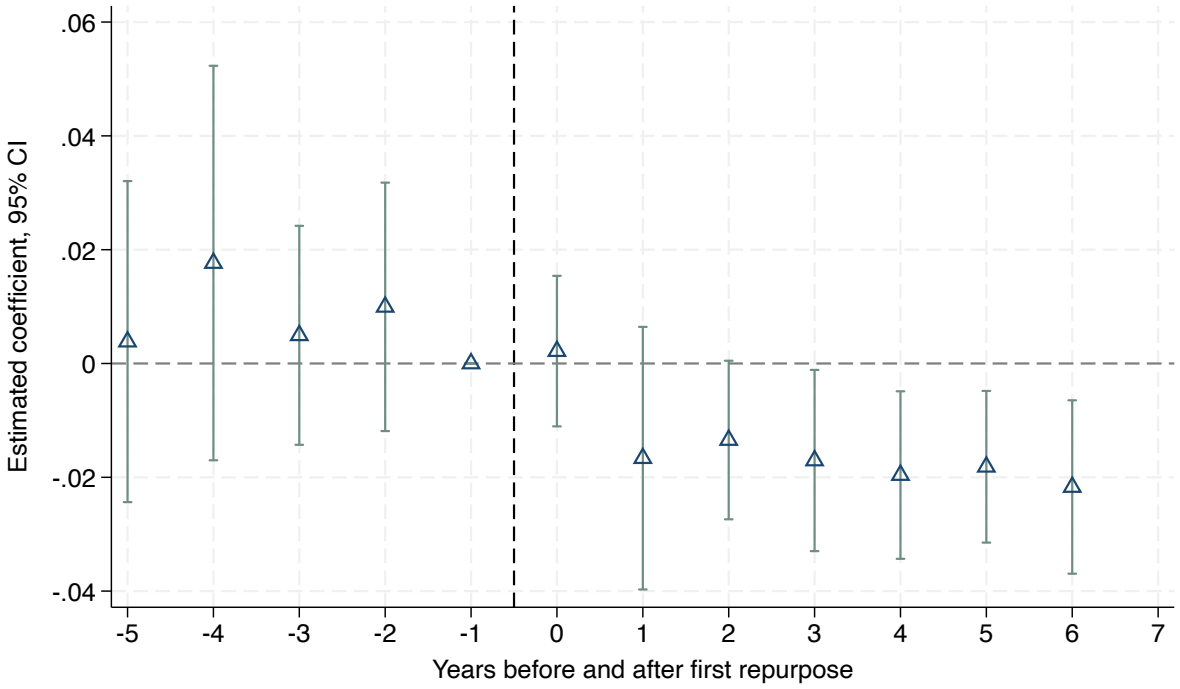


Figure A.3 Estimated effects of reusing practices on dropout rate before and after the first reuse starts - event study of areas already hosting reallocated properties

Table A.3 Testing the effect of the treatment on the main controls.

	(1)	(2)	(3)	(4)
	Student migration		**Grade retention**	
Repurpose = 1	-0.0029	-0.0050	-0.0080	-0.0080
	(0.0121)	(0.0128)	(0.0099)	(0.0100)
clustered SE	yes	yes	yes	yes
School FE	yes	yes	yes	yes
Time FE	yes	yes	yes	yes
Student migration	no	no	no	yes
Grade retention	no	yes	no	no
Observations	1,393	1,393	1,393	1,393
Number of schools	251	251	251	251
Mean dep. var.	0.0812	0.0812	0.0610	0.0610

Notes: TWFE model. The treatment variable is equal to 1 whenever there is at least one repurposed Mafia property within the school catchment area. Student migration is the share of students in grades 9–10 at $t-1$ not found in grades 10–11 at t; grade retention is the mean of the retention rates in grades 11 and 12. All columns are estimated on the same sample. Standard errors, clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A.4 Effect of the treatment on selection in enrollment preferences.

	(1)	(2)	(3)	(4)
	Enrollment share			
Repurpose = 1 × Technical	-0.0234**	-0.0247**	-0.0215*	-0.0222*
	(0.0116)	(0.0115)	(0.0120)	(0.0121)
Repurpose = 1 × Academic	0.0151*	0.0119	0.0146*	0.0111
	(0.0089)	(0.0087)	(0.0088)	(0.0085)
Repurpose = 1 × Vocational	-0.0665***	-0.0570**	-0.0636***	-0.0532**
	(0.0240)	(0.0236)	(0.0236)	(0.0233)
clustered SE	yes	yes	yes	yes
school FE	yes	yes	yes	yes
time FE	yes	yes	yes	yes
migration	no	yes	no	yes
retention	no	no	yes	yes
Observations	1,383	1,383	1,383	1,383
Number of schools	251	251	251	251
Mean dep. var.	0.2315	0.2315	0.2315	0.2315

Notes: TWFE model. The dependent variable is first-year enrollment as a share of the school's students. Each row reports the effect of repurposing for schools of the given track (academic: licei; technical: istituti tecnici; vocational: istituti professionali). All columns are estimated on the same sample. Standard errors, clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

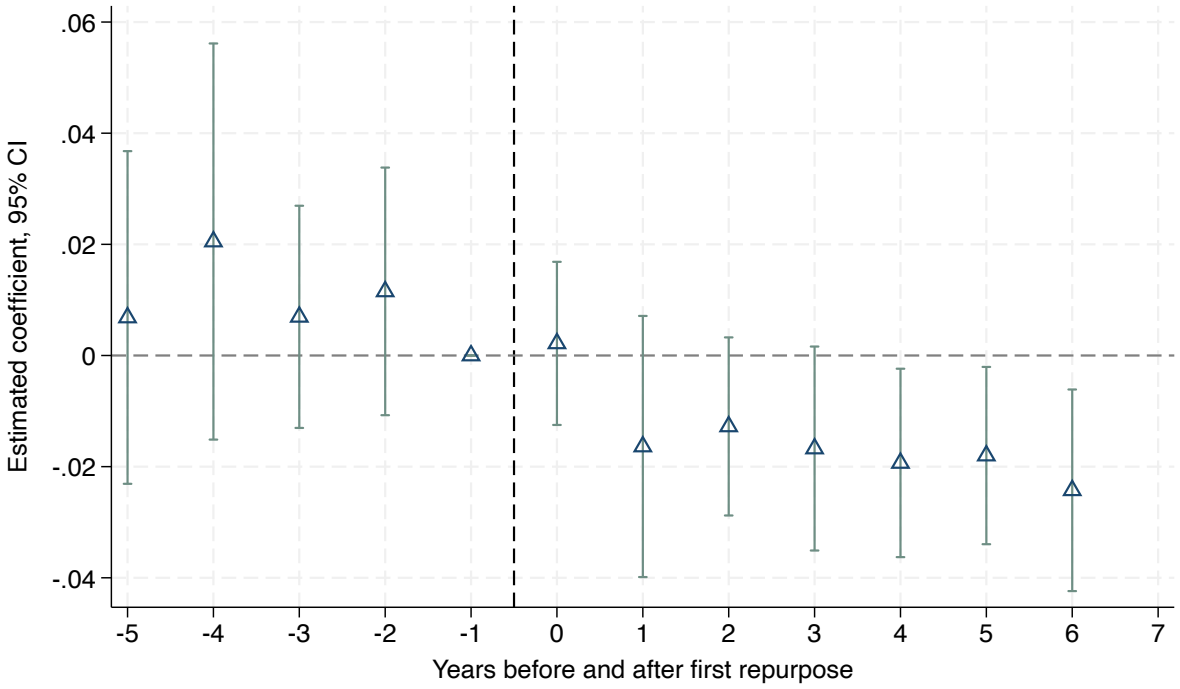


Figure A.4 Estimated effects of reusing practices on dropout rate before and after the first reuse starts - event study controlling for the presence of educational- and urban-related programmes at the municipality level

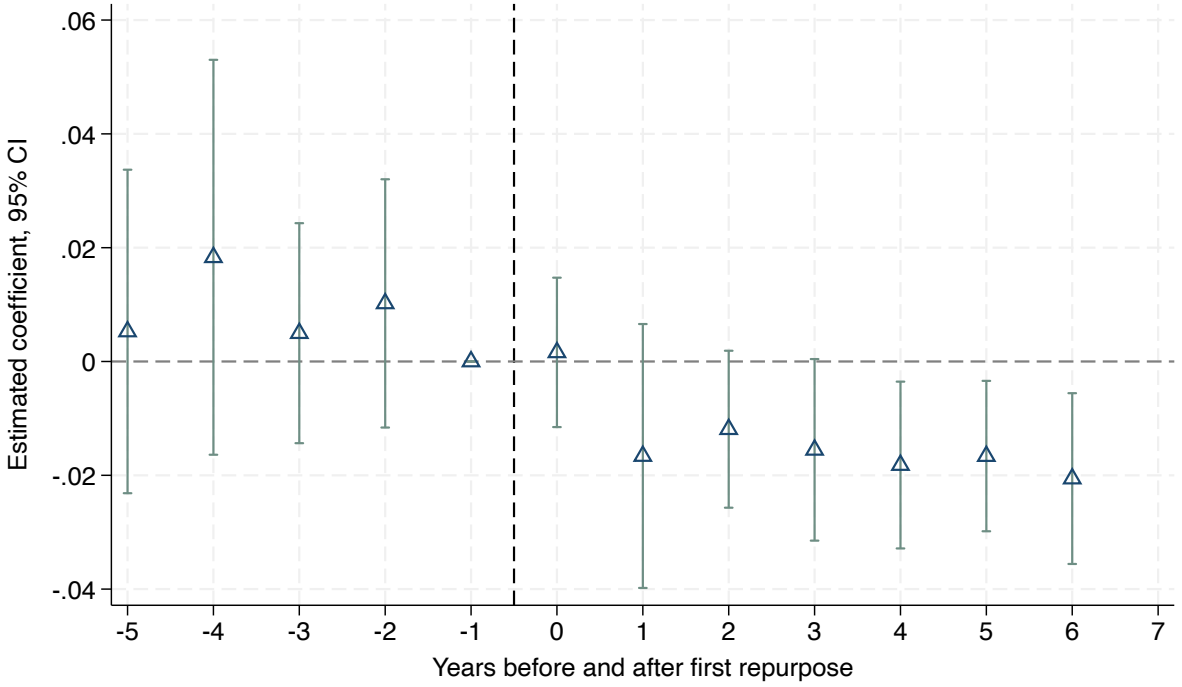


Figure A.5 Estimated effects of reusing practices on dropout rate before and after the first reuse starts - event study controlling for the presence of NGOs

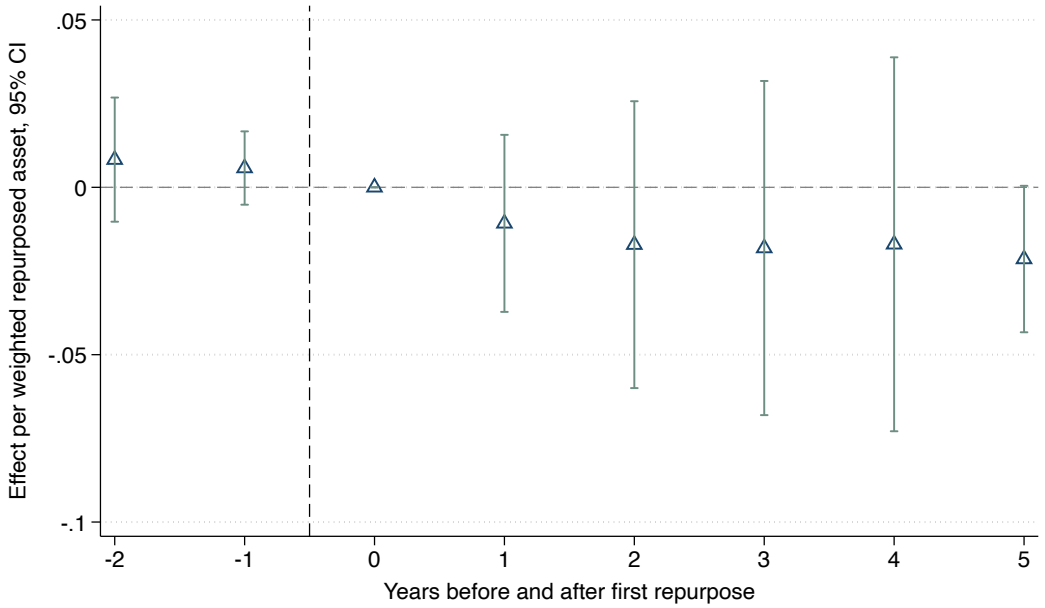


Figure A.6 Event study using the count of reused assets weighted by the average distance of properties from the population-weighted centroids of schools' catchment areas

Table A.5 Impact of repurposing Mafia real estate on dropout rates for G11-G13, by type of activity hosted.

	(1)	(2)	(3)	(4)
Types of repurpose		**Dropout rate G11-G13**		
Education = 1	-0.0112			
	(0.0100)			
Education = 0	-0.0199**			
	(0.0089)			
Employment = 1		-0.0082		
		(0.0125)		
Employment = 0		-0.0161**		
		(0.0081)		
Culture = 1			-0.0144**	
			(0.0063)	
Culture = 0			-0.0177	
			(0.0127)	
Welfare = 1				-0.0160*
				(0.0082)
Welfare = 0				-0.0145
				(0.0117)
clustered SE	yes	yes	yes	yes
school FE	yes	yes	yes	yes
time FE	yes	yes	yes	yes
migration	yes	yes	yes	yes
retention	yes	yes	yes	yes
Observations	1,393	1,393	1,393	1,393
R-squared (within)	0.206	0.205	0.205	0.205
Number of schools	251	251	251	251
Mean dep. var.	0.0580	0.0580	0.0580	0.0580

Notes: TWFE model. In each column, Category = 1 equals 1 whenever at least one property repurposed for that type of activity is located within the school catchment area; Category = 0 equals 1 whenever the catchment area hosts repurposed properties, none of which is used for that activity. The omitted group is catchment areas with no repurposed property. All columns control for students' migration and grade retention rates in grades 11 and 12, and are estimated on the same sample. Standard errors, clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A.6 Impact of reusing Mafia real estate on students' perceptions on useful and useless channels to find a job in their town

Dep. Variable	(1) Vocational training	(8) Civil service	(3) Ask the Mafia	(4) Ask politician	(5) Job centre	(6) Ask family	(7) Ask friend
Panel A: more useful choices							
Repurpose = 1	0.161**	0.136	-0.110	-0.0754	0.0144	0.0371	0.0492
	(0.0804)	(0.0827)	(0.0793)	(0.0863)	(0.0719)	(0.0405)	(0.0308)
Observations	161	161	161	161	161	161	161
R-squared	0.400	0.548	0.425	0.484	0.472	0.443	0.562
clustered SE	yes	yes	yes	yes	yes	yes	yes
time FE	yes	yes	yes	yes	yes	yes	yes
school FE	yes	yes	yes	yes	yes	yes	yes
Mean dep. var.	0.480	0.426	0.280	0.280	0.400	0.303	0.273
Panel B: more useless choices							
Repurpose = 1	-0.0516	-0.0772	0.136*	0.117*	-0.00241	-0.0788	-0.0575
	(0.0597)	(0.0469)	(0.0780)	(0.0662)	(0.0612)	(0.0614)	(0.0385)
Observations	161	161	161	161	161	161	161
R-squared	0.433	0.346	0.592	0.527	0.558	0.375	0.393
clustered SE	yes	yes	yes	yes	yes	yes	yes
time FE	yes	yes	yes	yes	yes	yes	yes
school FE	yes	yes	yes	yes	yes	yes	yes
Mean dep. var.	0.159	0.163	0.564	0.453	0.196	0.197	0.208

Notes:

Table A.7 Impact of reusing Mafia real estate on students' perceptions of why people turn to the Mafia

	(1)	(2)	(3)	(4)
Dep. Variable	Family ties	Neighbourhood	Lack of culture of legality	Looking for job
Repurpose = 1	0.0995***	0.0378	-0.0384	-0.0816***
	(0.0330)	(0.0391)	(0.0712)	(0.0278)
Observations	161	161	161	161
R-squared	0.498	0.415	0.768	0.636
clustered SE	yes	yes	yes	yes
time FE	yes	yes	yes	yes
school FE	yes	yes	yes	yes
Mean dep. var.	0.160	0.109	0.206	0.130

Dep. Variable	Absent institutions	Looking for easy money	Looking for power	I don't know
Repurpose = 1	-0.00753	0.107***	0.00948	-0.0112
	(0.0221)	(0.0375)	(0.0250)	(0.0141)
Observations	161	138	138	161
R-squared	0.625	0.593	0.489	0.483
clustered SE	yes	yes	yes	yes
time FE	yes	yes	yes	yes
school FE	yes	yes	yes	yes
Mean dep. var.	0.0324	0.270	0.107	0.0401

Notes:

Table A.8 Impact of reusing Mafia real estate on students' perceptions of who is stronger between the Mafia and the State

Dep. Variable	(1) State strong: we are the State	(2) Mafia strong:infiltrates the State	(3) Mafia strong: still exists	(4) State strong: greater resources	(5) State does too little
Repurpose =1	0.322*	-0.109**	-0.132*	-0.111	-0.00382
	(0.178)	(0.0447)	(0.0666)	(0.0859)	(0.101)
Observations	164	164	164	164	164
R-squared	0.675	0.705	0.845	0.659	0.468
clustered SE	yes	yes	yes	yes	yes
time FE	yes	yes	yes	yes	yes
school FE	yes	yes	yes	yes	yes
Mean dep. var.	1.968	2.734	1.914	2.237	1.483

Dep. Variable	Mafia strong: any means to its ends	State strong: defends democracy	Mafia strong: instils fear	State and Mafia coincide
Repurpose =1	-0.0990	0.0792	-0.0951	-0.0872
	(0.0711)	(0.163)	(0.127)	(0.0818)
Observations	164	164	164	164
R-squared	0.509	0.708	0.656	0.802
clustered SE	yes	yes	yes	yes
time FE	yes	yes	yes	yes
school FE	yes	yes	yes	yes
Mean dep. var.	2.741	1.752	2.387	2.117

Notes:

Table A.9 Impact of reusing Mafia real estate on students' perceptions of who they can trust more

Dep. Variable	(1) Magistrates	(2) Police	(3) Teachers	(4) Journalists	(5) Bankers	(6) Priests	(7) Local politicians	(8) Trade unionists
Panel A: most trusted								
Repurpose = 1	0.123	0.0107	-0.0160	-0.0548	-0.0267	0.0221	0.00150	0.0285
	(0.113)	(0.0212)	(0.127)	(0.0496)	(0.0512)	(0.0376)	(0.0166)	(0.0610)
Observations	164	164	164	164	164	164	164	164
R-squared	0.780	0.689	0.693	0.597	0.663	0.681	0.559	0.528
clustered SE	yes	yes	yes	yes	yes	yes	yes	yes
time FE	yes	yes	yes	yes	yes	yes	yes	yes
school FE	yes	yes	yes	yes	yes	yes	yes	yes
Mean dep. var.	0.273	0.0402	0.377	0.114	0.0724	0.129	0.0277	0.253
Panel B: least trusted								
Repurpose = 1	0.00747	-0.123*	0.00194	-0.0153	0.00380	0.0237	-0.167***	-0.0207
	(0.0355)	(0.0727)	(0.0326)	(0.0361)	(0.0542)	(0.0584)	(0.0476)	(0.0537)
Observations	164	164	164	164	164	164	164	164
R-squared	0.819	0.892	0.715	0.816	0.761	0.422	0.826	0.506
clustered SE	yes	yes	yes	yes	yes	yes	yes	yes
time FE	yes	yes	yes	yes	yes	yes	yes	yes
school FE	yes	yes	yes	yes	yes	yes	yes	yes
Mean dep. var.	0.0946	0.405	0.0345	0.0928	0.122	0.213	0.390	0.0698

Notes:

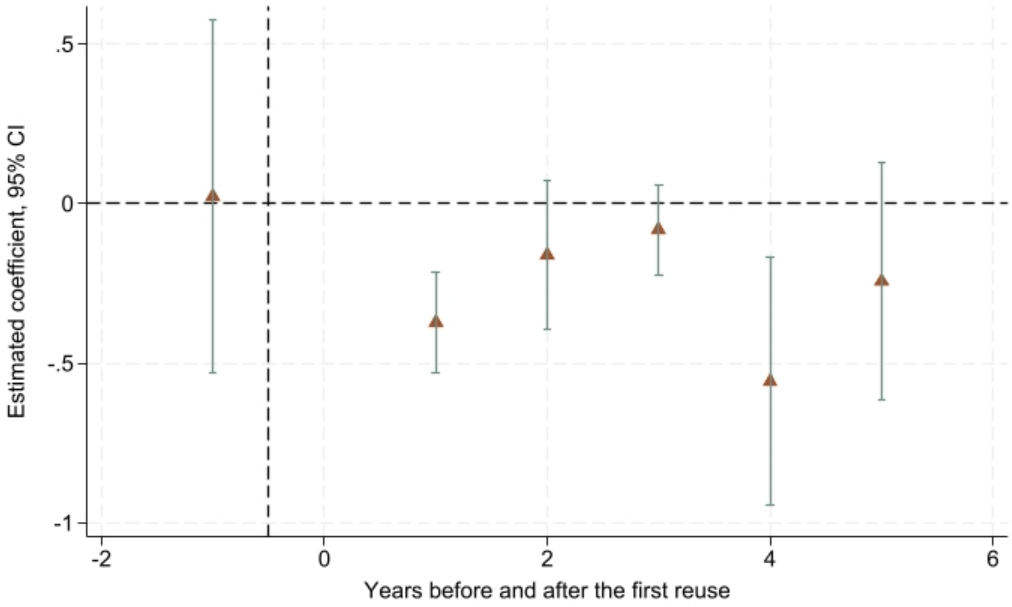


Figure A.7 Event study of the Szyuzhet scores

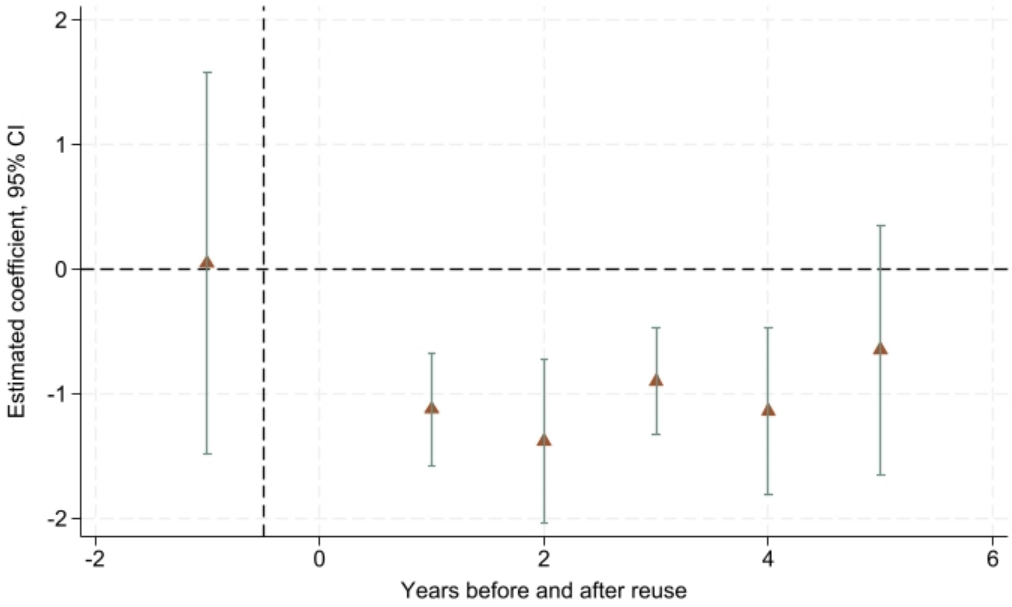


Figure A.8 Event study of the AFINN sentiment scores

Table A.10 Top 25 terms for positive and negative sentiments

Panel A: top 25% negative words	**Anger**	**Disgust**	**Fear**	**Sadness**
	Electoral	Ruthless	Forced	Violence
	Laundering	Struggle	Eradicate	Kill
	Murder	Threathening	Hide	Cruel
	Resorting	Inconvenient	Manifests	Unfair
	Conflict	Involve	Struggle	Evil
Observations	638	393	327	274
Panel B: top 25% positive words	**Joy**	**Trust**	**Surprise**	**Anticipation**
	Money	Money	Money	Public
	Respect	Association	Violent	Respect
	Achieve	Structure	Protection	Protection
	Gain	Respect	Deal	Gain
	Freedom	Achieve	Hope	Poweful
Observations	227	569	740	234

Those words you listed are the most frequent words appearing in sentences that scored high (top 25%) for each specific emotion dummy.

Table A.11 Examples of students' definitions of the Mafia

Rank	Definition	Italian word count	CSI
0	I don't know	2	-0.35
1	They are organizations that ruin our society	7	-0.22
2	Organizations that follow, therefore adopt, a behavior that may seem beneficial in some aspects such as the economic one and employment	21	0.12
3	Mafia-type organizations are a group of people who act illegally, sometimes hiding behind legal activities. Very often mafia organizations push people towards the wrong path, very often due to people's desperation	40	0.59

B Attributes of repurposed properties and local context

The repurposed properties in the sample display a fairly homogeneous profile in terms of real estate type. The large majority are residential estates (68%), followed by land (20%), while commercial and industrial properties account for only 10% and a residual share falls into other categories. Consistent with this, the most common prior use before confiscation was villas and houses (45%), followed by other uses (26%) and land (19%), with shops, garages, and commercial deposits representing marginal shares. In terms of reallocation, virtually all properties (99%) were transferred to local authorities, with only one asset sold to a private buyer. Regarding the activities carried out in the repurposed properties, the distribution is relatively balanced between welfare and cultural activities — roughly half of the properties host welfare-oriented activities (50.6%) and a slightly larger share host cultural ones (56.4%) — while employment and other activities remain much less common, at 11.6% and 12.1% respectively. Education activities are present in about one fifth of cases (19.4%).

Table B.1 Repurposed properties attributes

Type of real estate	N	%	**Reallocation type**	N	%
Commercial and industrial estate	35	10	Local authorities	345	99
Residential estate	236	68	Sold to private	1	1
Land	67	20	Total	346	100
Other	8	2			
Total	346	100			

Usage before confiscation	N	%	**All types of activities**	N	%
Other	92	26	Education	67	19.4
Box, garage	18	5	Welfare	175	50.6
Fabbricate	3	1	Culture	195	56.4
Commercial deposit	3	1	Employment	40	11.6
Shop	11	3	Other	42	12.1
Land	65	19			
Villas and Houses	155	45			
Total	346	100			

Turning to the local context, the data paint a picture of small, tight-knit, predominantly native residential communities. With an average of just 361 residents and 136 households per

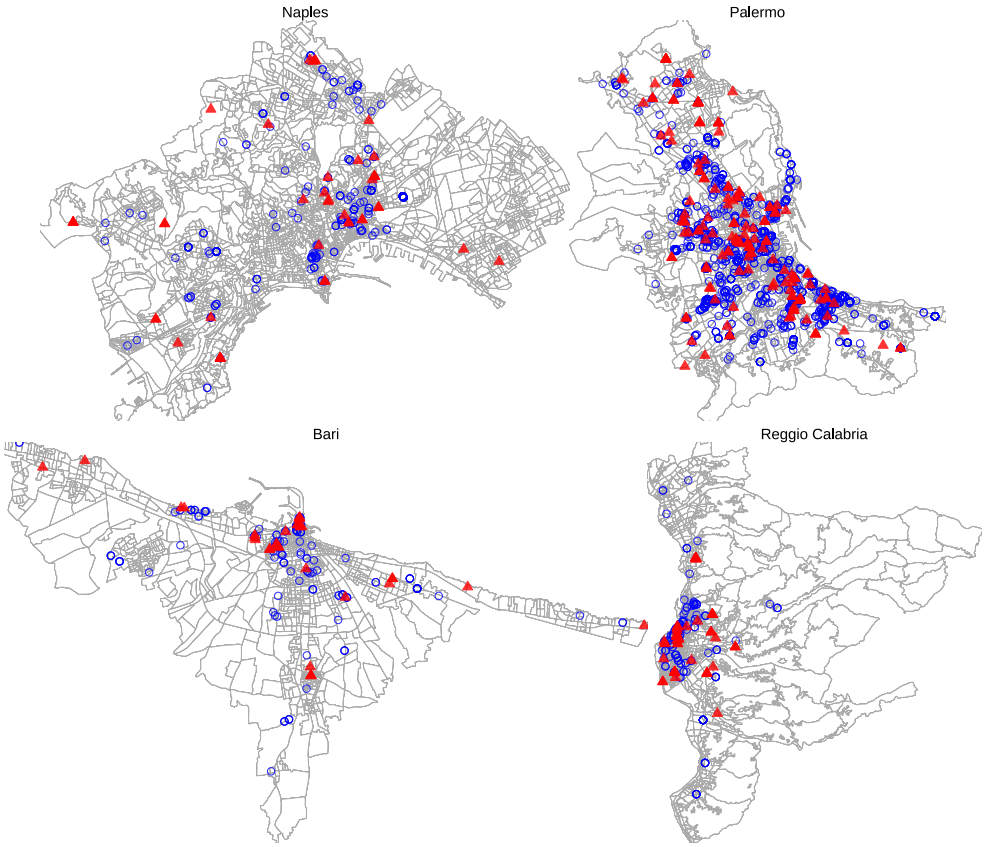
census section, these are not urban neighbourhoods in any dense sense — they are closer to village-scale settlements or peripheral urban pockets, the kind of places where social dynamics are likely shaped by proximity and long-standing local ties rather than anonymity. The very low share of non-native residents (under 4% on average) reinforces this characterisation of socially homogeneous, relatively closed communities. Household composition skews toward smaller units, with single- and two-person households together accounting for over half the average section, which may reflect an ageing population rather than a young or dynamic one. Educational attainment is modest: high school and secondary diplomas dominate, university graduates make up only 13% of residents, and a non-trivial share holds only a primary diploma or is merely literate. This suggests these are working- and lower-middle-class communities, not areas of high human capital concentration. Labour market conditions are consistent with this reading — while the employment rate averages around 77%, unemployment sits at roughly 11%, pointing to communities that are economically functional but under strain, without the slack of prosperous areas. The local real estate stock is overwhelmingly residential (81%), mostly in good or decent physical condition, with a limited commercial footprint (15%), further reinforcing the image of settled, predominantly domestic environments rather than economically active or commercially vibrant ones. Civic infrastructure is present but uneven: the average of 43 registered NGOs per area suggests some organised social fabric, but participation levels vary enormously across locations, indicating that civil society engagement is far from uniform and likely concentrated in a subset of better-resourced communities.

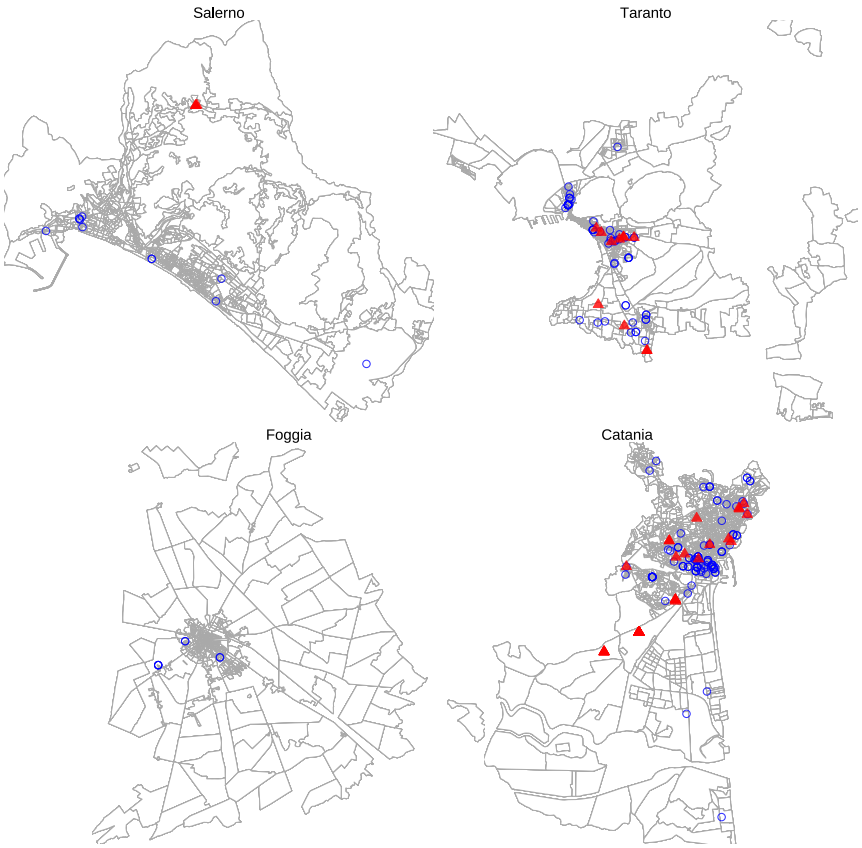
Table B.2 Census data

	(1) N	(2) mean	(3) sd	(4) min	(5) max
Population					
Residents	276	360.7	169.8	4	904.7
Residents 15-19 yo	276	0.0568	0.00919	0	0.104
Non-native residents	276	0.0376	0.0434	0	0.438
Number of households	276	136.3	62.78	1	366
Households: 1 component	276	0.264	0.0613	0	0.522
Households: 2 component	276	0.242	0.0297	0	0.341
Households: 3 component	276	0.212	0.0299	0	0.303
Households: 4 component	276	0.203	0.0590	0.0825	1
Households: 5 component	276	0.0589	0.0204	0	0.150
Households: 6 plus component	276	0.0206	0.0160	0	0.146
Education					
University degree	276	0.134	0.0602	0	0.319
High school diploma	276	0.298	0.0668	0.133	0.750
Secondary diploma	276	0.293	0.0586	0	0.461
Primary diploma	276	0.184	0.0383	0	0.289
Literarates	276	0.0789	0.0272	0.0263	0.322
Illiterates	276	0.0131	0.00756	0	0.0551
Labour market					
Workforce	276	142.2047	67.46701	2	368.1952
Employed	276	0.7686306	0.0796366	0.4690265	0.9230769
Unemployed	276	0.1073502	0.0214117	0	0.175686
Area characteristics					
Number of reale estate	276	33.63	35.62	1	330
Share of residential real estate	276	0.812	0.101	0.182	1
Share of commercial real estate	276	0.154	0.1000	0	0.776
Real estate: perfect conditions	276	0.153	0.146	0	1
Real estate: good conditions	276	0.416	0.150	0	1
Real estate: decent conditions	276	0.206	0.108	0	0.629
Real estate: bad conditions	276	0.0374	0.0502	0	0.484
Number of NGOs	209	42.80	77.69	0	342
Number of NGOs participants	276	3.848	39.65	0	651

Figure B.1 provides a comprehensive spatial overview of confiscated Mafia properties across all municipalities part of the sample, distinguishing between reused (red triangles) and not-yet-reused properties (empty blue circles). While confiscated properties are pre-

dominantly located in highly populated areas and regions with greater economic activity, particularly near coastal zones, the maps reveal that reuse patterns do not follow the same geographic clustering. The reused Mafia properties show considerable spatial heterogeneity within these areas, being distributed across diverse urban contexts rather than being concentrated in specific neighbourhoods or zones. This spatial variation is particularly important for the main identification assumptions, as it demonstrates that school catchment areas containing reused Mafia properties are distributed across different geographic and socioeconomic contexts within the broader areas of Mafia presence. Furthermore, the presence of both reused and unreused properties across various municipalities provides a rich spatial framework for comparison, which I exploit for robustness checks in Section **??**.




Salerno
Taranto
Foggia
Catania

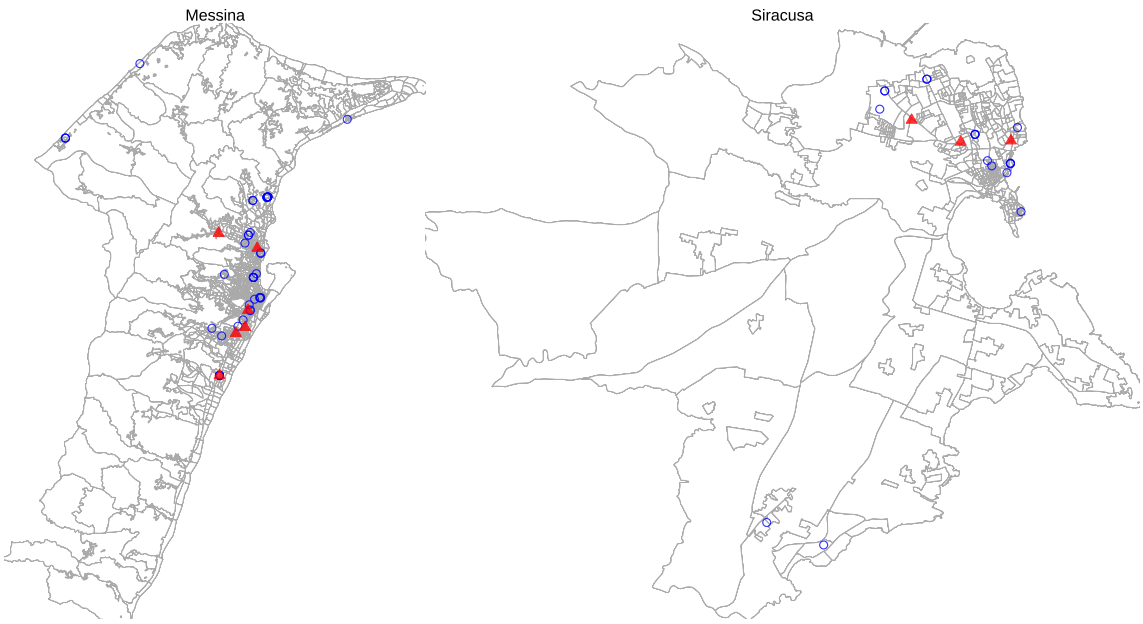


Figure B.1 The distribution of Mafia real estate in the 10 metropolitan areas of the historical Mafia-ridden regions. Empty blue circles locate reallocated but not reused real estate, while red triangles indicate the real estate under reusing practices.



Figure B.2 Word cloud of the descriptions of activities in reused Mafia properties illustrating the 100 most common terms used.

C Endogeneity Concerns

In this Appendix I investigate the determinants of the repurposing of Mafia properties and whether they threaten the identification strategy.

Endogenous treatment assignment would violate the parallel trends assumption underlying the DiD design if the decision to repurpose confiscated Mafia properties were driven by factors that also affect the evolution of educational outcomes. Selection into treatment does not by itself bias the estimates as long as parallel trends hold, but it is important to identify the factors driving repurposing in order to rule out context-specific confounders. Under the CRR policy, local authorities are responsible for identifying the NGOs that manage reallocated Mafia properties and for pursuing their reuse for social activities. Table 3 shows that treated areas host more NGOs and have higher grade retention than control areas, and treated schools might also be located in municipalities with stronger institutional capacity, higher civic engagement, or a greater commitment to implementing the policy. These features could independently influence educational outcomes and generate differential trends between treated and control areas.

I first address the baseline differences in NGO presence and grade retention by reweighting control schools by the inverse of their propensity score of being treated, estimated on baseline NGO presence and grade retention (Hirano et al., 2003). Table C.1 shows that the reweighted estimate is -1.6 percentage points in all specifications, almost identical to the main result. Because few control areas have NGO presence comparable to the most NGO-dense treated areas, the weights concentrate on a small number of control schools; trimming observations with propensity scores outside $[0.1, 0.9]$ (Crump et al., 2009) reduces the estimate to -1.0 percentage points, which is no longer statistically significant. The direction of the effect is therefore robust to reweighting, while its precision depends on the comparison with the few control areas most similar to the treated ones.

Table C.1 Impact of repurposing Mafia real estate on the share of students dropping out from grades 11 to 13: inverse probability weighting.

	Dropout rates G11-G13			
	(1)	(2)	(3)	(4)
Repurpose = 1	-0.0161*	-0.0162*	-0.0162*	-0.0163**
	(0.0085)	(0.0084)	(0.0082)	(0.0081)
clustered SE	yes	yes	yes	yes
school FE	yes	yes	yes	yes
time FE	yes	yes	yes	yes
migration	no	yes	no	yes
retention	no	no	yes	yes
IPW	yes	yes	yes	yes
Observations	1,393	1,393	1,393	1,393
Number of schools	251	251	251	251
Mean dep. var.	0.0544	0.0544	0.0544	0.0544

Notes: TWFE model weighted by inverse probability weights. The treatment variable is equal to 1 whenever there is at least one repurposed Mafia property within the school catchment area. The propensity score of being ever treated is estimated by logit on baseline NGO presence and grade retention rates in grades 11 and 12; treated schools receive weight 1 and control schools $p/(1-p)$, and weights are constant within school. Column (1) represents the baseline accounting for school and time fixed effects, while in Columns (2) and (3) I control for students' migration and grade retention rates in grades 11 and 12, respectively. Column (4) reports the complete specification. All columns are estimated on the same sample. Standard errors, clustered at the school level.

To investigate the determinants of repurposing directly, I estimate a linear probability model of the probability that a catchment area hosts a repurposed Mafia property on NGO presence in the previous year and on baseline municipal characteristics not controlled for in the main specification. Since repurposing is recorded in the year in which the municipality and a local NGO sign the management agreement, NGO presence in the preceding year is the relevant measure, because municipalities must identify suitable partner organisations beforehand. I collect several measures of municipal characteristics that could influence both repurposing decisions and educational outcomes. First, I record whether municipalities had been dissolved for Mafia infiltration at baseline, from Avviso Pubblico.[31] Second, I collect

[31] Avviso Pubblico is an Italian national association created in 1996 to promote transparency in the public administration.

baseline municipal expenditure data from the Ministry of the Interior for four policy areas: youth policies, education, urban and territorial planning, and municipal police. These categories capture municipal priorities and capacity in areas related to property reuse and community development. Finally, I construct an indicator equal to 1 for municipalities that did not publish the list of confiscated properties available for reuse on their websites, following the classification of Falcone, Illustrazione, Giannone, Mennella, Ferrante and Martone (2021, 2022), which rates municipalities on the transparency of the information they publish on reallocated Mafia properties, as required by law. The model is:

$$P(Repurpose = 1)_{cmt} = \beta_0 + \beta_1 NGO_{c,t-1} + \beta_2 X'_m + \epsilon_{cmt} \tag{4}$$

where $P(Repurpose = 1)_{cmt}$ is the probability that school catchment area c in municipality m hosts a repurposed Mafia property in year t, $NGO_{c,t-1}$ is the number of NGOs in the catchment area in the previous year, and X'_m includes the baseline municipal characteristics described above.

Table C.2 reports the results. Column (3) shows that one additional NGO in the previous year increases the probability of repurposing by 0.13 percentage points, or 0.19% relative to the mean. Although small, this suggests that areas with stronger organisational capacity are slightly more likely to host repurposed properties. Municipal characteristics also matter. Municipalities that did not publish the list of confiscated properties are substantially less likely to repurpose them, by 50 percentage points. A doubling of youth-policy expenditure increases the probability of repurposing by 7.1 percentage points (10.1% relative to the mean), and a doubling of urban-planning expenditure by 13.0 percentage points (18.5%), while higher municipal police expenditure is associated with less repurposing. Dissolution for Mafia infiltration at baseline does not predict repurposing, suggesting that treatment assignment reflects current institutional capacity rather than past Mafia infiltration of the municipal administration. Because municipal characteristics vary across only ten municipalities, these coefficients should be read as descriptive.

Table C.2 Predictors of the repurposing of Mafia properties: linear probability model.

	(1)	(2)	(3)
	P(Repurpose = 1)		
NGOs $t-1$	0.0012***		0.0013***
	(0.0002)		(0.0002)
Dissolution = 1		0.1265*	-0.0490
		(0.0692)	(0.0635)
No transparency = 1		-0.5569***	-0.5033***
		(0.0699)	(0.0632)
Exp youth policies		0.0818**	0.1025***
		(0.0319)	(0.0283)
Exp education		0.0523	-0.0118
		(0.0614)	(0.0552)
Exp urban planning		0.0668	0.1878*
		(0.1101)	(0.1009)
Exp municipal police		-0.0762	-0.1800**
		(0.0950)	(0.0878)
clustered SE	yes	yes	yes
Observations	1,387	1,387	1,387
Number of schools	251	251	251
Mean dep. var.	0.702	0.702	0.702

Notes: LPM. The outcome is equal to 1 whenever there is at least one repurposed Mafia property within the school catchment area. NGOs $t-1$ is the number of NGOs in the catchment area in the previous year. Municipal characteristics are measured at baseline (2015): dissolution for Mafia infiltration; no transparency = 1 if the municipality did not publish the list of confiscated properties; expenditures are logs of municipal payments. Municipal characteristics vary across the 10 municipalities of the sample. All columns are estimated on the same sample. Standard errors, clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

To address these concerns, I re-estimate the main specification controlling for NGO presence in the previous year and for linear time trends interacted with the baseline municipal predictors of repurposing identified above: transparency, and expenditure on youth policies, urban planning and municipal police. This allows areas that differ in these characteristics to

follow different dropout trends. Table C.3 reports the results. The coefficient is remarkably stable across specifications: -1.58 percentage points in the baseline (Column 1), -1.50 when controlling for NGO presence (Column 2), -1.58 with the trends (Column 3), and -1.47 with both (Column 4), significant at the 10% level throughout. NGO presence in the previous year shows no independent association with dropout. Allowing instead for a separate linear trend for each municipality yields a similar point estimate, -1.3 percentage points, which is less precisely estimated ($p = 0.12$); with municipality $\times$ year fixed effects, the estimate is -1.2 percentage points ($p = 0.16$), as about 30% of school-years are in municipality-year cells with no variation in treatment status, so that these fixed effects absorb most of the identifying variation. The stability of the point estimate across these specifications indicates that the effect is not driven by systematic treatment assignment based on local organisational capacity or municipal characteristics.

Table C.3 Impact of repurposing Mafia real estate on dropout rates for G11-G13, controlling for NGO presence and trends in the municipal predictors of repurposing.

	(1)	(2)	(3)	(4)
	Dropout rate G11-G13			
Repurpose = 1	-0.0158*	-0.0150*	-0.0158*	-0.0147*
	(0.0081)	(0.0081)	(0.0081)	(0.0082)
NGOs $t-1$		0.00014		0.00015
		(0.00010)		(0.00011)
clustered SE	yes	yes	yes	yes
school FE	yes	yes	yes	yes
time FE	yes	yes	yes	yes
migration	yes	yes	yes	yes
retention	yes	yes	yes	yes
Trends: transparency, youth, urban planning, police expenditure	no	no	yes	yes
Observations	1,387	1,387	1,387	1,387
Number of schools	251	251	251	251
Mean dep. var.	0.0576	0.0576	0.0576	0.0576

Notes: TWFE model. The treatment variable is equal to 1 whenever there is at least one repurposed Mafia property within the school catchment area. NGOs $t-1$ is the number of NGOs in the catchment area in the previous year. Columns (3) and (4) include linear time trends interacted with the baseline (2015) municipal predictors of repurposing: no transparency in publishing the list of confiscated properties, and log expenditure on youth policies, urban planning and municipal police. All columns control for students' migration and grade retention rates in grades 11 and 12, and are estimated on the school-years for which the NGO lag is observed. Standard errors, clustered at the school level.

D Construction of Additional Variables

This appendix provides additional information on the data and variables used in this paper.

Schools' catchment areas. To test whether the treatment allocation and results are sensitive to changes in the construction of school catchment areas, I build an alternative version where I weight the Euclidean distance used to assign census blocks to schools based on schools' capacity. I proceed in three steps: first, I aggregate census blocks to their closest school based on simple Euclidean distance. Second, I multiply the distance by the average enrollment rate of each school and divide the result by each school's enrollment rate at the baseline as follows:

$$d_{ic}^{*} = d_{ic} \times \frac{\bar{E}}{E_i}$$

This way, schools with higher capacity, which show an above-average enrollment, will display larger catchment areas, accounting for supply-side factors in the allocation of census blocks. Third, I repeat this process for each of the 14 schools' specialisations to consistently align with a location-allocation approach. Only 17 of the sample schools change their treatment status under these new assumptions, relaxing the concern about artificial treatment allocation. Figure ?? compares schools' catchment areas obtained using Euclidean and weighted-Euclidean distances for language academic high schools in Naples.

As expected, capacity-weighted catchment areas show slightly different boundaries. I employ this alternative specification solely for robustness testing in Section **??**, where I examine treatment effects only in areas with unchanged boundaries to assess whether the main results are driven by schools with ambiguous treatment allocation across the two construction methods.

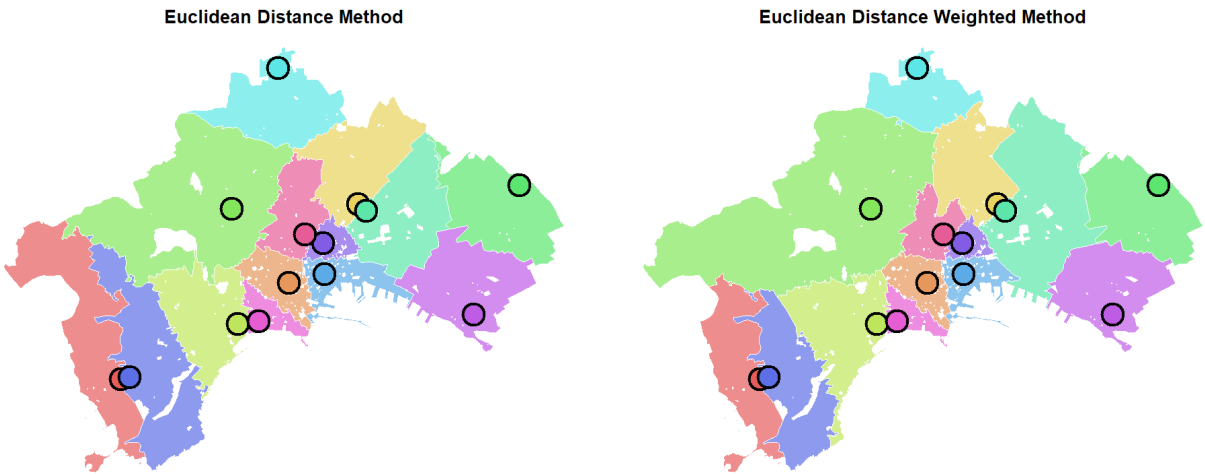


Figure D.1 Schools' catchment areas measured with Euclidean distance (left) and capacity-weighted distance (right)

Students' migration. I report below the measure I compute to control for school-specific students' migration patterns, as explained in Section **??**. I rely on the assumption that students who disappear from enrollment registries before turning 16 are more likely to be transferring to other schools rather than genuinely dropping out, given that education is compulsory until the age of 16. I build this measure following the same construction methodology as the main outcome variable, but capturing the share of students who do not complete grades 9 and 10 as their first and second years of high school. The measure is reported in Equation 6. Starting from the cohort of students who completed grades from 9 to 11 in year $t-1$, I subtract those who progressed to complete grade 12 in year t, and those who completed grades 10 and 11 in year t, expressing this as a share of the initial cohort who completed grades from 9 to 11 in year $t-1$.

$$Dropout_{G9-11_t} = \frac{Completed_{G9-11_{t-1}} - Completed_{G12_t} - Completed_{G10-11_t}}{Completed_{G9-11_{t-1}}} \tag{5}$$

Grade retention rate. It is possible that students repeating years due to inadequate academic performance will confound my measure of dropout by appearing enrolled in the same grade for both time $t-1$ and time t. To address this, I construct a proxy for grade retention rates for students enrolled on grades 11, 12, and 13. Figure 3 in Section **??** shows

that students following the standard educational timeline are expected to be 16 or 17 years old in grade 11, and 17 or 18 years old in grade 12.

I calculate the share of students repeating grade 11 as the number of students aged 18 or older as a share of all students completing grade 11 in time t. Similarly, the share of students repeating grade 12 is calculated as the number of students aged 19 or older as a share of all students completing grade 12 in time t. Equations 7 and 8 show the respective grade retention rates. Due to data limitations, I cannot compute the retention rate for grade 13, as I cannot observe whether students are older than 19 in grade 13. However, since the average retention rate declines from 1.2% in grade 11 to zero in grade 12, the grade 13 retention rate would likely be negligible.

$$Retention_{G11_t} = \frac{aged18_{G11_t}}{Completed_{G11_t}} \tag{6}$$

$$Retention_{G12_t} = \frac{aged19+_{G12_t}}{Completed_{G12_t}} \tag{7}$$

Schools' quality index. I construct a composite baseline indicator of school quality, drawing on two dimensions evaluated through a structured assessment framework used by the Italian schools, namely the annual Self-Evaluation Report (RAV). The first dimension, focusing on the school-level learning outcomes, encompasses four criteria: school grades, standardised test scores, civic competences, and long-term outcomes. The second dimension, which focuses on learning process, covers seven criteria related to the organisational and pedagogical functioning of the school: module planning, learning environment, inclusion and diversity, student guidance, school organisation, human resources development, and integration with the local community. For each dimension, I construct a sub-index by taking the row mean of the standardised scores assigned to each criterion of learning outcomes performance and learning process performance, respectively. The overall school performance indicator is then computed as the simple average of these two sub-indices, giving equal weight to outcomes and processes. This produces a composite score at the school level that captures both what students achieve and how the school operates to support that achievement, and serves as the baseline measure of school quality in the analysis.

Table D.1 RAV School Quality Indicators

Learning Outcomes		Learning Process	
ID criteria	**Description**	**ID criteria**	**Description**
21	School grades	31	Modules planning
22	Standardised test scores	32	Learning environment
23	Civic competences	33	Inclusion and diversity
24	Long-term outcomes	34	Student giudance
		35	School organisation
		36	HR development
		37	Integration with the local community

Heterogeneous effects of repurposing. To assess whether the estimated effects are driven by a specific type of activity offered in the repurposed properties, I decompose the treatment effect by activity category (education, employment, welfare, culture). Because these categories are not mutually exclusive — a single school catchment area may contain several repurposed properties offering different types of activity — I estimate a separate regression for each category rather than including all categories jointly. Within each regression, I split treated areas into two groups according to whether they offer the activity of interest, and compare both against the same set of never- and not-yet-treated control areas. Formally, for a given activity category I estimate

$$DropoutG11\text{-}G13_{cnmt} = \beta_1(Category = 1)_{ct} + \beta_2(Category = 0)_{ct} + \beta_3 X'ct + \delta c + \eta_t + \epsilon_{cnmt} \tag{8}$$

where $(Category = 1)_{ct}$ equals 1 when the school catchment area contains at least one repurposed property offering activities of the relevant type, and $(Category = 0)_{ct}$ equals 1 when area c is treated but none of its repurposed properties offer that type of activity. Control areas take the value zero for both indicators and serve as the omitted reference group. The vector X'_{ct} contains the same controls as in the baseline specification, and δ_c and η_t denote catchment-area and time fixed effects, respectively. Standard errors are clustered at the school level. The coefficient β_1 captures the dropout effect in treated areas that provide the activity in question, while β_2 captures the effect in treated areas that do not. Comparing the two isolates the marginal contribution of each activity type: if a category were the main driver of the overall effect, its β_1 should be substantially larger in magnitude, and more precisely estimated, than the corresponding β_2. The estimates are reported in Figure 9.

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